
ChronoMorph: Dual-View Generative Classification for Few-Shot Time Series

Anonymous Author(s)

Affiliation

Address

email

Abstract

Few-shot time series classification (TSC) is challenging because each class provides only a few labeled sequences, yet reliable prediction often requires capturing both temporal dynamics and waveform morphology, as human analysts do when combining numerical evolution with shape-based evidence. Recent pretrained causal language models offer a promising interface for data-efficient transfer, but directly applying them to TSC exposes two mismatches: continuous time-series signals are not naturally aligned with the language-token space, and unconstrained autoregressive decoding is poorly suited to closed-set classification. We propose **ChronoMorph**, a dual-view generative transfer framework for few-shot TSC. ChronoMorph encodes each sequence through a temporal branch that captures ordered dependencies and a morphology branch that converts each waveform into a morphology map. The resulting continuous tokens are fused and projected into a pretrained causal language model via stage-wise transfer pretraining. Instead of unconstrained decoding, ChronoMorph performs supervised one-pass scoring over dataset-specific class tokens, which can be initialized and regularized by label-verbalizer prototypes when class names are semantically meaningful. Extensive few-shot experiments on the UCR benchmark and additional external datasets show that ChronoMorph consistently outperforms strong task-specific, Transformer-style, and LLM/foundation baselines. Further analyses validate the contributions of dual-view modeling, transfer pretraining, and the class-token interface.

1 Introduction

Time series classification (TSC) is a fundamental problem in healthcare monitoring, industrial diagnosis, sensor analytics, and scientific discovery [Ismail Fawaz et al., 2019]. In many practical scenarios, however, labeled time series are difficult to obtain: annotation often requires domain expertise, new categories may appear with only a few examples, and class distributions can be highly imbalanced. Few-shot TSC therefore requires more than fitting a classifier on a small support set. A successful model must retain transferable temporal evidence while inducing a stable decision rule from very limited supervision [Tang et al., 2020].

A key challenge is that discriminative evidence in time series is inherently heterogeneous. Some classes are distinguished by chronological dynamics, such as local trends, amplitudes, phase shifts, and long-range evolution. Others depend more strongly on waveform morphology, such as recurring peaks, valleys, cycles, motifs, or discriminative subsequences. This dual nature is reflected in the history of TSC methods: shapelet-based approaches search for local waveform patterns predictive of class membership, convolutional and patch-based models capture ordered temporal context through receptive fields, and time-series imaging methods transform sequences into visual structures to expose global morphology and cross-local relations [Ye and Keogh, 2009b, Dempster et al., 2020, Ismail Fawaz et al., 2020, Wang and Oates, 2015a]. Under few-shot supervision, this heterogeneity

38 becomes especially problematic. With only a few labeled examples, a model that emphasizes a
39 single form of evidence may latch onto unstable cues and miss the complementary temporal or
40 morphological structure needed to separate fine-grained classes.

41 Recent studies have explored adapting pretrained language models to time-series analysis through
42 numerical serialization, reprogramming, embedding alignment, semantic anchors, and parameter-
43 efficient fine-tuning [Brown et al., 2020, Gruver et al., 2023, Zhou et al., 2023, Jin et al., 2024b, Sun
44 et al., 2023, Jin et al., 2024a, Chang et al., 2025, Zhang et al., 2024, Jiang et al., 2024]. These models
45 are attractive for few-shot TSC because they provide reusable pretrained computation, conditional-
46 generation objectives, and token-level scoring interfaces. However, many language-model-based
47 time-series methods are designed around forecasting, reconstruction, or unconstrained text generation,
48 while few-shot TSC requires closed-set discrimination over dataset-specific labels. Moreover, time-
49 series evidence is continuous, domain-specific, and structurally different from the discrete token
50 sequences on which language models are pretrained. As a result, the central challenge is not merely to
51 adapt a language model to time series, but to design an interface that preserves task-relevant temporal
52 evidence, aligns it with the language-model input space, and supports reliable closed-set prediction.

53 This perspective suggests three requirements for LLM-based few-shot TSC: (i) the representation
54 should preserve both chronological dynamics and morphology-oriented evidence, since either form
55 may carry the decisive class signal; (ii) continuous time-series features should be aligned with the
56 embedding space expected by the pretrained language model before downstream adaptation; and (iii)
57 prediction should compare valid class candidates directly, rather than relying on unconstrained text
58 generation [Raffel et al., 2020, Hokamp and Liu, 2017, Koo et al., 2024]. Failing to satisfy any of
59 these requirements can weaken the role of the language model as a reliable few-shot classifier and
60 reduce it to a loosely connected generative component.

61 We propose **ChronoMorph**, a dual-view generative transfer framework for few-shot TSC. For each
62 sequence, ChronoMorph constructs two complementary views. A temporal branch models ordered
63 local relations and long-range dynamics, while a morphology branch transforms the waveform
64 into a two-dimensional morphology map that makes shape-level structure accessible to a frozen
65 vision encoder. The two views are fused into continuous tokens and projected into a pretrained
66 causal language model through stage-wise transfer pretraining, which reduces the mismatch between
67 time-series representations and the language-model input space. For downstream classification,
68 ChronoMorph replaces unconstrained decoding with supervised one-pass scoring over dataset-
69 specific class tokens, turning the generative interface into a discriminative closed-set classifier. When
70 class names carry meaningful semantics, label verbalizers initialize and regularize the corresponding
71 class-token embeddings, allowing semantic information to help the same single-token decision rule
72 without requiring multi-token phrase decoding.

73 Our main contributions are: (i) we formulate LLM-based few-shot TSC as an interface-design
74 problem that jointly considers complementary time-series evidence, continuous-to-language align-
75 ment, and closed-set prediction; (ii) we introduce a dual-view time-series-to-language encoder that
76 preserves chronological and morphology-oriented evidence before transferring continuous tokens
77 into a pretrained causal language model through stage-wise pretraining; and (iii) we propose a
78 supervised class-token decision interface for one-pass scoring over valid labels, and extend it with
79 label-verbalizer initialization and regularization when class names are semantically meaningful.
80 Across 128 UCR datasets, ChronoMorph achieves the best average accuracy and average rank un-
81 der 1-, 2-, 5-, and 10-shot settings; additional semantic-label experiments further show when label
82 semantics can improve the same class-token interface.

83 2 Related Work

84 **LLMs and foundation models for time series.** Recent surveys show that foundation-model
85 research for time series has expanded from adapting pretrained language models to building native
86 temporal foundation models Zhang et al. [2024], Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025].
87 Representative adaptation-based methods include TEST Sun et al. [2023], which aligns time-series
88 embeddings with the LLM embedding space through instance-, feature-, and text-prototype-aware
89 contrastive learning; Time-LLM Jin et al. [2024a], which reprograms time-series patches into a
90 frozen LLM via text prototypes; and LLM4TS Chang et al. [2025], which uses a two-stage alignment-
91 and-task fine-tuning strategy. OpenTSLM Langer et al. [2025] moves toward time-series language

models that treat time series as a native modality through soft-prompt or cross-attention interfaces. At the same time, recent ablation studies caution that LLM components do not automatically improve time-series performance when the modality interface is poorly designed Tan et al. [2024], Zheng et al. [2024]. The closest few-shot classification line is LLMFew Chen et al. [2025], which introduces a patch-wise temporal convolution encoder and LoRA-tuned decoder for few-shot multivariate TSC. In contrast, our work focuses on dual-view temporal evidence, explicit time-series-to-language transfer, and generative label-token scoring rather than an external classifier head.

Temporal and morphological evidence in TSC. Classical and deep TSC methods support the view that discriminative evidence can appear at multiple structural levels. Shapelet methods identify local subsequences with high class-discriminative power Ye and Keogh [2009a], Tang et al. [2020], convolutional methods such as InceptionTime and ROCKET exploit local receptive fields and kernel responses at multiple scales Ismail Fawaz et al. [2020], Dempster et al. [2020], and patch-based Transformers preserve local subseries information while enabling longer-context attention Nie et al. [2022]. These methods primarily operate in the chronological-dependency space. In parallel, time-series imaging methods transform one-dimensional signals into two-dimensional structures such as Gramian angular fields and Markov transition fields, enabling vision models to capture global shape and transition patterns Wang and Oates [2015b]. Recent work further shows that visualization can improve LLM reasoning over time series and that frozen vision transformers applied to imaged time series can produce strong classification representations Liu et al. [2025], Roschmann et al. [2025]. These two lines motivate our dual-space hypothesis: few-shot TSC labels may depend on both chronological dependencies and morphological structures.

Generative classification and semantic-prior label spaces. Casting classification as generation is widely used in text-to-text learning, where labels are generated as target tokens under the same sequence objective as other tasks Raffel et al. [2020]. However, unconstrained decoding is mismatched with closed-set classification because many generated strings are irrelevant to the valid label set. Prior work on constrained decoding motivates restricting generation to valid lexical or structural outputs Hokamp and Liu [2017], Koo et al. [2024]. ChronoMorph adopts the same closed-set principle but implements the final decision as supervised one-pass class-token scoring: the model learns dataset-specific class-token rows and compares their next-token logits rather than performing unconstrained generation. For datasets with meaningful label names, we further use label verbalizers as semantic priors over the same class-token rows. This keeps the decision aligned with the autoregressive interface while avoiding the length bias, shared-prefix artifacts, and language-prior sensitivity of direct phrase-likelihood classification.

3 Method

3.1 Unified Time-Series-to-Language Formulation

Given a time series $x \in \mathbb{R}^T$, ChronoMorph casts time-series prediction as conditional language modeling with continuous time-series tokens inserted into a textual prompt. Let $\tau(\cdot)$ be the tokenizer, E_{lm} the LLM input embedding matrix, P_ψ the projector into the LLM hidden space, and $Z_x \in \mathbb{R}^{N_x \times d_e}$ the token sequence produced by the time-series encoder. For a prefix p^{pre} , an optional time-series description p^{ts} , and a suffix p^{post} , the mixed input embeddings are

$$H_x = [E_{\text{lm}}(\tau(p^{\text{pre}})); E_{\text{lm}}(\tau(p^{\text{ts}})); P_\psi(Z_x); E_{\text{lm}}(\tau(p^{\text{post}}))] . \quad (1)$$

For an answer sequence $a = (a_1, \dots, a_M)$, ChronoMorph optimizes the standard causal objective but masks the loss outside the answer span:

$$\mathcal{L}_{\text{LM}}(a \mid x, p) = - \sum_{m=1}^M \log p_\Theta(a_m \mid H_x, E_{\text{lm}}(a_{<m})) . \quad (2)$$

Thus question answering, captioning, and classification share the same conditional-generation interface; classification is implemented as label-token prediction rather than as a separate discriminative head. Figure 1 summarizes the pipeline.

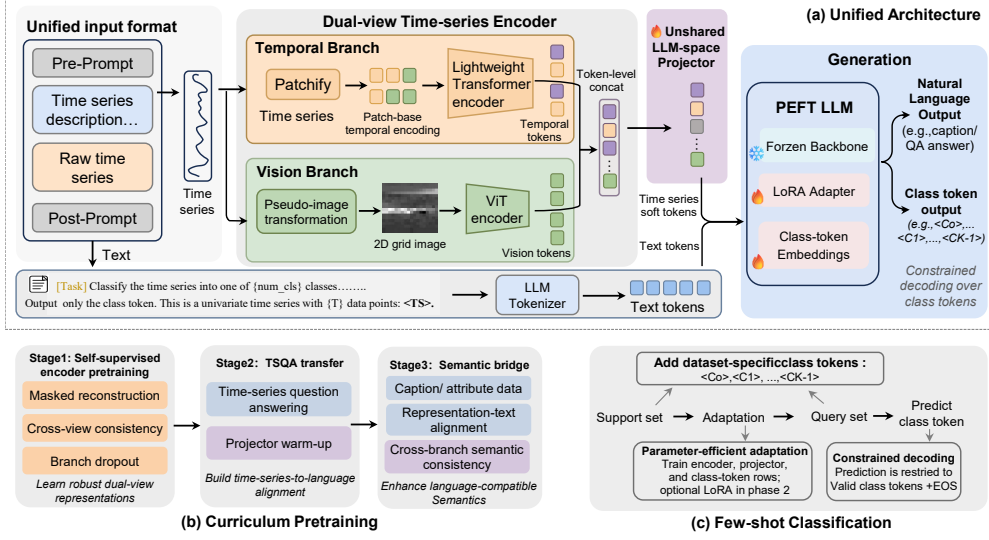


Figure 1: Overview of ChronoMorph. A temporal branch and a morphology-map visual branch produce dual-view time-series tokens, which are fused and projected into a pretrained causal LLM. Stage I learns dual-view representations, Stage II learns time-series-conditioned generation, and optional Stage III adds semantic enrichment; the main UCR benchmark uses the Stage II checkpoint. Downstream few-shot classification uses dataset-specific class tokens and one-pass scoring over the valid label set.

3.2 Dual-View Time-Series Tokens

ChronoMorph preserves two complementary evidence spaces before language conditioning. The temporal branch first extracts overlapping patches

$$R_i = x_{(i-1)s+1:(i-1)s+\ell}, \quad i = 1, \dots, N_t, \quad N_t = \left\lfloor \frac{T-\ell}{s} \right\rfloor + 1, \quad (3)$$

where ℓ and s denote patch length and stride. A dynamic PatchTST-style encoder with sinusoidal positions then maps the patch sequence to temporal tokens,

$$Z^t = G_t([W_t R_1 + \pi_1; \dots; W_t R_{N_t} + \pi_{N_t}]), \quad (4)$$

using three Transformer blocks, hidden size 128, eight attention heads, and patch length/stride 16/8 in the main experiments. This branch targets ordered local relations, phase shifts, and long-range chronological dependencies.

The visual branch is designed to expose waveform morphology to a pretrained vision encoder without using dataset-level image statistics. We first apply a deterministic per-sequence transform. Each sequence is normalized by its own median and interquartile range,

$$\bar{x}_t = \frac{x_t - \text{median}(x)}{\text{IQR}(x) + \epsilon}, \quad (5)$$

where the statistics are computed only from the current sequence. To retain local waveform continuity while making non-adjacent windows jointly visible, we set the window length to $q = \lfloor \sqrt{T} \rfloor$ and the window stride to $\delta = \max(1, \lfloor \rho q \rfloor)$, with $\rho = 0.1$. The normalized signal is unfolded into an overlapping window grid, contrast-normalized, resized to the vision-backbone resolution, and replicated across RGB channels:

$$I(x) = \text{RGB}(\text{Resize}(\gamma(\text{Unfold}_{q,\delta}(\bar{x})))), \quad (6)$$

where $\gamma(\cdot)$ denotes per-sample image-grid normalization and contrast adjustment. The transform is fixed and the vision backbone remains frozen; adaptation is handled by the lightweight projectors introduced below. We feed $I(x)$ to facebook/dinov2-base and use the patch tokens from its fourth hidden layer, excluding the class token, as morphology-oriented tokens:

$$Z^v = G_v(I(x)). \quad (7)$$

157 The two streams are mapped to the same encoder dimension by lightweight MLP projectors and
 158 fused by token concatenation in the default configuration,

$$\tilde{Z}^t = Q_t(Z^t), \quad \tilde{Z}^v = Q_v(Z^v), \quad Z_x = [\tilde{Z}^t; \tilde{Z}^v]. \quad (8)$$

159 This late token-level fusion avoids prematurely collapsing chronological and morphological cues.
 160 Pooled temporal, visual, and fused representations are also used by auxiliary pretraining objectives.

161 3.3 Curriculum Bridging

162 The few-shot support set is too small to learn the time-series encoder, the time-series-to-LLM interface,
 163 and the decision rule from scratch. ChronoMorph therefore uses a staged transfer curriculum. Stage I
 164 trains the dual-view encoder with masked patch reconstruction, VICReg losses on two augmented
 165 views of the temporal, visual, and fused pooled representations, and branch dropout. For two
 166 augmentations $x^{(1)}$ and $x^{(2)}$, let $h_b^{(r)} = \text{Pool}_b(Z_b(x^{(r)}))$ denote the pooled representation of branch
 167 $b \in \{t, v, f\}$, where f denotes the fused branch. We use

$$\mathcal{L}_{\text{ssl}}^b = \alpha \|h_b^{(1)} - h_b^{(2)}\|_2^2 + \beta \sum_{r=1}^2 \sum_j \max(0, 1 - \sigma_j(h_b^{(r)})) + \gamma \sum_{r=1}^2 \|\text{offdiag}(\text{Cov}(h_b^{(r)}))\|_2^2, \quad (9)$$

168 which combines invariance, variance, and covariance terms. The masked reconstruction loss predicts
 169 temporal patches from a masked input,

$$\mathcal{L}_{\text{rec}} = \frac{1}{2} \sum_{r=1}^2 \|R_\omega(Z_t(\mathcal{M}(x^{(r)}))) - \mathcal{P}_{\ell,s}(x^{(r)})\|_2^2, \quad (10)$$

170 where $\mathcal{M}$ masks a contiguous subsequence, $\mathcal{P}_{\ell,s}$ extracts temporal patches, and R_ω is the reconstruc-
 171 tion head. The Stage I objective is

$$\mathcal{L}_I = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_t \mathcal{L}_{\text{ssl}}^t + \lambda_v \mathcal{L}_{\text{ssl}}^v + \lambda_f \mathcal{L}_{\text{ssl}}^f. \quad (11)$$

172 This stage uses raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split
 173 sequences from a fixed 98-dataset UCR pool; no UCR labels or official TEST split are used. Because
 174 this pool is drawn from the same archive as the UCR benchmark, the main results should be interpreted
 175 as unlabeled-source transfer rather than strict leave-dataset-out pretraining.

176 Stage II attaches the projector and frozen causal LLM and trains time-series question answering,

$$\mathcal{L}_{\text{II}} = \mathbb{E}_{(x,p,a) \sim \mathcal{D}_{\text{TSQA}}} [\mathcal{L}_{\text{LM}}(a | x, p)], \quad (12)$$

177 first warming up the projector and then jointly optimizing the encoder and projector. The Stage II
 178 checkpoint is the default initialization for UCR few-shot classification. Stage III is optional and mixes
 179 TSQA, M4 captioning, and synthetic attribute supervision with representation-text alignment and
 180 branch-consistency losses; it is used only for semantic enrichment, not for the default UCR results.
 181 Appendix B.1 gives the stage-wise data details.

182 3.4 Class-Token Decision Interface

183 For a dataset D with C classes, we introduce class tokens $\mathcal{C}_D = \{\langle c_0 \rangle, \dots, \langle c_{C-1} \rangle\}$ with vocabulary
 184 indices $\mathcal{V}_D = \{v_0, \dots, v_{C-1}\}$. Let $h_\Theta(x, p)$ be the final hidden state at the last prompt position after
 185 conditioning on H_x , and let W_{out} be the LLM output embedding matrix. The one-pass score for
 186 class c is

$$s_c(x) = [W_{\text{out}} h_\Theta(x, p)]_{v_c}, \quad \hat{y} = \arg \max_{c \in \{0, \dots, C-1\}} s_c(x). \quad (13)$$

187 The supervised target answer is the single class token $\langle c_y \rangle$, giving

$$\mathcal{L}_{\text{cls-token}} = -\log p_\Theta(v_y | H_x). \quad (14)$$

188 This interface differs from attaching a conventional classifier head to the final representation. The
 189 decision is still made through the autoregressive output distribution of the LLM, but the candidate
 190 space is restricted to supervised class tokens that are valid for the current dataset. It also differs from
 191 phrase-likelihood classification with natural-language labels: each class is represented by exactly one

learned token, so the score is not affected by label length, shared subword prefixes, or uncontrolled continuations.

Only the input embeddings and output rows of the new class tokens are trainable; the rest of the vocabulary remains fixed. This design turns generative classification into supervised decoding over a compact, dataset-specific label space: during training, the target answer is the class token for the labeled support example, and during inference the model is allowed to score only valid class tokens. For anonymous-label datasets such as UCR, these tokens are closed-set identifiers. For semantic-label datasets, each class has a label card with a canonical name and template-consistent verbalizers $\mathcal{S}_c$. We average their token embeddings to form a prototype,

$$p_c = \frac{1}{|\mathcal{S}_c|} \sum_{r \in \mathcal{S}_c} \frac{1}{|\tau(r)|} \sum_{u \in \tau(r)} E_{\text{lm}}(u), \quad (15)$$

rescale it to the average vocabulary-row norm, and initialize the corresponding class-token rows from this prototype. During adaptation, we regularize the input row e_{v_c} and output row w_{v_c} toward the detached prototype $\tilde{p}_c$:

$$\mathcal{L}_{\text{sem}} = \frac{1}{C} \sum_{c=0}^{C-1} [1 - \cos(e_{v_c}, \tilde{p}_c) + 1 - \cos(w_{v_c}, \tilde{p}_c)], \quad \mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{cls-token}} + \lambda_{\text{row}} \mathcal{L}_{\text{sem}}. \quad (16)$$

This injects label semantics as a prior while preserving a stable single-token decision space.

3.5 Downstream Adaptation

Few-shot adaptation uses a two-phase schedule. In the warm-up phase, we optimize the time-series encoder, projector, and dataset-specific class-token rows,

$$\Theta_{\text{warm}} = \{\theta_{\text{enc}}, \psi, e_{\mathcal{V}_D}, w_{\mathcal{V}_D}\}, \quad \min_{\Theta_{\text{warm}}} \mathcal{L}_{\text{adapt}}. \quad (17)$$

In the joint phase, we additionally enable LoRA adapters on the language side,

$$\Theta_{\text{joint}} = \Theta_{\text{warm}} \cup \{\Delta\Theta_{\text{LoRA}}\}, \quad \Delta W_l = B_l A_l, \quad \text{rank}(\Delta W_l) \leq r. \quad (18)$$

The main benchmark uses LoRA rank 16 and alpha 32, applied to the LLM adaptation modules exposed by the implementation. This schedule keeps the early optimization problem compact while still allowing limited language-side adaptation after the time-series interface has stabilized.

4 Experiments

4.1 Experimental Setup

Protocol. We evaluate on all 128 datasets in the UCR Archive 2018 with a fixed full-label-space few-shot protocol. For each dataset D and support size $S \in \{1, 2, 5, 10\}$, all classes are included; we sample up to S labeled examples per class from the official training split and evaluate on all examples in the official test split. All methods use the same sampled support sets within each of five independent runs.

Pretraining data and default initialization. The main benchmark uses the Stage II checkpoint: Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pool; Stage II is trained on the TSQA training split. Stage III is not used for Table 1. No pretraining stage uses any official UCR TEST split or UCR class label. For UCR datasets included in the 98-dataset pool, ChronoMorph may have seen unlabeled official TRAIN sequences before downstream adaptation; after this pretraining, each target run uses only the sampled support labels for supervised adaptation.

Baselines and metrics. We compare against CNN/prototype baselines (RESNET, TAPNET, COSCO-RESNET), Transformer-style or decomposition-based baselines (PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER), and the LLM/foundation-style GPT4TS baseline. Accuracy is averaged over five runs and then macro-averaged over datasets. Avg averages the four shot settings, and AvgRank averages dataset-level ranks across shots; lower rank is better. Standard deviations and full per-dataset tables are in Appendix A.

Table 1: Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. We report the mean over five runs. Avg and AvgRank average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN/prototype	ResNet	47.79	53.98	61.27	66.24	57.32	4.89
CNN/prototype	TapNet	45.40	50.79	59.47	64.93	55.15	5.45
CNN/prototype	COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear	44.21	48.01	52.30	55.43	49.99	7.24
Transformer/TS	PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
Transformer/TS	TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
Transformer/TS	Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
Transformer/TS	Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
Transformer/TS	FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
Transformer/TS	Informer	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	ChronoMorph	48.31	55.55	64.14	69.16	59.29	4.24

Fairness and training details. All methods share the same official TRAIN support pool, TEST query split, and sampled support indices. We do not add a validation split by default; each method follows its runner’s checkpoint-selection rule, and some baselines select checkpoints using support-set performance. Appendix B.8 lists the training recipes and hyperparameters.

Implementation details. ChronoMorph uses meta-llama/Llama-3.2-1B, facebook/dinov2-base, temporal patch length 16, stride 8, hidden size 128, and the Stage II checkpoint. Downstream adaptation uses 5 warm-up epochs, 55 joint-training epochs, and LoRA with rank 16 and alpha 32. Additional details are in Appendix B.

4.2 Main Results

Benchmark-level comparison. Table 1 shows that ChronoMorph obtains the highest mean accuracy under all four support sizes: 48.31, 55.55, 64.14, and 69.16 for 1-, 2-, 5-, and 10-shot. The strongest non-ChronoMorph baseline is COSCO-RESNET, with 47.88, 54.95, 63.55, and 67.18. Thus the improvement over the strongest specialized CNN/prototype baseline is modest in accuracy, especially in the lowest-shot regimes, but it is consistent across support sizes and is also reflected in the best AvgRank (4.24 vs. 4.70 for COSCO-RESNET). The gap is more substantial against Transformer-style and LLM/foundation baselines: ChronoMorph improves Avg by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results suggest that the proposed interface is most clearly beneficial relative to generic sequence and LLM-style baselines, while remaining competitive with strong specialized CNN/prototype methods.

Dataset-level robustness. Table 2 reports Win/Tie/Loss counts against representative baselines. ChronoMorph wins substantially more often than it loses against PATCHTST, TIMESNET, and GPT4TS. Against the strongest CNN/prototype baselines, the picture is closer: ChronoMorph is nearly balanced with COSCO-RESNET and RESNET at several shot levels, which is consistent with the modest average-accuracy margins in Table 1.

Additional external datasets. We also evaluate MIT-BIH, SleepEDF, and CinC2017 under full-label-space 10-, 20-, and 30-shot protocols, using 10 runs for each external setting. Table 3 reports the average over these shot settings, with the full table in Appendix B.5. ChronoMorph obtains the best average accuracy on MIT-BIH and CinC2017 and the second-best overall external average, but it is clearly behind RESNET and TAPNET on SleepEDF. We therefore treat these experiments as evidence of transfer beyond UCR, not as a claim of uniform dominance across external domains.

Table 2: Win/Tie/Loss of ChronoMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

Table 3: Average few-shot accuracy (%) on three external datasets, computed over 10-, 20-, and 30-shot settings with 10 runs per setting. Best is bold and second-best is underlined.

Model	MIT-BIH	SleepEDF	CinC2017	Avg.
ChronoMorph	61.04	47.16	34.63	<u>47.61</u>
ResNet	50.23	66.84	<u>33.63</u>	50.23
TapNet	38.73	<u>66.70</u>	31.29	45.57
COSCO-ResNet	36.19	62.74	32.48	43.80
PatchTST	50.46	28.92	29.21	36.20
OneFitsAll	43.42	27.05	29.19	33.22
TimesNet	53.79	23.23	28.36	35.13
DLinear	<u>57.29</u>	19.81	24.87	33.99
Autoformer	41.64	26.48	27.45	31.86
Crossformer	56.76	24.93	25.62	35.77
FEDformer	56.93	21.79	25.05	34.59
Informer	53.25	22.45	28.39	34.70

4.3 Computational Efficiency and Adaptation Cost

Because ChronoMorph reuses a pretrained causal LLM, total parameters and adapted parameters must be reported separately. Table 4 shows the main trade-off: ChronoMorph updates only 12.3M parameters per target dataset, but the frozen LLM backbone dominates memory and inference cost. ChronoMorph is therefore most appropriate when transfer accuracy and reusable adaptation matter more than compact deployment. The complete efficiency table, including throughput and additional baselines, is in Appendix B.7.

4.4 Ablation Studies

Table 5 summarizes the main top-level UCR ablations on all 128 datasets. Each ablation follows the same five-run full-label-space few-shot protocol and modifies only the component under study. Focused compact views with the corresponding drops are provided in Tables 20–22. Because UCR labels are anonymous, the decision ablation tests closed-set class-token scoring rather than semantic label priors. The no-LLM ablation has a different completed coverage set and is therefore reported separately in Appendix Table 23.

Effect of curriculum pretraining. Removing curriculum pretraining drops Avg from 59.29 to 55.94, with shot-level accuracies decreasing from 48.31/55.55/64.14/69.16 to 44.70/51.81/60.65/66.60 for 1-/2-/5-/10-shot, respectively. The focused view in Appendix Table 20 reports a 3.35-point average drop, suggesting that the aggregate Stage I+II initialization helps when downstream supervision is too limited to learn the representation and LLM interface from scratch. This ablation removes the full pretraining pipeline at once and should not be read as isolating each substage.

Effect of dual-view time-series-to-language modeling. Temporal-only and visual-only variants reach Avg scores of 54.21 and 55.09, below the full model’s 59.29. Appendix Table 21 shows that removing the visual branch yields a 5.08-point average drop, while removing the temporal branch yields a 4.20-point drop. The temporal-only variant is weaker at 1-, 2-, and 5-shot, whereas the

Table 4: Compact efficiency comparison under the 10-shot UCR protocol. Runtime and memory are measured on the representative UCR runtime subset. Total parameters count the deployed model; updated parameters count downstream-adapted parameters only.

Model	Total params	Updated params	Peak GB	Infer ms/ex.
ChronoMorph	1.28B	12.3M	13.75	156.85
COSCO-ResNet	662.1K	662.1K	0.58	0.25
ResNet	629.1K	629.1K	0.33	0.26
PatchTST	665.4K	598.3K	1.54	0.75
GPT4TS	82.2M	1.1M	0.56	0.86

Table 5: Summary of top-level ablations on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined. Since UCR labels are anonymous, the label-space ablation isolates closed-set class-token scoring rather than semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg
ChronoMorph	48.31	55.55	64.14	69.16	59.29
w/o Curriculum Pretraining	44.70	<u>51.81</u>	60.65	<u>66.60</u>	<u>55.94</u>
Temporal only	43.51	50.55	58.40	<u>64.39</u>	54.21
Visual only	<u>45.88</u>	<u>50.02</u>	<u>60.86</u>	63.58	55.09
w/o Class-Token Label-Space Scoring	44.46	50.44	<u>60.08</u>	65.59	55.14

visual-only variant is weaker at 2- and 10-shot. Neither single view dominates across all shots, which supports the intended complementarity between chronological and morphological evidence.

Effect of class-token label-space scoring. Disabling class-token label-space scoring reduces Avg from 59.29 to 55.14. As detailed in Appendix Table 22, the corresponding shot-level accuracies decrease from 48.31/55.55/64.14/69.16 to 44.46/50.44/60.08/65.59, giving a 4.15-point average drop. This indicates that, under the same adaptation budget, unconstrained generative adaptation is less stable for closed-set few-shot classification than directly scoring the dataset-specific label tokens.

Effect of semantic label priors. For external datasets with meaningful label names, Table 6 compares anonymous class-token rows with rows initialized and regularized from label-verbalizer prototypes. Semantic priors improve the 10/20/30-shot average on MIT-BIH, SleepEDF, and CinC2017, increasing the overall average from 47.61 to 50.43. The full appendix table shows that the benefit is not uniform at every shot, with a decrease on CinC2017 20-shot, but the averaged result supports the semantic-prior class-token space as a useful extension of supervised closed-set decoding.

Effect of the LLM backbone. The no-LLM variant is evaluated on the 120 UCR datasets covered by both the full and no-LLM runs, so we report it separately in Appendix D, with the compact summary in Table 23. On this shared subset, replacing the language pathway with a lightweight linear classifier reduces Avg from 58.67 to 52.31, a 6.36-point drop. The decrease is consistent across support sizes: 47.93 to 42.95 at 1-shot, 54.98 to 48.20 at 2-shot, 63.27 to 57.06 at 5-shot, and 68.48 to 61.01 at 10-shot. This suggests that, under our implementation and adaptation schedule, the language pathway is beneficial; however, this ablation does not isolate whether the gain comes from pretrained knowledge, additional capacity, or the token-scoring interface.

Overall, the ablations support the central design: curriculum pretraining improves transferability, dual-view modeling preserves more complete evidence, closed-set class-token scoring stabilizes the final decision, and semantic-prior row initialization and regularization further improve semantic-label transfer.

Table 6: Semantic-prior class-token ablation on external datasets. Values are average accuracy (%) over 10-, 20-, and 30-shot settings with 10 runs per setting. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH	SleepEDF	CinC2017	Avg.
Anonymous class tokens	61.04	47.16	34.63	47.61
Semantic-prior class tokens	64.18	50.83	36.27	50.43
Δ	+3.14	+3.67	+1.64	+2.82

5 Limitations

ChronoMorph improves average accuracy and average rank, but its gains over the strongest CNN/prototype baselines are modest in extremely low-shot settings. The main empirical message is therefore not that LLM-based TSC now universally dominates specialized CNN-style few-shot classifiers, but that a carefully designed time-series-to-language interface can substantially outperform previous Transformer-style and LLM/foundation-style baselines while remaining competitive with strong task-specific methods. ChronoMorph also has a larger inference footprint than compact CNN baselines because it reuses a pretrained causal LLM, even though only a small fraction of parameters is updated during downstream adaptation. In our efficiency probe, this makes ChronoMorph two to three orders of magnitude slower at inference than compact CNN/prototype baselines. The current UCR pretraining setting also uses unlabeled official TRAIN sequences from a fixed subset of UCR datasets, so it does not establish strict leave-dataset-out transfer for every UCR target dataset. The external semantic-prior results improve the averaged performance on all three semantic-label datasets, but individual shot-level effects still vary, so label semantics should be viewed as a prior over the class-token rows rather than as a replacement for supervised adaptation. ChronoMorph may benefit applications where labeled time-series data are scarce, such as physiological monitoring or industrial inspection, but the present study is not a deployment-ready clinical or safety-critical system. Any deployment in such domains would require domain-specific validation, privacy review, and failure analysis beyond the scope of this paper.

References

- T. B. Brown, B. Mann, B. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.
- C. Chang, W.-Y. Wang, W.-C. Peng, and T.-F. Chen. Llm4ts: Aligning pre-trained llms as data-efficient time-series forecasters. *ACM Transactions on Intelligent Systems and Technology*, 16(3): 1–20, 2025.
- Y. Chen, Z. Li, C. Yang, X. Wang, and G. Xu. Large language models are few-shot multivariate time series classifiers. *Data Mining and Knowledge Discovery*, 39(5):66, 2025.
- A. Dempster, F. Petitjean, and G. I. Webb. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery*, 34(5): 1454–1495, 2020.
- N. Gruver, M. Finzi, S. Qiu, and A. G. Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- C. Hokamp and Q. Liu. Lexically constrained decoding for sequence generation using grid beam search. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1535–1546, 2017.
- H. Ismail Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Muller. Deep learning for time series classification: a review. *Data mining and knowledge discovery*, 33(4):917–963, 2019.
- H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6):1936–1962, 2020.

353 Y. Jiang, Z. Pan, X. Zhang, S. Garg, A. Schneider, Y. Nevmyvaka, and D. Song. Empowering time
354 series analysis with large language models: A survey. *arXiv preprint arXiv:2402.03182*, 2024.

355 M. Jin, S. Wang, L. Ma, Z. Chu, J. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-f. Li, S. Pan, et al.
356 Time-llm: Time series forecasting by reprogramming large language models. In *International*
357 *Conference on Learning Representations*, 2024a.

358 M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, and Q. Wen.
359 Time-LLM: Time series forecasting by reprogramming large language models. In *International*
360 *Conference on Learning Representations*, 2024b.

361 T. Koo, F. Liu, and L. He. Automata-based constraints for language model decoding. *arXiv preprint*
362 *arXiv:2407.08103*, 2024.

363 P. Langer, T. Kaar, M. Rosenblattl, M. A. Xu, W. Chow, M. Maritsch, R. Jakob, N. Wang, J. Liu,
364 A. Verma, et al. Opentslm: Time-series language models for reasoning over multivariate medical
365 text-and time-series data. *arXiv preprint arXiv:2510.02410*, 2025.

366 H. Liu, C. Liu, and B. A. Prakash. A picture is worth a thousand numbers: Enabling llms reason about
367 time series via visualization. In *Proceedings of the 2025 Conference of the Nations of the Americas*
368 *Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume*
369 *1: Long Papers)*, pages 7486–7518, 2025.

370 Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-
371 term forecasting with transformers. In *The Eleventh International Conference on Learning*
372 *Representations*, 2022.

373 C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu.
374 Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine*
375 *learning research*, 21(140):1–67, 2020.

376 S. Roschmann, Q. Bouniot, V. Feofanov, I. Redko, and Z. Akata. Time series representations for
377 classification lie hidden in pretrained vision transformers. *arXiv preprint arXiv:2506.08641*, 2025.

378 F. Shi, X. Yin, K. Wang, W. Tu, Q. Sun, and H. Ning. Large language models for time series analysis:
379 Techniques, applications, and challenges. *arXiv preprint arXiv:2506.11040*, 2025.

380 C. Sun, H. Li, Y. Li, and S. Hong. Test: Text prototype aligned embedding to activate llm’s ability for
381 time series. *arXiv preprint arXiv:2308.08241*, 2023.

382 M. Tan, M. A. Merrill, V. Gupta, T. Althoff, and T. Hartvigsen. Are language models actually useful
383 for time series forecasting? *Advances in Neural Information Processing Systems*, 37:60162–60191,
384 2024.

385 W. Tang, L. Liu, and G. Long. Interpretable time-series classification on few-shot samples. In *2020*
386 *International joint conference on neural networks (IJCNN)*, pages 1–8. IEEE, 2020.

387 Z. Wang and T. Oates. Encoding time series as images for visual inspection and classification using
388 tiled convolutional neural networks. In *Workshops at the Twenty-Ninth AAAI Conference on*
389 *Artificial Intelligence*, 2015a.

390 Z. Wang and T. Oates. Imaging time-series to improve classification and imputation. In *Proceedings*
391 *of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, pages 3939–3945,
392 2015b.

393 J. Ye, W. Zhang, K. Yi, Y. Yu, Z. Li, J. Li, and F. Tsung. A survey of time series foundation
394 models: Generalizing time series representation with large language model. *arXiv e-prints*, pages
395 arXiv–2405, 2024.

396 L. Ye and E. Keogh. Time series shapelets: a new primitive for data mining. In *Proceedings of the*
397 *15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages
398 947–956, 2009a.

- 399 L. Ye and E. Keogh. Time series shapelets: A new primitive for data mining. In *Proceedings of the*
400 *15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages
401 947–956, 2009b.
- 402 X. Zhang, R. R. Chowdhury, R. K. Gupta, and J. Shang. Large language models for time series: A
403 survey. *arXiv preprint arXiv:2402.01801*, 2024.
- 404 L. N. Zheng, C. G. Dong, W. E. Zhang, L. Yue, M. Xu, O. Maennel, and W. Chen. Revisited large lan-
405 guage model for time series analysis through modality alignment. *arXiv preprint arXiv:2410.12326*,
406 2024.
- 407 T. Zhou, P. Niu, X. Wang, L. Sun, and R. Jin. One fits all: Power general time series analysis by
408 pretrained lm. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

409 **A Full UCR Benchmark Results**

410 This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets.
411 Each cell is shown as mean $\pm$ standard deviation over five runs, and the summary rows at the end of
412 each table report average accuracy, average rank, and the number of datasets attaining the best mean
413 accuracy (counting ties).

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	53.80 $\pm$ 4.97	42.00 $\pm$ 4.95	48.40 $\pm$ 7.30	24.00 $\pm$ 4.24	35.60 $\pm$ 3.21	27.67 $\pm$ 7.02	24.33 $\pm$ 9.71	28.33 $\pm$ 2.52	27.80 $\pm$ 7.09	31.67 $\pm$ 9.81	18.60 $\pm$ 4.51	44.00 $\pm$ 5.20
Adiac	33.81 $\pm$ 1.23	12.84 $\pm$ 3.09	33.81 $\pm$ 1.23	8.85 $\pm$ 2.36	11.97 $\pm$ 1.69	21.40 $\pm$ 4.24	34.53 $\pm$ 1.77	8.10 $\pm$ 1.92	40.82 $\pm$ 3.61	24.21 $\pm$ 3.24	3.02 $\pm$ 1.05	45.17 $\pm$ 5.28
AllGestureWiimoteX	25.71 $\pm$ 4.01	26.86 $\pm$ 2.52	24.43 $\pm$ 2.55	15.57 $\pm$ 2.95	21.86 $\pm$ 4.59	20.71 $\pm$ 2.87	11.33 $\pm$ 1.19	16.76 $\pm$ 2.52	13.43 $\pm$ 1.74	18.90 $\pm$ 1.08	21.86 $\pm$ 4.59	25.11 $\pm$ 4.65
AllGestureWiimoteY	26.37 $\pm$ 2.89	26.17 $\pm$ 2.88	22.71 $\pm$ 4.62	17.49 $\pm$ 2.02	25.17 $\pm$ 5.50	21.81 $\pm$ 2.78	12.29 $\pm$ 1.27	20.29 $\pm$ 1.98	13.06 $\pm$ 1.34	21.33 $\pm$ 2.38	25.17 $\pm$ 5.50	25.54 $\pm$ 4.42
AllGestureWiimoteZ	27.11 $\pm$ 1.95	22.66 $\pm$ 4.13	<u>25.49 $\pm$ 2.57</u>	18.26 $\pm$ 2.01	19.71 $\pm$ 2.87	19.24 $\pm$ 1.57	13.29 $\pm$ 0.43	18.76 $\pm$ 2.79	14.14 $\pm$ 1.72	19.71 $\pm$ 1.79	19.71 $\pm$ 2.87	22.26 $\pm$ 3.96
ArrowHead	47.66 $\pm$ 8.94	46.63 $\pm$ 9.45	48.00 $\pm$ 7.74	52.00 $\pm$ 6.38	49.71 $\pm$ 1.40	47.62 $\pm$ 9.46	38.29 $\pm$ 4.00	50.86 $\pm$ 1.71	35.89 $\pm$ 2.60	50.86 $\pm$ 6.05	41.49 $\pm$ 12.34	52.57 $\pm$ 3.43
BME	49.60 $\pm$ 2.85	50.80 $\pm$ 2.13	49.73 $\pm$ 3.42	56.67 $\pm$ 3.43	62.93 $\pm$ 5.35	55.33 $\pm$ 7.86	51.56 $\pm$ 3.15	61.33 $\pm$ 4.06	55.87 $\pm$ 7.71	58.89 $\pm$ 6.19	35.73 $\pm$ 3.22	66.40 $\pm$ 3.88
Beef	36.00 $\pm$ 6.41	31.33 $\pm$ 5.06	27.33 $\pm$ 7.96	38.67 $\pm$ 9.89	32.00 $\pm$ 7.67	40.00 $\pm$ 8.82	34.44 $\pm$ 6.94	41.11 $\pm$ 1.92	30.67 $\pm$ 6.83	40.00 $\pm$ 6.67	31.33 $\pm$ 9.31	34.00 $\pm$ 2.79
BeetleFly	55.00 $\pm$ 24.24	60.00 $\pm$ 25.00	61.00 $\pm$ 24.85	67.00 $\pm$ 4.47	61.00 $\pm$ 9.62	60.00 $\pm$ 13.23	55.00 $\pm$ 5.00	61.67 $\pm$ 11.55	48.00 $\pm$ 2.74	60.00 $\pm$ 13.23	70.00 $\pm$ 11.18	70.00 $\pm$ 11.73
BirdChicken	62.00 $\pm$ 8.37	71.00 $\pm$ 9.62	<u>70.00 $\pm$ 17.68</u>	56.00 $\pm$ 4.18	56.00 $\pm$ 4.18	51.67 $\pm$ 2.89	51.67 $\pm$ 12.58	53.33 $\pm$ 5.77	55.00 $\pm$ 5.00	51.67 $\pm$ 7.64	59.00 $\pm$ 9.62	68.00 $\pm$ 14.83
CBF	81.67 $\pm$ 6.47	87.62 $\pm$ 2.73	71.71 $\pm$ 7.91	54.02 $\pm$ 3.44	66.38 $\pm$ 7.40	55.85 $\pm$ 2.16	33.63 $\pm$ 1.20	58.00 $\pm$ 5.61	36.89 $\pm$ 2.37	54.78 $\pm$ 1.68	35.20 $\pm$ 3.75	72.48 $\pm$ 4.85
Car	37.00 $\pm$ 4.62	39.00 $\pm$ 4.80	41.67 $\pm$ 9.28	38.67 $\pm$ 11.63	27.33 $\pm$ 9.90	42.22 $\pm$ 5.09	24.44 $\pm$ 3.47	43.89 $\pm$ 5.85	27.00 $\pm$ 4.15	40.00 $\pm$ 7.26	30.00 $\pm$ 9.65	55.67 $\pm$ 8.94
Chinatown	66.53 $\pm$ 19.14	63.73 $\pm$ 26.98	62.51 $\pm$ 27.20	67.41 $\pm$ 23.86	67.35 $\pm$ 11.32	77.07 $\pm$ 16.93	81.83 $\pm$ 2.19	77.94 $\pm$ 9.20	77.61 $\pm$ 8.88	75.41 $\pm$ 7.92	56.68 $\pm$ 20.52	93.59 $\pm$ 3.92
ChlorineConcentration	34.61 $\pm$ 2.26	35.41 $\pm$ 4.36	34.04 $\pm$ 2.01	33.26 $\pm$ 2.96	35.54 $\pm$ 3.20	33.64 $\pm$ 3.96	32.92 $\pm$ 4.26	33.42 $\pm$ 5.09	33.78 $\pm$ 2.74	34.11 $\pm$ 3.50	39.93 $\pm$ 15.64	32.61 $\pm$ 3.30
CinCECGTorso	37.20 $\pm$ 9.73	36.03 $\pm$ 6.56	36.51 $\pm$ 8.82	40.06 $\pm$ 3.86	35.72 $\pm$ 7.09	44.86 $\pm$ 6.03	35.36 $\pm$ 5.77	50.53 $\pm$ 4.66	34.97 $\pm$ 4.68	45.68 $\pm$ 5.50	42.49 $\pm$ 6.76	48.74 $\pm$ 6.47
Coffee	78.57 $\pm$ 9.78	92.86 $\pm$ 8.38	84.29 $\pm$ 13.02	72.86 $\pm$ 14.42	74.29 $\pm$ 18.80	63.10 $\pm$ 10.91	65.48 $\pm$ 7.43	71.43 $\pm$ 16.37	76.43 $\pm$ 4.07	64.29 $\pm$ 7.14	51.43 $\pm$ 4.07	75.71 $\pm$ 13.69
Computers	57.52 $\pm$ 7.64	60.40 $\pm$ 6.13	<u>59.12 $\pm$ 4.47</u>	47.12 $\pm$ 5.87	52.64 $\pm$ 10.26	50.93 $\pm$ 1.29	48.80 $\pm$ 0.69	51.87 $\pm$ 2.57	48.96 $\pm$ 2.79	52.27 $\pm$ 3.23	45.20 $\pm$ 5.79	56.40 $\pm$ 5.77
CricketX	26.15 $\pm$ 3.29	26.10 $\pm$ 3.67	20.56 $\pm$ 4.67	17.28 $\pm$ 1.96	18.15 $\pm$ 4.50	20.34 $\pm$ 0.39	8.89 $\pm$ 0.82	17.52 $\pm$ 1.26	11.38 $\pm$ 2.35	20.68 $\pm$ 0.74	7.28 $\pm$ 1.05	26.26 $\pm$ 5.29
CricketY	26.97 $\pm$ 3.03	24.46 $\pm$ 4.25	22.72 $\pm$ 1.74	18.56 $\pm$ 4.13	17.90 $\pm$ 3.61	22.82 $\pm$ 1.56	10.34 $\pm$ 1.89	20.94 $\pm$ 3.72	11.23 $\pm$ 0.80	21.20 $\pm$ 3.98	9.54 $\pm$ 2.33	23.33 $\pm$ 5.19
CricketZ	32.51 $\pm$ 2.32	28.72 $\pm$ 2.15	26.56 $\pm$ 4.19	19.64 $\pm$ 2.73	22.26 $\pm$ 3.91	20.34 $\pm$ 4.11	11.03 $\pm$ 1.54	20.00 $\pm$ 2.31	12.26 $\pm$ 1.25	20.34 $\pm$ 2.57	7.64 $\pm$ 1.05	<u>29.28 $\pm$ 4.82</u>
Crop	25.10 $\pm$ 4.94	28.14 $\pm$ 2.90	25.10 $\pm$ 4.94	10.23 $\pm$ 2.66	17.27 $\pm$ 1.43	27.73 $\pm$ 2.89	24.10 $\pm$ 4.30	26.19 $\pm$ 2.96	27.27 $\pm$ 3.58	30.31 $\pm$ 5.00	3.80 $\pm$ 0.58	36.48 $\pm$ 3.38
DiatomSizeReduction	79.67 $\pm$ 5.51	55.36 $\pm$ 19.14	84.12 $\pm$ 4.82	89.22 $\pm$ 8.91	49.61 $\pm$ 24.86	75.71 $\pm$ 7.74	55.01 $\pm$ 11.77	56.43 $\pm$ 20.40	50.07 $\pm$ 12.66	76.47 $\pm$ 11.70	41.44 $\pm$ 12.35	85.82 $\pm$ 8.55
DistalPhalanxOutlineAgeGroup	43.02 $\pm$ 17.16	43.02 $\pm$ 18.34	43.60 $\pm$ 14.92	42.73 $\pm$ 19.01	46.76 $\pm$ 16.49	45.80 $\pm$ 19.81	46.28 $\pm$ 21.27	56.59 $\pm$ 9.36	50.94 $\pm$ 19.66	46.28 $\pm$ 20.78	35.68 $\pm$ 14.61	42.01 $\pm$ 16.44
DistalPhalanxOutlineCorrect	62.10 $\pm$ 1.32	57.32 $\pm$ 9.22	63.04 $\pm$ 2.44	57.54 $\pm$ 11.13	53.55 $\pm$ 11.63	53.99 $\pm$ 14.78	53.26 $\pm$ 12.27	53.62 $\pm$ 11.97	55.14 $\pm$ 10.52	53.26 $\pm$ 12.97	56.74 $\pm$ 7.30	60.58 $\pm$ 2.98
DistalPhalanxTW	51.22 $\pm$ 5.15	44.03 $\pm$ 11.70	53.38 $\pm$ 4.02	43.74 $\pm$ 5.17	51.22 $\pm$ 6.58	45.32 $\pm$ 9.89	42.69 $\pm$ 4.40	38.13 $\pm$ 7.51	48.20 $\pm$ 5.84	47.48 $\pm$ 5.62	18.56 $\pm$ 14.51	54.53 $\pm$ 1.72
DodgerLoopDay	32.50 $\pm$ 7.34	25.00 $\pm$ 4.05	29.75 $\pm$ 7.68	41.50 $\pm$ 2.05	26.00 $\pm$ 6.58	40.42 $\pm$ 4.73	20.42 $\pm$ 1.91	32.92 $\pm$ 3.15	21.50 $\pm$ 4.45	40.42 $\pm$ 7.94	26.00 $\pm$ 6.58	34.50 $\pm$ 4.38
DodgerLoopGame	52.17 $\pm$ 6.30	55.07 $\pm$ 17.38	49.13 $\pm$ 8.90	60.87 $\pm$ 12.35	59.86 $\pm$ 22.05	51.45 $\pm$ 9.96	51.21 $\pm$ 5.14	56.52 $\pm$ 10.82	50.43 $\pm$ 2.44	51.45 $\pm$ 9.59	59.86 $\pm$ 22.05	56.96 $\pm$ 11.69
DodgerLoopWeekend	84.49 $\pm$ 8.37	74.78 $\pm$ 12.27	83.19 $\pm$ 10.24	93.62 $\pm$ 3.81	96.23 $\pm$ 1.19	95.65 $\pm$ 1.26	63.77 $\pm$ 4.41	94.44 $\pm$ 0.42	68.41 $\pm$ 4.57	93.72 $\pm$ 5.86	96.23 $\pm$ 1.19	85.36 $\pm$ 4.30
ECG200	63.00 $\pm$ 12.61	72.00 $\pm$ 6.75	66.20 $\pm$ 8.32	71.80 $\pm$ 4.44	61.60 $\pm$ 12.82	61.33 $\pm$ 10.97	65.00 $\pm$ 11.79	63.00 $\pm$ 20.66	64.20 $\pm$ 14.29	68.67 $\pm$ 7.09	51.40 $\pm$ 12.93	60.60 $\pm$ 6.91
ECG5000	54.22 $\pm$ 21.00	50.05 $\pm$ 20.59	61.53 $\pm$ 13.66	62.80 $\pm$ 19.19	53.35 $\pm$ 20.14	69.07 $\pm$ 12.52	54.44 $\pm$ 9.79	57.20 $\pm$ 16.09	62.35 $\pm$ 12.99	<u>68.84 $\pm$ 13.99</u>	19.70 $\pm$ 25.30	67.18 $\pm$ 10.57
ECGFiveDays	58.65 $\pm$ 7.69	57.24 $\pm$ 6.51	54.80 $\pm$ 13.00	57.63 $\pm$ 2.63	62.21 $\pm$ 8.35	57.99 $\pm$ 4.84	54.86 $\pm$ 3.17	55.87 $\pm$ 1.56	54.08 $\pm$ 6.73	<u>58.46 $\pm$ 5.08</u>	51.54 $\pm$ 6.38	61.77 $\pm$ 6.61
EOGHorizontalSignal	20.11 $\pm$ 2.01	22.38 $\pm$ 1.86	22.60 $\pm$ 2.77	22.43 $\pm$ 4.01	21.22 $\pm$ 3.15	20.63 $\pm$ 4.63	15.29 $\pm$ 2.63	23.57 $\pm$ 5.45	10.39 $\pm$ 2.26	22.01 $\pm$ 4.37	27.24 $\pm$ 2.99	29.06 $\pm$ 4.16
EOGVerticalSignal	11.27 $\pm$ 3.67	14.09 $\pm$ 3.23	13.26 $\pm$ 3.09	24.20 $\pm$ 3.27	23.76 $\pm$ 3.73	23.39 $\pm$ 2.79	11.23 $\pm$ 0.70	24.86 $\pm$ 4.17	11.27 $\pm$ 3.83	23.11 $\pm$ 4.07	19.06 $\pm$ 6.08	27.35 $\pm$ 1.90
Earthquakes	50.07 $\pm$ 8.55	51.08 $\pm$ 6.33	50.94 $\pm$ 6.46	49.35 $\pm$ 5.05	45.61 $\pm$ 20.94	48.68 $\pm$ 19.16	49.40 $\pm$ 1.10	49.40 $\pm$ 14.03	47.05 $\pm$ 5.90	47.00 $\pm$ 24.65	46.04 $\pm$ 13.71	55.54 $\pm$ 11.57
ElectricDevices	28.83 $\pm$ 9.01	<u>32.82 $\pm$ 12.31</u>	28.35 $\pm$ 12.24	17.59 $\pm$ 1.67	27.21 $\pm$ 7.38	21.93 $\pm$ 4.54	17.11 $\pm$ 1.87	15.12 $\pm$ 2.81	17.04 $\pm$ 2.05	15.61 $\pm$ 7.10	17.06 $\pm$ 3.10	33.14 $\pm$ 10.96
EthanolLevel	24.44 $\pm$ 1.30	<u>25.40 $\pm$ 0.85</u>	25.28 $\pm$ 1.57	24.92 $\pm$ 1.50	23.80 $\pm$ 1.65	26.13 $\pm$ 0.58	26.53 $\pm$ 1.55	<u>26.67 $\pm$ 1.29</u>	25.32 $\pm$ 1.50	25.80 $\pm$ 1.39	24.76 $\pm$ 1.40	26.68 $\pm$ 1.86
FaceAll	42.17 $\pm$ 4.79	32.36 $\pm$ 9.47	42.17 $\pm$ 4.79	28.66 $\pm$ 9.09	19.36 $\pm$ 3.07	29.11 $\pm$ 4.19	33.12 $\pm$ 5.46	25.94 $\pm$ 1.75	34.05 $\pm$ 3.40	32.92 $\pm$ 2.10	8.64 $\pm$ 3.46	32.84 $\pm$ 6.86
FaceFour	68.64 $\pm$ 10.56	47.50 $\pm$ 11.12	56.36 $\pm$ 4.30	64.32 $\pm$ 4.73	54.09 $\pm$ 8.34	70.08 $\pm$ 4.30	52.65 $\pm$ 3.99	67.05 $\pm$ 7.10	51.36 $\pm$ 4.28	73.11 $\pm$ 3.99	42.95 $\pm$ 6.60	54.77 $\pm$ 8.25
FacesUCR	54.83 $\pm$ 5.21	37.63 $\pm$ 5.89	43.65 $\pm$ 3.88	35.69 $\pm$ 7.47	20.81 $\pm$ 1.95	36.24 $\pm$ 3.26	41.98 $\pm$ 2.95	40.05 $\pm$ 2.80	41.87 $\pm$ 3.09	42.63 $\pm$ 2.58	16.92 $\pm$ 7.54	50.37 $\pm$ 2.98
FiftyWords	13.27 $\pm$ 1.89	12.79 $\pm$ 3.45	13.27 $\pm$ 1.89	32.70 $\pm$ 4.33	19.60 $\pm$ 4.90	35.90 $\pm$ 3.31	3.44 $\pm$ 1.59	38.97 $\pm$ 6.44	11.30 $\pm$ 2.86	36.70 $\pm$ 4.41	3.74 $\pm$ 2.40	36.48 $\pm$ 2.66
Fish	49.03 $\pm$ 5.96	21.26 $\pm$ 3.60	<u>45.71 $\pm$ 1.71</u>	36.00 $\pm$ 3.98	23.20 $\pm$ 3.80	32.57 $\pm$ 4.57	16.95 $\pm$ 2.93	30.29 $\pm$ 2.62	21.71 $\pm$ 2.77	31.81 $\pm$ 4.76	20.80 $\pm$ 5.10	34.74 $\pm$ 4.91
FordA	59.23 $\pm$ 9.68	57.68 $\pm$ 10.00	59.83 $\pm$ 8.76	50.73 $\pm$ 1.01	56.70 $\pm$ 8.04	50.30 $\pm$ 0.33	49.57 $\pm$ 1.42	49.72 $\pm$ 0.63	49.68 $\pm$ 1.11	49.92 $\pm$ 0.80	50.61 $\pm$ 1.68	50.71 $\pm$ 0.98
FordB	52.54 $\pm$ 6.44	52.64 $\pm$ 6.88	52.40 $\pm$ 4.53	50.12 $\pm$ 1.83	53.60 $\pm$ 4.09	49.96 $\pm$ 0.94						

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	53.26 ± 6.97	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	57.47 ± 8.89	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	64.00 ± 8.08	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	61.92 ± 19.97	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	81.01 ± 10.12
GunPointOldVersusYoung	92.38 ± 10.44	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	55.81 ± 6.79	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	76.76 ± 10.41	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	26.49 ± 6.85	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	51.04 ± 2.39	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	80.84 ± 7.67	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	18.67 ± 3.56	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	45.61 ± 5.13	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	78.33 ± 24.61	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	54.88 ± 8.28	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	59.67 ± 8.87
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	56.16 ± 8.33	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	82.32 ± 3.73	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	77.67 ± 17.26	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	28.53 ± 8.23	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	51.29 ± 5.58	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	41.69 ± 18.94	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	58.42 ± 17.26	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	46.10 ± 3.62	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	56.06 ± 5.34
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	63.30 ± 3.34
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	68.66 ± 11.89	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	42.58 ± 4.62	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	49.89 ± 4.96	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	45.62 ± 7.45	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	65.33 ± 15.20	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	32.10 ± 5.72
PhalangesOutlinesCorrect	54.90 ± 6.98	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	10.43 ± 1.42	8.46 ± 1.00	10.43 ± 1.42	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	48.67 ± 8.33	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	38.65 ± 2.92	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	60.10 ± 2.28	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	50.29 ± 2.90	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	98.86 ± 2.56	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	80.22 ± 9.41	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	60.81 ± 40.85	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	58.76 ± 6.63	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	60.81 ± 11.71	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	41.16 ± 5.10	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	57.33 ± 5.03	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	34.29 ± 6.61	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	61.00 ± 15.57	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	29.87 ± 4.86	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95

Continued on next page

Table 7: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	28.89 $\pm$ 5.77	29.07 $\pm$ 4.60	28.00 $\pm$ 2.65	35.60 $\pm$ 4.19	29.91 $\pm$ 10.77	26.37 $\pm$ 1.89	23.11 $\pm$ 2.89	26.67 $\pm$ 4.87	21.56 $\pm$ 1.94	26.96 $\pm$ 3.68	27.91 $\pm$ 3.11	37.42 $\pm$ 8.82
ShakeGestureWiimoteZ	63.20 $\pm$ 7.69	58.00 $\pm$ 11.83	67.60 $\pm$ 11.61	40.40 $\pm$ 11.26	43.20 $\pm$ 9.55	45.33 $\pm$ 3.06	16.00 $\pm$ 7.21	40.00 $\pm$ 3.46	24.40 $\pm$ 4.98	46.00 $\pm$ 6.00	43.20 $\pm$ 9.55	60.40 $\pm$ 7.92
ShapeletSim	66.33 $\pm$ 12.25	87.00 $\pm$ 16.91	73.89 $\pm$ 23.85	50.11 $\pm$ 2.27	75.22 $\pm$ 8.30	53.89 $\pm$ 1.67	52.96 $\pm$ 1.40	48.89 $\pm$ 5.64	50.44 $\pm$ 1.82	51.11 $\pm$ 0.96	47.78 $\pm$ 3.07	62.44 $\pm$ 7.44
ShapesAll	23.63 $\pm$ 4.60	15.60 $\pm$ 1.98	23.63 $\pm$ 4.60	45.70 $\pm$ 2.59	19.27 $\pm$ 4.58	44.72 $\pm$ 2.58	6.00 $\pm$ 0.58	43.94 $\pm$ 2.14	8.07 $\pm$ 3.90	44.06 $\pm$ 1.11	8.13 $\pm$ 2.75	50.20 $\pm$ 3.03
SmallKitchenAppliances	40.37 $\pm$ 15.51	46.24 $\pm$ 15.42	44.32 $\pm$ 15.50	42.19 $\pm$ 3.16	38.13 $\pm$ 22.42	32.98 $\pm$ 5.23	32.80 $\pm$ 3.67	35.82 $\pm$ 4.27	35.41 $\pm$ 3.13	36.89 $\pm$ 7.74	38.88 $\pm$ 7.80	42.88 $\pm$ 15.27
SmoothSubspace	74.93 $\pm$ 5.59	74.40 $\pm$ 5.13	<u>74.80 $\pm$ 8.48</u>	31.73 $\pm$ 3.96	50.00 $\pm$ 10.52	46.67 $\pm$ 1.76	53.33 $\pm$ 9.45	50.22 $\pm$ 10.96	62.27 $\pm$ 6.56	60.67 $\pm$ 10.35	56.67 $\pm$ 5.19	68.53 $\pm$ 8.95
SonyAIBORobotSurface1	77.57 $\pm$ 10.30	80.80 $\pm$ 8.99	<u>80.37 $\pm$ 5.89</u>	59.80 $\pm$ 6.73	64.53 $\pm$ 18.83	52.86 $\pm$ 10.38	56.96 $\pm$ 5.82	55.35 $\pm$ 9.37	56.07 $\pm$ 4.43	57.68 $\pm$ 9.32	55.61 $\pm$ 9.09	57.20 $\pm$ 8.62
SonyAIBORobotSurface2	80.29 $\pm$ 6.43	84.09 $\pm$ 5.18	<u>80.90 $\pm$ 6.99</u>	67.39 $\pm$ 19.08	73.03 $\pm$ 13.78	66.18 $\pm$ 11.61	60.37 $\pm$ 27.05	61.35 $\pm$ 27.10	69.49 $\pm$ 15.13	68.07 $\pm$ 16.06	57.31 $\pm$ 15.03	66.09 $\pm$ 10.78
StarLightCurves	63.75 $\pm$ 3.45	62.30 $\pm$ 9.88	57.59 $\pm$ 9.08	57.39 $\pm$ 13.40	57.55 $\pm$ 15.28	67.54 $\pm$ 17.11	37.78 $\pm$ 3.53	57.92 $\pm$ 15.55	27.99 $\pm$ 0.00	67.36 $\pm$ 17.40	64.81 $\pm$ 14.61	63.03 $\pm$ 14.23
Strawberry	52.27 $\pm$ 4.26	51.84 $\pm$ 3.86	54.43 $\pm$ 4.33	54.86 $\pm$ 4.34	53.62 $\pm$ 3.45	<u>58.29 $\pm$ 0.95</u>	54.86 $\pm$ 13.21	60.18 $\pm$ 0.68	56.11 $\pm$ 9.26	57.93 $\pm$ 0.83	45.89 $\pm$ 11.15	51.46 $\pm$ 5.21
SwedishLeaf	65.95 $\pm$ 6.01	46.08 $\pm$ 6.36	65.95 $\pm$ 6.01	27.42 $\pm$ 3.44	40.83 $\pm$ 4.40	36.69 $\pm$ 1.99	27.20 $\pm$ 2.42	24.53 $\pm$ 4.08	31.65 $\pm$ 2.48	38.19 $\pm$ 0.81	8.10 $\pm$ 1.23	57.98 $\pm$ 2.72
Symbols	85.77 $\pm$ 4.83	70.17 $\pm$ 4.91	76.68 $\pm$ 7.53	79.60 $\pm$ 7.65	72.50 $\pm$ 6.12	81.54 $\pm$ 3.92	37.15 $\pm$ 0.97	82.18 $\pm$ 0.42	46.13 $\pm$ 9.06	80.57 $\pm$ 0.95	51.54 $\pm$ 6.72	80.94 $\pm$ 7.54
SyntheticControl	72.53 $\pm$ 6.61	76.40 $\pm$ 5.24	68.67 $\pm$ 10.59	38.47 $\pm$ 6.28	80.33 $\pm$ 9.12	47.78 $\pm$ 5.87	44.89 $\pm$ 3.20	48.22 $\pm$ 0.38	42.13 $\pm$ 2.87	50.00 $\pm$ 4.48	24.53 $\pm$ 7.68	84.20 $\pm$ 6.54
ToeSegmentation1	69.82 $\pm$ 7.39	69.65 $\pm$ 4.27	65.09 $\pm$ 8.05	51.67 $\pm$ 1.47	55.79 $\pm$ 6.85	49.27 $\pm$ 1.10	49.71 $\pm$ 3.29	46.64 $\pm$ 4.73	52.46 $\pm$ 2.82	48.39 $\pm$ 3.23	49.47 $\pm$ 2.47	58.16 $\pm$ 3.70
ToeSegmentation2	68.92 $\pm$ 8.63	70.31 $\pm$ 3.38	<u>70.92 $\pm$ 10.76</u>	50.31 $\pm$ 5.37	65.54 $\pm$ 9.84	50.26 $\pm$ 2.91	35.90 $\pm$ 10.81	50.51 $\pm$ 4.24	33.23 $\pm$ 16.47	52.05 $\pm$ 6.54	41.54 $\pm$ 14.95	75.23 $\pm$ 12.13
Trace	61.00 $\pm$ 4.64	65.20 $\pm$ 9.09	<u>65.80 $\pm$ 6.61</u>	49.60 $\pm$ 3.65	<u>74.40 $\pm$ 6.15</u>	51.33 $\pm$ 5.51	42.33 $\pm$ 4.51	47.67 $\pm$ 0.58	44.80 $\pm$ 7.69	51.33 $\pm$ 4.93	37.80 $\pm$ 6.42	99.20 $\pm$ 1.79
TwoLeadECG	69.48 $\pm$ 12.74	64.97 $\pm$ 10.26	<u>66.51 $\pm$ 13.23</u>	54.03 $\pm$ 2.33	57.82 $\pm$ 5.04	59.32 $\pm$ 3.05	53.32 $\pm$ 7.30	52.62 $\pm$ 3.32	54.47 $\pm$ 5.41	60.32 $\pm$ 1.06	50.73 $\pm$ 1.02	64.88 $\pm$ 11.28
TwoPatterns	48.77 $\pm$ 2.41	61.51 $\pm$ 6.85	<u>46.17 $\pm$ 11.65</u>	31.86 $\pm$ 3.24	36.86 $\pm$ 6.10	29.76 $\pm$ 2.56	25.52 $\pm$ 1.93	28.23 $\pm$ 0.74	26.06 $\pm$ 1.27	29.87 $\pm$ 1.68	26.32 $\pm$ 2.53	45.74 $\pm$ 2.64
UMD	43.06 $\pm$ 4.02	54.58 $\pm$ 9.53	45.56 $\pm$ 6.14	46.81 $\pm$ 7.70	50.14 $\pm$ 6.28	46.30 $\pm$ 5.02	45.60 $\pm$ 3.21	47.69 $\pm$ 4.19	48.61 $\pm$ 2.55	47.45 $\pm$ 6.56	31.39 $\pm$ 4.35	45.47 $\pm$ 8.73
UWaveGestureLibraryAll	29.36 $\pm$ 4.20	24.46 $\pm$ 5.19	28.34 $\pm$ 3.82	66.77 $\pm$ 5.89	41.63 $\pm$ 5.40	<u>69.19 $\pm$ 5.01</u>	25.78 $\pm$ 1.68	67.23 $\pm$ 2.91	26.67 $\pm$ 3.53	69.21 $\pm$ 6.24	50.70 $\pm$ 5.42	33.37 $\pm$ 0.26
UWaveGestureLibraryX	26.45 $\pm$ 4.19	26.60 $\pm$ 3.44	24.71 $\pm$ 1.83	47.27 $\pm$ 6.01	35.36 $\pm$ 5.04	49.30 $\pm$ 7.17	22.37 $\pm$ 0.14	47.22 $\pm$ 8.02	24.34 $\pm$ 5.30	49.17 $\pm$ 7.61	17.28 $\pm$ 8.20	30.33 $\pm$ 0.23
UWaveGestureLibraryY	25.37 $\pm$ 4.63	26.13 $\pm$ 4.45	24.62 $\pm$ 4.93	43.95 $\pm$ 4.51	38.35 $\pm$ 7.26	47.05 $\pm$ 4.02	24.58 $\pm$ 1.89	46.37 $\pm$ 4.93	26.81 $\pm$ 2.91	47.18 $\pm$ 5.20	20.37 $\pm$ 4.71	40.36 $\pm$ 0.25
UWaveGestureLibraryZ	24.77 $\pm$ 4.74	27.03 $\pm$ 3.18	24.45 $\pm$ 3.15	42.90 $\pm$ 1.96	36.37 $\pm$ 3.72	<u>44.61 $\pm$ 1.78</u>	22.47 $\pm$ 2.82	44.70 $\pm$ 2.64	25.10 $\pm$ 2.80	<u>44.75 $\pm$ 1.63</u>	22.77 $\pm$ 3.47	47.44 $\pm$ 2.61
Wafer	50.39 $\pm$ 4.56	44.94 $\pm$ 10.31	49.91 $\pm$ 13.57	67.87 $\pm$ 4.81	51.56 $\pm$ 17.59	63.90 $\pm$ 3.73	64.15 $\pm$ 3.76	63.99 $\pm$ 9.60	63.82 $\pm$ 9.82	<u>65.48 $\pm$ 1.03</u>	43.02 $\pm$ 26.21	46.27 $\pm$ 7.11
Wine	57.41 $\pm$ 14.70	54.07 $\pm$ 4.61	<u>54.81 $\pm$ 11.31</u>	52.96 $\pm$ 8.34	50.74 $\pm$ 6.23	48.77 $\pm$ 3.85	50.00 $\pm$ 0.00	50.00 $\pm$ 0.00	52.22 $\pm$ 4.97	37.65 $\pm$ 19.80	51.48 $\pm$ 3.31	51.11 $\pm$ 7.48
WordSynonyms	14.92 $\pm$ 3.03	13.70 $\pm$ 5.79	<u>14.92 $\pm$ 3.03</u>	19.78 $\pm$ 2.70	16.96 $\pm$ 4.46	19.75 $\pm$ 1.90	6.79 $\pm$ 1.46	21.16 $\pm$ 1.50	10.06 $\pm$ 3.54	20.53 $\pm$ 2.41	4.98 $\pm$ 1.34	20.91 $\pm$ 2.73
Worms	36.10 $\pm$ 9.87	33.25 $\pm$ 6.66	<u>35.06 $\pm$ 11.43</u>	17.14 $\pm$ 4.04	23.90 $\pm$ 9.39	19.91 $\pm$ 6.54	23.38 $\pm$ 4.50	18.61 $\pm$ 5.86	17.66 $\pm$ 7.03	19.05 $\pm$ 5.25	17.66 $\pm$ 2.36	24.68 $\pm$ 8.95
WormsTwoClass	44.79 $\pm$ 3.48	49.09 $\pm$ 6.96	<u>52.99 $\pm$ 6.71</u>	49.09 $\pm$ 3.72	50.39 $\pm$ 3.23	49.35 $\pm$ 3.44	48.92 $\pm$ 10.50	52.38 $\pm$ 1.98	53.51 $\pm$ 4.81	53.68 $\pm$ 3.75	52.99 $\pm$ 2.82	47.53 $\pm$ 10.69
Yoga	55.60 $\pm$ 4.57	50.68 $\pm$ 5.18	<u>53.54 $\pm$ 4.28</u>	49.99 $\pm$ 1.44	49.75 $\pm$ 4.01	49.48 $\pm$ 0.91	51.11 $\pm$ 3.62	49.62 $\pm$ 0.72	51.12 $\pm$ 3.30	48.62 $\pm$ 0.52	49.35 $\pm$ 2.85	50.42 $\pm$ 4.51
Avg. Acc.	47.79	45.40	<u>47.88</u>	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	48.31
Avg. Rank	<u>5.11</u>	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	4.58
#Best	25	16	14	7	5	11	2	11	2	15	4	26

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	61.60 $\pm$ 7.86	47.20 $\pm$ 6.30	59.20 $\pm$ 5.26	25.80 $\pm$ 4.38	42.00 $\pm$ 3.16	40.33 $\pm$ 2.31	30.67 $\pm$ 2.52	44.00 $\pm$ 6.24	35.80 $\pm$ 3.19	40.67 $\pm$ 5.69	24.80 $\pm$ 4.66	54.33 $\pm$ 4.16
Adiac	35.45 $\pm$ 2.60	14.27 $\pm$ 3.74	35.45 $\pm$ 2.60	18.87 $\pm$ 2.86	17.80 $\pm$ 5.08	27.79 $\pm$ 2.14	44.67 $\pm$ 0.39	16.62 $\pm$ 1.77	55.24 $\pm$ 4.67	32.91 $\pm$ 3.49	3.48 $\pm$ 1.27	56.01 $\pm$ 1.25
AllGestureWiimoteX	32.43 $\pm$ 2.94	27.69 $\pm$ 3.48	32.09 $\pm$ 3.58	19.46 $\pm$ 2.47	23.77 $\pm$ 3.63	24.43 $\pm$ 4.71	11.86 $\pm$ 1.00	22.67 $\pm$ 2.29	14.09 $\pm$ 1.28	21.10 $\pm$ 5.46	23.77 $\pm$ 3.63	26.51 $\pm$ 4.36
AllGestureWiimoteY	35.11 $\pm$ 5.43	28.57 $\pm$ 4.31	28.91 $\pm$ 5.81	20.54 $\pm$ 2.69	33.31 $\pm$ 2.65	24.33 $\pm$ 2.50	12.00 $\pm$ 1.38	21.90 $\pm$ 1.33	16.14 $\pm$ 1.67	22.57 $\pm$ 1.00	33.31 $\pm$ 2.65	29.43 $\pm$ 1.69
AllGestureWiimoteZ	29.80 $\pm$ 2.53	26.94 $\pm$ 2.84	29.31 $\pm$ 4.56	16.77 $\pm$ 2.02	22.97 $\pm$ 5.35	19.86 $\pm$ 4.12	12.76 $\pm$ 0.36	19.57 $\pm$ 2.48	14.97 $\pm$ 0.98	20.29 $\pm$ 3.48	22.97 $\pm$ 5.35	27.91 $\pm$ 5.21
ArrowHead	50.17 $\pm$ 5.13	44.80 $\pm$ 3.82	45.14 $\pm$ 5.25	57.03 $\pm$ 4.92	49.14 $\pm$ 13.02	53.14 $\pm$ 6.02	34.10 $\pm$ 6.27	50.67 $\pm$ 2.01	42.29 $\pm$ 4.92	55.43 $\pm$ 6.44	43.77 $\pm$ 8.76	57.03 $\pm$ 2.81
BME	48.40 $\pm$ 4.91	55.33 $\pm$ 7.02	54.40 $\pm$ 8.12	50.80 $\pm$ 5.11	67.33 $\pm$ 7.90	48.89 $\pm$ 2.14	64.00 $\pm$ 5.21	58.89 $\pm$ 3.91	67.33 $\pm$ 7.44	54.89 $\pm$ 9.20	50.80 $\pm$ 8.05	79.47 $\pm$ 11.40
Beef	42.00 $\pm$ 2.98	36.67 $\pm$ 8.50	36.70 $\pm$ 9.60	49.33 $\pm$ 7.23	39.33 $\pm$ 6.41	48.89 $\pm$ 5.09	40.00 $\pm$ 3.33	34.44 $\pm$ 5.09	42.67 $\pm$ 4.94	51.11 $\pm$ 7.70	32.67 $\pm$ 7.23	43.33 $\pm$ 5.27
BeetleFly	50.00 $\pm$ 9.35	67.00 $\pm$ 18.23	55.00 $\pm$ 12.75	52.00 $\pm$ 11.51	57.00 $\pm$ 16.43	45.00 $\pm$ 10.00	48.33 $\pm$ 12.58	38.33 $\pm$ 7.64	45.00 $\pm$ 9.35	56.67 $\pm$ 7.64	57.00 $\pm$ 4.47	75.00 $\pm$ 18.03
BirdChicken	71.00 $\pm$ 6.52	<u>72.00 $\pm$ 4.47</u>	76.00 $\pm$ 7.42	52.00 $\pm$ 14.40	65.00 $\pm$ 10.61	45.00 $\pm$ 8.66	51.67 $\pm$ 7.64	46.67 $\pm$ 5.77	53.00 $\pm$ 2.74	50.00 $\pm$ 13.23	49.00 $\pm$ 2.24	69.00 $\pm$ 22.19
CBF	85.76 $\pm$ 5.83	90.62 $\pm$ 7.16	82.18 $\pm$ 10.67	64.78 $\pm$ 6.67	82.69 $\pm$ 10.89	68.07 $\pm$ 6.00	34.89 $\pm$ 2.41	73.78 $\pm$ 6.88	36.78 $\pm$ 1.75	70.00 $\pm$ 6.65	42.84 $\pm$ 7.09	85.91 $\pm$ 4.23
Car	42.00 $\pm$ 15.43	41.67 $\pm$ 13.49	52.67 $\pm$ 7.87	55.33 $\pm$ 10.63	43.00 $\pm$ 11.45	54.44 $\pm$ 6.74	32.78 $\pm$ 0.96	52.78 $\pm$ 10.84	37.00 $\pm$ 7.76	55.56 $\pm$ 4.19	31.00 $\pm$ 10.71	63.33 $\pm$ 7.45
Chinatown	95.74 $\pm$ 1.26	96.73 $\pm$ 1.19	96.73 $\pm$ 0.81	54.52 $\pm$ 24.75	91.08 $\pm$ 5.39	85.33 $\pm$ 8.50	86.10 $\pm$ 2.71	84.55 $\pm$ 3.36	80.70 $\pm$ 5.41	82.80 $\pm$ 4.37	45.01 $\pm$ 24.65	91.84 $\pm$ 4.85
ChlorineConcentration	34.61 $\pm$ 4.95	35.24 $\pm$ 5.79	35.19 $\pm$ 5.79	34.54 $\pm$ 2.93	34.28 $\pm$ 2.22	31.22 $\pm$ 6.21	33.24 $\pm$ 2.95	33.84 $\pm$ 2.69	33.91 $\pm$ 2.07	33.69 $\pm$ 2.59	39.08 $\pm$ 9.47	32.47 $\pm$ 3.03
CinCECGTorso	46.52 $\pm$ 12.75	48.35 $\pm$ 9.58	51.06 $\pm$ 10.06	39.77 $\pm$ 4.46	<u>54.91 $\pm$ 12.73</u>	48.14 $\pm$ 2.63	36.18 $\pm$ 3.15	53.94 $\pm$ 2.51	50.30 $\pm$ 12.26	48.89 $\pm$ 2.81	49.00 $\pm$ 5.40	62.95 $\pm$ 6.66
Coffee	94.29 $\pm$ 7.41	88.57 $\pm$ 12.73	94.29 $\pm$ 3.19	92.14 $\pm$ 3.91	90.71 $\pm$ 10.89	73.81 $\pm$ 8.99	63.10 $\pm$ 7.43	85.71 $\pm$ 3.57	84.29 $\pm$ 10.29	77.38 $\pm$ 4.12	65.71 $\pm$ 15.07	88.57 $\pm$ 4.66
Computers	52.24 $\pm$ 10.15	54.40 $\pm$ 10.81	52.72 $\pm$ 12.14	48.08 $\pm$ 7.70	53.28 $\pm$ 11.25	48.00 $\pm$ 10.51	48.67 $\pm$ 4.60	48.00 $\pm$ 11.45	50.40 $\pm$ 1.72	46.67 $\pm$ 11.22	50.16 $\pm$ 7.53	57.73 $\pm$ 17.44
CricketX	36.51 $\pm$ 3.27	34.10 $\pm$ 3.98	33.03 $\pm$ 3.86	20.62 $\pm$ 2.07	30.56 $\pm$ 3.87	27.35 $\pm$ 2.06	8.12 $\pm$ 1.41	26.15 $\pm$ 1.85	12.15 $\pm$ 1.24	27.44 $\pm$ 2.96	11.54 $\pm$ 1.86	35.98 $\pm$ 3.13
CricketY	37.54 $\pm$ 5.48	30.62 $\pm$ 2.69	33.28 $\pm$ 3.86	22.56 $\pm$ 4.44	24.21 $\pm$ 2.69	26.50 $\pm$ 7.27	9.06 $\pm$ 0.39	25.90 $\pm$ 6.29	12.67 $\pm$ 0.94	27.52 $\pm$ 6.80	10.00 $\pm$ 3.24	30.43 $\pm$ 5.23
CricketZ	40.41 $\pm$ 4.15	<u>35.49 $\pm$ 2.72</u>	33.69 $\pm$ 5.21	20.36 $\pm$ 1.13	24.92 $\pm$ 3.09	26.75 $\pm$ 2.72	8.12 $\pm$ 1.50	24.44 $\pm$ 0.78	12.97 $\pm$ 2.47	26.50 $\pm$ 2.74	11.18 $\pm$ 2.42	33.33 $\pm$ 3.02
Crop	31.31 $\pm$ 3.87	37.82 $\pm$ 2.09	31.31 $\pm$ 3.87	14.67 $\pm$ 0.98	24.66 $\pm$ 2.54	36.33 $\pm$ 2.63	35.06 $\pm$ 3.15	37.89 $\pm$ 1.28	35.83 $\pm$ 2.39	38.66 $\pm$ 2.99	4.63 $\pm$ 0.75	43.94 $\pm$ 2.11
DiatomSizeReduction	78.43 $\pm$ 9.98	84.44 $\pm$ 4.38	89.74 $\pm$ 8.40	83.73 $\pm$ 5.41	63.01 $\pm$ 13.25	84.42 $\pm$ 6.89	59.91 $\pm$ 2.78	84.97 $\pm$ 5.38	59.28 $\pm$ 18.44	82.35 $\pm$ 7.47	35.10 $\pm$ 12.79	90.39 $\pm$ 5.62
DistalPhalanxOutlineAgeGroup	66.91 $\pm$ 6.08	67.77 $\pm$ 3.31	65.76 $\pm$ 6.53	55.54 $\pm$ 12.17	58.42 $\pm$ 11.15	57.07 $\pm$ 19.57	50.84 $\pm$ 14.56	56.12 $\pm$ 13.32	56.26 $\pm$ 12.92	57.07 $\pm$ 14.47	38.99 $\pm$ 8.80	67.77 $\pm$ 3.39
DistalPhalanxOutlineCorrect	52.32 $\pm$ 12.15	51.59 $\pm$ 11.49	52.54 $\pm$ 12.44	49.57 $\pm$ 11.83	51.45 $\pm$ 15.36	57.49 $\pm$ 13.09	56.76 $\pm$ 10.57	55.80 $\pm$ 11.93	51.38 $\pm$ 12.34	56.64 $\pm$ 12.97	50.65 $\pm$ 9.31	55.80 $\pm$ 14.95
DistalPhalanxTW	49.50 $\pm$ 8.83	55.54 $\pm$ 4.21	54.10 $\pm$ 4.92	39.28 $\pm$ 8.61	53.96 $\pm$ 7.56	41.97 $\pm$ 11.56	42.45 $\pm$ 10.74	39.81 $\pm$ 18.49	47.05 $\pm$ 8.79	45.80 $\pm$ 16.17	22.45 $\pm$ 8.85	51.37 $\pm$ 8.97
DodgerLoopDay	28.75 $\pm$ 2.34	23.75 $\pm$ 2.50	28.00 $\pm$ 5.70	44.00 $\pm$ 4.09	38.50 $\pm$ 7.42	39.58 $\pm$ 4.02	23.75 $\pm$ 2.50	42.50 $\pm$ 7.60	19.75 $\pm$ 4.09	42.50 $\pm$ 3.31	38.50 $\pm$ 7.42	35.50 $\pm$ 8.78
DodgerLoopGame	54.93 $\pm$ 6.45	63.04 $\pm$ 8.02	58.41 $\pm$ 11.09	53.62 $\pm$ 8.51	65.65 $\pm$ 18.08	67.87 $\pm$ 7.26	48.55 $\pm$ 4.53	71.50 $\pm$ 0.84	50.14 $\pm$ 3.81	68.36 $\pm$ 3.42	65.65 $\pm$ 18.08	61.01 $\pm$ 13.03
DodgerLoopWeekend	92.75 $\pm$ 2.40	87.68 $\pm$ 5.59	92.03 $\pm$ 3.69	94.64 $\pm$ 2.74	96.09 $\pm$ 1.41	96.38 $\pm$ 1.26	64.98 $\pm$ 4.71	97.34 $\pm$ 0.42	65.22 $\pm$ 8.45	96.14 $\pm$ 0.84	96.09 $\pm$ 1.41	84.64 $\pm$ 2.73
ECG200	69.40 $\pm$ 7.77	75.20 $\pm$ 7.05	71.80 $\pm$ 5.54	76.80 $\pm$ 3.56	71.80 $\pm$ 4.38	78.33 $\pm$ 1.53	74.33 $\pm$ 5.69	75.67 $\pm$ 3.21	73.60 $\pm$ 3.65	75.33 $\pm$ 0.58	52.20 $\pm$ 18.20	73.00 $\pm$ 3.46
ECG5000	61.90 $\pm$ 13.09	69.05 $\pm$ 11.38	61.86 $\pm$ 10.65	74.23 $\pm$ 10.96	58.61 $\pm$ 14.82	72.42 $\pm$ 4.03	65.61 $\pm$ 11.76	69.40 $\pm$ 4.84	67.93 $\pm$ 11.62	66.36 $\pm$ 9.81	30.14 $\pm$ 19.40	67.68 $\pm$ 9.76
ECGFiveDays	68.78 $\pm$ 9.46	56.79 $\pm$ 4.91	71.73 $\pm$ 8.76	57.12 $\pm$ 2.23	67.29 $\pm$ 3.58	66.55 $\pm$ 6.20	74.64 $\pm$ 13.99	75.18 $\pm$ 2.37	68.76 $\pm$ 10.24	68.52 $\pm$ 6.54	57.79 $\pm$ 7.24	69.12 $\pm$ 9.82
EOGHorizontalSignal	20.66 $\pm$ 1.70	23.43 $\pm$ 5.48	21.38 $\pm$ 4.60	25.86 $\pm$ 4.54	28.07 $\pm$ 2.90	30.66 $\pm$ 2.91	12.06 $\pm$ 1.96	31.68 $\pm$ 2.61	9.56 $\pm$ 1.50	29.01 $\pm$ 2.49	29.83 $\pm$ 3.23	37.35 $\pm$ 4.07
EOGVerticalSignal	13.37 $\pm$ 1.32	16.57 $\pm$ 3.25	14.86 $\pm$ 2.77	26.80 $\pm$ 1.98	29.56 $\pm$ 4.11	26.98 $\pm$ 4.28	11.88 $\pm$ 1.54	26.34 $\pm$ 4.16	10.83 $\pm$ 1.45	26.61 $\pm$ 6.09	20.88 $\pm$ 2.94	31.82 $\pm$ 3.47
Earthquakes	52.95 $\pm$ 7.17	57.84 $\pm$ 11.11	49.35 $\pm$ 10.38	53.24 $\pm$ 4.93	63.88 $\pm$ 6.42	63.31 $\pm$ 10.45	49.16 $\pm$ 1.10	54.92 $\pm$ 17.17	56.55 $\pm$ 8.34	60.43 $\pm$ 21.88	57.12 $\pm$ 15.34	55.25 $\pm$ 14.33
ElectricDevices	37.63 $\pm$ 9.44	41.80 $\pm$ 2.08	34.70 $\pm$ 5.95	21.27 $\pm$ 5.73	37.69 $\pm$ 10.00	28.63 $\pm$ 3.21	22.56 $\pm$ 2.12	20.63 $\pm$ 2.87	22.76 $\pm$ 4.51	29.88 $\pm$ 3.14	14.36 $\pm$ 4.97	47.14 $\pm$ 3.99
EthanolLevel	25.32 $\pm$ 0.64	<u>25.52 $\pm$ 1.56</u>	25.12 $\pm$ 2.09	25.76 $\pm$ 1.71	26.96 $\pm$ 0.82	26.87 $\pm$ 2.61	29.20 $\pm$ 1.91	25.67 $\pm$ 3.18	25.24 $\pm$ 0.98	26.47 $\pm$ 2.08	25.12 $\pm$ 0.81	<u>28.00 $\pm$ 3.11</u>
FaceAll	55.98 $\pm$ 6.19	45.22 $\pm$ 8.18	55.98 $\pm$ 6.19	40.27 $\pm$ 5.76	24.12 $\pm$ 4.85	38.56 $\pm$ 3.11	38.84 $\pm$ 3.81	36.71 $\pm$ 2.55	38.89 $\pm$ 3.64	41.40 $\pm$ 3.79	11.35 $\pm$ 5.01	39.09 $\pm$ 1.28
FaceFour	77.73 $\pm$ 4.99	52.05 $\pm$ 10.37	81.82 $\pm$ 7.14	74.09 $\pm$ 5.98	59.32 $\pm$ 5.76	79.55 $\pm$ 9.02	57.58 $\pm$ 7.31	80.68 $\pm$ 3.41	62.50 $\pm$ 7.14	84.85 $\pm$ 2.37	40.45 $\pm$ 14.16	68.55 $\pm$ 4.91
FacesUCR	67.20 $\pm$ 6.42	51.90 $\pm$ 6.85	<u>62.04 $\pm$ 8.47</u>	41.67 $\pm$ 3.36	25.42 $\pm$ 6.93	44.55 $\pm$ 2.52	45.95 $\pm$ 4.27	49.07 $\pm$ 5.45	48.56 $\pm$ 3.79	49.27 $\pm$ 6.47	10.96 $\pm$ 5.26	55.08 $\pm$ 2.32
FiftyWords	19.03 $\pm$ 1.51	13.49 $\pm$ 1.95	19.03 $\pm$ 1.51	40.09 $\pm$ 3.22	28.35 $\pm$ 3.05	44.69 $\pm$ 2.97	5.05 $\pm$ 0.44	46.74 $\pm$ 2.43	16.53 $\pm$ 1.75	44.18 $\pm$ 3.70	15.16 $\pm$ 3.75	47.30 $\pm$ 2.89
Fish	50.97 $\pm$ 10.77	26.97 $\pm$ 9.88	<u>50.51 $\pm$ 16.65</u>	50.17 $\pm$ 6.36	31.54 $\pm$ 5.18	45.90 $\pm$ 8.88	20.00 $\pm$ 4.12	39.05 $\pm$ 5.64	22.97 $\pm$ 3.62	45.71 $\pm$ 10.47	22.86 $\pm$ 2.83	49.91 $\pm$ 10.25
FordA	64.98 $\pm$ 8.59	62.27 $\pm$ 9.82	<u>62.82 $\pm$ 9.33</u>	49.92 $\pm$ 1.84	60.48 $\pm$ 7.29	50.30 $\pm$ 1.31	50.66 $\pm$ 0.90	50.15 $\pm$ 0.79	50.27 $\pm$ 1.01	50.58 $\pm$ 1.43	50.76 $\pm$ 2.11	53.35 $\pm$ 2.84
FordB	61.19 $\pm$ 3.67	58.10 $\pm$ 1.40	<u>58.67 $\pm$ 2.63</u>	49.09 $\pm$ 1.35	56.89 $\pm$ 2.99	49.84 $\pm$ 0.						

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	64.88 ± 2.42	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	66.58 ± 5.17	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	79.33 ± 16.25
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	68.61 ± 1.37	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	87.66 ± 5.66
GunPointOldVersusYoung	91.94 ± 11.58	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	58.73 ± 5.25	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	69.89 ± 13.52	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	29.33 ± 7.26	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	56.88 ± 7.13	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	76.30 ± 9.04	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	19.93 ± 2.98	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.07 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	51.03 ± 4.83	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.83 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	83.15 ± 15.39	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	54.19 ± 10.41	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 10.67	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	63.28 ± 8.25
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	59.36 ± 11.07	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	91.78 ± 0.99	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	89.67 ± 7.85	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	34.53 ± 1.25	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	63.62 ± 4.54	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 1.71	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	50.39 ± 7.58	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	49.42 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	54.07 ± 11.30	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	46.36 ± 2.32	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	67.98 ± 4.06
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	68.35 ± 3.26
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	83.65 ± 4.68	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	55.96 ± 3.30	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	62.09 ± 2.37	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	56.03 ± 5.89	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	80.67 ± 3.65	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	40.19 ± 5.06
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	60.64 ± 3.81	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	12.22 ± 1.88	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	62.40 ± 1.67	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	49.52 ± 0.76	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	69.42 ± 3.86	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	61.92 ± 1.78	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	100.00 ± 0.00	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	83.78 ± 3.63	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	68.10 ± 13.07	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	65.43 ± 5.55	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	67.12 ± 8.12	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	41.39 ± 9.49	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	70.00 ± 11.14	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	40.21 ± 6.72	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	62.87 ± 8.65	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	36.00 ± 6.56	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70

Continued on next page

Table 8: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	32.00 $\pm$ 3.88	30.71 $\pm$ 4.13	33.56 $\pm$ 5.04	45.91 $\pm$ 5.66	40.67 $\pm$ 4.39	29.56 $\pm$ 4.02	23.56 $\pm$ 2.47	32.52 $\pm$ 7.62	24.09 $\pm$ 3.07	30.81 $\pm$ 0.84	29.47 $\pm$ 4.28	43.20 $\pm$ 6.72
ShakeGestureWiimoteZ	76.00 $\pm$ 6.63	76.80 $\pm$ 7.43	81.60 $\pm$ 0.89	47.20 $\pm$ 7.95	58.40 $\pm$ 5.18	54.00 $\pm$ 6.00	26.00 $\pm$ 4.00	54.67 $\pm$ 4.62	32.40 $\pm$ 2.61	56.00 $\pm$ 5.29	58.40 $\pm$ 5.18	81.60 $\pm$ 2.61
ShapeletSim	78.89 $\pm$ 14.16	98.33 $\pm$ 2.08	92.11 $\pm$ 5.26	50.11 $\pm$ 2.87	80.22 $\pm$ 16.23	47.59 $\pm$ 3.16	50.74 $\pm$ 3.06	48.15 $\pm$ 1.79	49.22 $\pm$ 2.93	46.48 $\pm$ 3.06	49.67 $\pm$ 1.50	67.44 $\pm$ 10.92
ShapesAll	36.80 $\pm$ 3.60	28.17 $\pm$ 3.75	36.80 $\pm$ 3.60	50.73 $\pm$ 1.71	32.23 $\pm$ 5.06	55.78 $\pm$ 1.50	6.11 $\pm$ 0.63	53.17 $\pm$ 1.00	18.33 $\pm$ 5.95	53.33 $\pm$ 2.62	27.63 $\pm$ 3.10	63.03 $\pm$ 2.22
SmallKitchenAppliances	48.85 $\pm$ 7.91	50.40 $\pm$ 8.87	50.19 $\pm$ 7.71	41.49 $\pm$ 3.33	47.15 $\pm$ 6.04	49.60 $\pm$ 6.54	35.20 $\pm$ 2.93	40.09 $\pm$ 2.14	42.03 $\pm$ 4.49	44.09 $\pm$ 8.36	34.61 $\pm$ 4.49	55.68 $\pm$ 8.74
SmoothSubspace	78.93 $\pm$ 10.17	77.87 $\pm$ 5.55	80.13 $\pm$ 5.80	33.33 $\pm$ 0.00	51.20 $\pm$ 5.65	58.22 $\pm$ 4.73	68.89 $\pm$ 7.31	59.11 $\pm$ 4.23	70.67 $\pm$ 4.88	81.33 $\pm$ 6.36	48.27 $\pm$ 9.05	85.07 $\pm$ 5.20
SonyAIBORobotSurface1	90.42 $\pm$ 4.34	94.08 $\pm$ 1.77	89.38 $\pm$ 5.35	64.83 $\pm$ 12.29	70.05 $\pm$ 3.77	62.90 $\pm$ 13.91	61.90 $\pm$ 7.89	66.61 $\pm$ 10.89	59.10 $\pm$ 5.18	62.73 $\pm$ 9.45	52.65 $\pm$ 8.42	71.85 $\pm$ 8.28
SonyAIBORobotSurface2	88.00 $\pm$ 2.82	84.87 $\pm$ 6.58	83.67 $\pm$ 9.41	74.59 $\pm$ 3.28	78.70 $\pm$ 2.68	69.60 $\pm$ 7.04	77.26 $\pm$ 1.79	78.49 $\pm$ 0.86	78.30 $\pm$ 1.49	79.29 $\pm$ 1.17	56.81 $\pm$ 17.39	68.21 $\pm$ 5.28
StarLightCurves	76.49 $\pm$ 11.97	66.10 $\pm$ 9.91	66.22 $\pm$ 10.70	67.92 $\pm$ 13.52	71.57 $\pm$ 12.97	63.90 $\pm$ 14.70	35.37 $\pm$ 11.93	65.31 $\pm$ 14.87	34.69 $\pm$ 14.98	63.78 $\pm$ 14.15	68.45 $\pm$ 12.08	74.47 $\pm$ 15.08
Strawberry	62.81 $\pm$ 6.83	57.14 $\pm$ 6.89	60.22 $\pm$ 9.39	51.84 $\pm$ 8.28	56.59 $\pm$ 2.41	53.33 $\pm$ 7.10	66.58 $\pm$ 2.89	52.97 $\pm$ 8.18	59.62 $\pm$ 11.05	54.59 $\pm$ 4.47	51.03 $\pm$ 11.13	59.41 $\pm$ 6.38
SwedishLeaf	72.45 $\pm$ 3.37	54.30 $\pm$ 6.89	72.45 $\pm$ 3.37	34.02 $\pm$ 3.00	43.26 $\pm$ 7.26	48.91 $\pm$ 2.97	33.28 $\pm$ 3.94	46.24 $\pm$ 1.25	38.85 $\pm$ 4.16	47.52 $\pm$ 2.78	7.94 $\pm$ 0.74	64.99 $\pm$ 3.83
Symbols	90.49 $\pm$ 1.42	75.88 $\pm$ 7.72	88.48 $\pm$ 4.41	81.23 $\pm$ 0.99	76.18 $\pm$ 6.87	81.37 $\pm$ 1.84	37.42 $\pm$ 3.12	82.04 $\pm$ 0.58	58.95 $\pm$ 6.86	80.00 $\pm$ 1.74	51.50 $\pm$ 7.93	82.69 $\pm$ 3.43
SyntheticControl	79.67 $\pm$ 4.05	85.07 $\pm$ 4.63	82.60 $\pm$ 6.01	41.40 $\pm$ 6.37	83.80 $\pm$ 5.24	55.00 $\pm$ 3.38	44.11 $\pm$ 1.39	59.89 $\pm$ 3.15	47.93 $\pm$ 4.58	62.89 $\pm$ 3.15	20.80 $\pm$ 4.95	87.93 $\pm$ 3.95
ToeSegmentation1	68.42 $\pm$ 7.02	60.79 $\pm$ 8.14	61.58 $\pm$ 7.54	53.95 $\pm$ 2.19	58.60 $\pm$ 10.97	51.90 $\pm$ 4.30	53.80 $\pm$ 2.21	53.95 $\pm$ 0.44	52.37 $\pm$ 4.44	51.61 $\pm$ 3.08	50.70 $\pm$ 2.53	60.26 $\pm$ 8.54
ToeSegmentation2	76.15 $\pm$ 9.44	79.69 $\pm$ 9.24	72.15 $\pm$ 7.68	48.92 $\pm$ 5.64	68.62 $\pm$ 12.68	66.41 $\pm$ 15.71	51.54 $\pm$ 5.38	63.08 $\pm$ 15.78	47.08 $\pm$ 8.33	63.59 $\pm$ 15.79	54.31 $\pm$ 14.96	76.15 $\pm$ 12.44
Trace	76.00 $\pm$ 3.00	76.40 $\pm$ 3.05	73.80 $\pm$ 2.95	52.40 $\pm$ 4.04	76.60 $\pm$ 4.39	47.33 $\pm$ 3.21	51.00 $\pm$ 4.00	49.00 $\pm$ 3.00	55.60 $\pm$ 4.62	49.00 $\pm$ 5.00	46.00 $\pm$ 6.78	98.40 $\pm$ 3.05
TwoLeadECG	82.53 $\pm$ 6.47	79.19 $\pm$ 7.45	70.63 $\pm$ 4.92	55.68 $\pm$ 3.84	59.95 $\pm$ 3.74	62.36 $\pm$ 5.86	56.60 $\pm$ 1.27	55.90 $\pm$ 1.72	55.49 $\pm$ 2.80	60.14 $\pm$ 7.03	48.48 $\pm$ 5.21	74.63 $\pm$ 8.09
TwoPatterns	52.71 $\pm$ 4.44	71.75 $\pm$ 5.17	65.30 $\pm$ 11.45	32.62 $\pm$ 1.84	44.61 $\pm$ 9.45	31.66 $\pm$ 1.59	24.88 $\pm$ 1.30	28.57 $\pm$ 1.14	25.14 $\pm$ 0.61	29.88 $\pm$ 2.34	27.30 $\pm$ 2.15	78.23 $\pm$ 0.41
UMD	54.86 $\pm$ 7.95	71.67 $\pm$ 9.36	68.75 $\pm$ 16.74	54.17 $\pm$ 4.55	62.50 $\pm$ 12.00	52.31 $\pm$ 6.22	47.69 $\pm$ 9.93	60.19 $\pm$ 1.45	53.06 $\pm$ 4.83	56.25 $\pm$ 6.05	47.50 $\pm$ 4.16	78.86 $\pm$ 14.11
UWaveGestureLibraryAll	32.50 $\pm$ 3.38	28.30 $\pm$ 5.01	31.51 $\pm$ 4.14	69.44 $\pm$ 6.55	43.65 $\pm$ 6.39	66.54 $\pm$ 6.96	26.72 $\pm$ 0.50	66.58 $\pm$ 7.48	30.82 $\pm$ 2.10	66.61 $\pm$ 7.59	54.22 $\pm$ 7.86	42.54 $\pm$ 0.28
UWaveGestureLibraryX	31.94 $\pm$ 2.72	28.97 $\pm$ 3.44	30.99 $\pm$ 3.77	50.42 $\pm$ 3.39	42.01 $\pm$ 2.19	51.69 $\pm$ 1.72	21.35 $\pm$ 1.20	51.05 $\pm$ 1.76	28.72 $\pm$ 2.91	50.93 $\pm$ 1.76	22.53 $\pm$ 5.80	42.54 $\pm$ 0.28
UWaveGestureLibraryY	28.00 $\pm$ 4.17	26.42 $\pm$ 2.97	27.96 $\pm$ 5.26	43.36 $\pm$ 3.42	39.43 $\pm$ 5.20	42.54 $\pm$ 7.86	19.88 $\pm$ 2.66	43.61 $\pm$ 6.37	28.02 $\pm$ 2.78	41.75 $\pm$ 7.40	28.22 $\pm$ 7.42	42.54 $\pm$ 0.28
UWaveGestureLibraryZ	31.19 $\pm$ 2.60	30.19 $\pm$ 2.10	28.42 $\pm$ 4.12	44.60 $\pm$ 3.60	41.59 $\pm$ 3.78	42.17 $\pm$ 5.87	23.93 $\pm$ 2.40	43.23 $\pm$ 7.30	29.59 $\pm$ 1.71	40.53 $\pm$ 6.44	23.38 $\pm$ 7.02	59.88 $\pm$ 2.03
Wafer	47.08 $\pm$ 18.17	57.95 $\pm$ 22.06	52.16 $\pm$ 16.40	56.77 $\pm$ 13.38	49.59 $\pm$ 17.01	67.22 $\pm$ 1.00	55.02 $\pm$ 6.93	66.32 $\pm$ 1.28	56.21 $\pm$ 9.73	67.64 $\pm$ 1.29	54.99 $\pm$ 20.43	55.01 $\pm$ 15.62
Wine	60.74 $\pm$ 8.43	50.74 $\pm$ 7.92	60.48 $\pm$ 12.52	53.70 $\pm$ 8.88	47.41 $\pm$ 11.54	53.70 $\pm$ 3.21	53.09 $\pm$ 5.35	50.00 $\pm$ 0.00	50.74 $\pm$ 1.66	54.94 $\pm$ 26.28	50.74 $\pm$ 1.66	48.89 $\pm$ 5.34
WordSynonyms	20.63 $\pm$ 4.03	16.05 $\pm$ 1.67	20.63 $\pm$ 4.03	21.35 $\pm$ 4.96	20.06 $\pm$ 3.25	29.05 $\pm$ 0.77	6.11 $\pm$ 1.90	29.73 $\pm$ 1.73	13.92 $\pm$ 2.48	29.57 $\pm$ 1.41	6.43 $\pm$ 4.20	34.92 $\pm$ 2.64
Worms	54.03 $\pm$ 11.78	44.94 $\pm$ 11.42	52.73 $\pm$ 7.32	22.60 $\pm$ 2.99	28.57 $\pm$ 6.43	19.48 $\pm$ 1.30	17.32 $\pm$ 5.41	17.75 $\pm$ 5.25	15.58 $\pm$ 4.00	20.35 $\pm$ 3.75	18.70 $\pm$ 5.00	38.70 $\pm$ 4.63
WormsTwoClass	55.32 $\pm$ 9.03	52.47 $\pm$ 8.50	55.84 $\pm$ 3.31	48.57 $\pm$ 2.69	49.09 $\pm$ 4.63	49.78 $\pm$ 2.70	51.52 $\pm$ 8.84	47.19 $\pm$ 3.97	47.27 $\pm$ 4.37	46.32 $\pm$ 1.50	48.83 $\pm$ 7.82	54.03 $\pm$ 7.72
Yoga	52.99 $\pm$ 4.56	50.68 $\pm$ 2.33	50.48 $\pm$ 3.16	50.86 $\pm$ 0.91	51.49 $\pm$ 3.75	50.12 $\pm$ 2.91	51.48 $\pm$ 2.93	50.70 $\pm$ 2.90	51.46 $\pm$ 3.58	50.58 $\pm$ 2.12	51.69 $\pm$ 4.15	50.64 $\pm$ 4.90
Avg. Acc.	53.98	50.79	<u>54.95</u>	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	55.55
Avg. Rank	4.84	5.63	<u>4.83</u>	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	4.28
#Best	33	13	19	10	2	9	4	6	3	7	2	<u>32</u>

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	76.00 $\pm$ 2.92	64.60 $\pm$ 5.18	72.80 $\pm$ 1.64	36.20 $\pm$ 3.49	50.40 $\pm$ 2.61	60.00 $\pm$ 2.00	40.00 $\pm$ 1.00	62.67 $\pm$ 2.52	46.80 $\pm$ 3.70	58.67 $\pm$ 1.53	30.60 $\pm$ 4.83	68.33 $\pm$ 2.89
Adiac	56.21 $\pm$ 4.18	42.81 $\pm$ 4.46	56.21 $\pm$ 4.18	35.29 $\pm$ 1.32	34.58 $\pm$ 2.33	43.82 $\pm$ 1.54	46.97 $\pm$ 3.07	38.62 $\pm$ 1.68	64.50 $\pm$ 5.47	43.99 $\pm$ 3.86	7.11 $\pm$ 1.15	69.51 $\pm$ 2.42
AllGestureWiimoteX	40.46 $\pm$ 4.50	34.14 $\pm$ 3.58	38.71 $\pm$ 5.08	23.63 $\pm$ 2.19	35.40 $\pm$ 4.31	32.24 $\pm$ 3.43	13.57 $\pm$ 1.57	29.95 $\pm$ 4.86	18.17 $\pm$ 3.28	29.38 $\pm$ 3.89	35.40 $\pm$ 4.31	39.40 $\pm$ 2.86
AllGestureWiimoteY	43.20 $\pm$ 1.70	40.60 $\pm$ 1.51	36.20 $\pm$ 3.42	27.89 $\pm$ 2.07	40.97 $\pm$ 3.06	36.76 $\pm$ 2.07	15.38 $\pm$ 0.81	34.43 $\pm$ 2.02	19.49 $\pm$ 2.37	35.43 $\pm$ 1.55	40.97 $\pm$ 3.06	48.11 $\pm$ 2.89
AllGestureWiimoteZ	41.46 $\pm$ 4.33	35.71 $\pm$ 5.12	36.46 $\pm$ 5.30	20.20 $\pm$ 1.22	27.11 $\pm$ 2.51	26.57 $\pm$ 0.25	13.86 $\pm$ 1.71	26.67 $\pm$ 2.65	17.37 $\pm$ 2.23	26.86 $\pm$ 0.99	27.11 $\pm$ 2.51	36.14 $\pm$ 0.97
ArrowHead	66.40 $\pm$ 5.44	61.03 $\pm$ 5.46	64.23 $\pm$ 7.62	59.43 $\pm$ 5.27	58.86 $\pm$ 12.66	60.19 $\pm$ 7.46	42.67 $\pm$ 0.87	60.76 $\pm$ 1.19	48.11 $\pm$ 2.87	63.43 $\pm$ 8.42	54.06 $\pm$ 9.79	73.60 $\pm$ 3.71
BME	55.60 $\pm$ 7.80	56.93 $\pm$ 3.90	68.00 $\pm$ 9.90	58.53 $\pm$ 7.20	89.60 $\pm$ 10.65	58.89 $\pm$ 5.55	73.33 $\pm$ 8.11	76.89 $\pm$ 7.34	80.00 $\pm$ 6.70	73.33 $\pm$ 5.93	53.07 $\pm$ 5.63	97.73 $\pm$ 1.53
Beef	54.67 $\pm$ 4.47	47.33 $\pm$ 2.79	55.33 $\pm$ 2.98	52.67 $\pm$ 7.60	43.33 $\pm$ 4.71	64.44 $\pm$ 3.85	50.00 $\pm$ 3.33	52.22 $\pm$ 8.39	62.00 $\pm$ 8.03	84.44 $\pm$ 1.92	35.33 $\pm$ 4.47	51.33 $\pm$ 6.91
BeetleFly	72.00 $\pm$ 7.58	60.00 $\pm$ 11.73	79.00 $\pm$ 10.84	71.00 $\pm$ 14.32	57.00 $\pm$ 8.37	70.00 $\pm$ 8.66	51.67 $\pm$ 2.89	66.67 $\pm$ 16.07	49.00 $\pm$ 6.52	70.00 $\pm$ 10.00	59.00 $\pm$ 18.51	91.67 $\pm$ 2.89
BirdChicken	69.00 $\pm$ 12.94	77.00 $\pm$ 9.75	90.00 $\pm$ 3.54	57.00 $\pm$ 5.70	61.00 $\pm$ 16.73	63.33 $\pm$ 7.64	36.67 $\pm$ 2.89	65.00 $\pm$ 10.00	54.00 $\pm$ 8.94	68.33 $\pm$ 15.28	54.00 $\pm$ 4.18	85.00 $\pm$ 11.73
CBF	95.87 $\pm$ 2.59	95.58 $\pm$ 3.67	95.89 $\pm$ 3.27	61.47 $\pm$ 7.78	95.67 $\pm$ 4.48	74.52 $\pm$ 5.66	39.04 $\pm$ 0.80	80.11 $\pm$ 4.97	42.29 $\pm$ 1.09	80.59 $\pm$ 4.09	41.20 $\pm$ 4.62	93.76 $\pm$ 3.90
Car	59.33 $\pm$ 9.02	51.00 $\pm$ 7.96	65.67 $\pm$ 5.73	52.33 $\pm$ 6.41	45.67 $\pm$ 2.79	68.89 $\pm$ 6.31	31.67 $\pm$ 5.00	48.89 $\pm$ 4.19	41.00 $\pm$ 12.56	57.78 $\pm$ 5.85	36.00 $\pm$ 4.50	69.67 $\pm$ 3.98
Chinatown	96.15 $\pm$ 2.16	96.97 $\pm$ 0.49	97.43 $\pm$ 0.38	50.96 $\pm$ 33.10	96.03 $\pm$ 2.24	84.94 $\pm$ 15.12	91.06 $\pm$ 1.94	93.00 $\pm$ 5.85	89.56 $\pm$ 2.00	87.95 $\pm$ 8.75	73.00 $\pm$ 0.76	93.46 $\pm$ 10.56
ChlorineConcentration	35.97 $\pm$ 2.09	35.63 $\pm$ 3.63	36.15 $\pm$ 2.53	35.57 $\pm$ 3.02	35.80 $\pm$ 3.16	35.02 $\pm$ 2.94	36.15 $\pm$ 1.24	35.49 $\pm$ 4.56	37.02 $\pm$ 1.05	35.96 $\pm$ 1.50	38.60 $\pm$ 5.59	39.58 $\pm$ 3.20
CinCECGTorso	49.41 $\pm$ 5.43	57.77 $\pm$ 5.88	56.17 $\pm$ 3.91	43.81 $\pm$ 1.84	69.26 $\pm$ 4.67	63.41 $\pm$ 2.73	37.80 $\pm$ 1.42	64.95 $\pm$ 2.40	64.83 $\pm$ 11.83	68.41 $\pm$ 5.18	54.58 $\pm$ 3.46	82.56 $\pm$ 7.22
Coffee	86.43 $\pm$ 14.81	97.86 $\pm$ 1.96	100.00 $\pm$ 0.00	92.86 $\pm$ 3.57	97.86 $\pm$ 1.96	86.90 $\pm$ 2.06	76.19 $\pm$ 2.06	82.14 $\pm$ 9.45	92.86 $\pm$ 10.41	85.71 $\pm$ 0.00	50.00 $\pm$ 6.19	94.29 $\pm$ 1.96
Computers	58.80 $\pm$ 7.81	61.60 $\pm$ 5.76	60.96 $\pm$ 6.58	46.64 $\pm$ 4.37	54.48 $\pm$ 7.54	48.87 $\pm$ 5.33	50.53 $\pm$ 1.29	48.00 $\pm$ 8.43	49.04 $\pm$ 4.09	48.67 $\pm$ 4.31	46.56 $\pm$ 5.23	62.67 $\pm$ 4.74
CricketX	45.90 $\pm$ 4.41	43.08 $\pm$ 3.97	43.08 $\pm$ 2.15	25.23 $\pm$ 1.80	36.10 $\pm$ 5.19	34.87 $\pm$ 4.89	11.79 $\pm$ 1.43	30.77 $\pm$ 0.51	16.92 $\pm$ 1.32	33.68 $\pm$ 3.44	11.33 $\pm$ 1.41	49.06 $\pm$ 3.21
CricketY	42.67 $\pm$ 4.49	43.08 $\pm$ 3.40	43.28 $\pm$ 6.01	30.41 $\pm$ 3.78	36.92 $\pm$ 6.92	37.86 $\pm$ 2.42	9.06 $\pm$ 1.21	36.58 $\pm$ 2.06	15.23 $\pm$ 1.85	37.26 $\pm$ 2.44	12.05 $\pm$ 2.18	49.66 $\pm$ 1.41
CricketZ	51.59 $\pm$ 3.02	49.64 $\pm$ 3.11	47.79 $\pm$ 6.26	26.31 $\pm$ 2.67	36.31 $\pm$ 4.35	37.18 $\pm$ 1.18	10.77 $\pm$ 0.51	34.87 $\pm$ 1.43	15.44 $\pm$ 1.54	37.18 $\pm$ 0.77	10.92 $\pm$ 1.59	53.68 $\pm$ 2.06
Crop	49.93 $\pm$ 1.45	46.93 $\pm$ 2.27	49.93 $\pm$ 1.45	31.91 $\pm$ 2.21	38.13 $\pm$ 2.76	46.92 $\pm$ 2.69	44.47 $\pm$ 0.97	47.82 $\pm$ 2.55	47.76 $\pm$ 1.51	48.69 $\pm$ 2.69	8.15 $\pm$ 3.22	52.18 $\pm$ 2.24
DiatomSizeReduction	85.75 $\pm$ 6.98	70.26 $\pm$ 16.84	95.95 $\pm$ 1.26	61.57 $\pm$ 2.74	65.49 $\pm$ 10.17	87.69 $\pm$ 0.50	68.63 $\pm$ 2.55	66.34 $\pm$ 8.52	73.53 $\pm$ 9.37	87.25 $\pm$ 0.00	41.83 $\pm$ 13.83	93.66 $\pm$ 1.97
DistalPhalanxOutlineAgeGroup	65.76 $\pm$ 4.54	68.63 $\pm$ 3.96	67.19 $\pm$ 5.75	58.99 $\pm$ 14.81	63.31 $\pm$ 4.46	64.75 $\pm$ 3.74	67.63 $\pm$ 2.16	69.30 $\pm$ 3.24	68.20 $\pm$ 2.35	68.59 $\pm$ 2.31	53.67 $\pm$ 12.57	69.35 $\pm$ 1.20
DistalPhalanxOutlineCorrect	66.81 $\pm$ 3.92	61.96 $\pm$ 7.85	68.62 $\pm$ 6.57	56.23 $\pm$ 10.81	60.07 $\pm$ 6.07	62.44 $\pm$ 3.65	62.68 $\pm$ 3.32	58.94 $\pm$ 3.37	63.41 $\pm$ 7.40	59.30 $\pm$ 3.08	55.87 $\pm$ 8.58	63.77 $\pm$ 11.63
DistalPhalanxTW	58.99 $\pm$ 2.69	57.12 $\pm$ 1.40	58.56 $\pm$ 3.40	49.64 $\pm$ 7.01	60.58 $\pm$ 2.61	57.31 $\pm$ 0.42	51.32 $\pm$ 3.62	59.47 $\pm$ 4.40	54.10 $\pm$ 4.24	56.59 $\pm$ 5.05	31.94 $\pm$ 15.91	56.40 $\pm$ 4.87
DodgerLoopDay	37.75 $\pm$ 3.35	40.50 $\pm$ 4.11	37.00 $\pm$ 3.26	46.75 $\pm$ 4.01	44.25 $\pm$ 2.09	52.50 $\pm$ 5.45	22.08 $\pm$ 0.72	45.42 $\pm$ 2.60	24.25 $\pm$ 3.49	50.83 $\pm$ 4.73	44.25 $\pm$ 2.09	45.75 $\pm$ 3.26
DodgerLoopGame	66.52 $\pm$ 4.02	82.32 $\pm$ 3.38	72.17 $\pm$ 3.64	76.09 $\pm$ 5.15	84.49 $\pm$ 1.89	75.36 $\pm$ 5.02	50.00 $\pm$ 2.61	78.99 $\pm$ 7.63	53.33 $\pm$ 2.09	77.29 $\pm$ 4.82	84.49 $\pm$ 1.89	72.61 $\pm$ 10.25
DodgerLoopWeekend	95.51 $\pm$ 1.65	92.46 $\pm$ 3.46	94.78 $\pm$ 1.88	97.39 $\pm$ 0.40	97.25 $\pm$ 0.61	96.62 $\pm$ 0.84	66.43 $\pm$ 4.25	97.83 $\pm$ 0.00	72.90 $\pm$ 1.82	96.86 $\pm$ 0.42	97.25 $\pm$ 0.61	93.62 $\pm$ 2.37
ECG200	76.40 $\pm$ 3.58	77.80 $\pm$ 6.26	76.20 $\pm$ 4.60	75.00 $\pm$ 1.58	62.80 $\pm$ 16.36	77.67 $\pm$ 3.06	76.67 $\pm$ 3.79	78.00 $\pm$ 2.00	77.60 $\pm$ 3.21	79.33 $\pm$ 4.93	69.60 $\pm$ 4.51	67.40 $\pm$ 8.91
ECG5000	78.85 $\pm$ 9.03	75.51 $\pm$ 9.29	75.56 $\pm$ 6.88	84.88 $\pm$ 3.51	71.69 $\pm$ 7.99	82.01 $\pm$ 5.12	76.33 $\pm$ 7.90	83.48 $\pm$ 5.73	74.58 $\pm$ 8.80	82.23 $\pm$ 5.84	18.64 $\pm$ 16.84	73.89 $\pm$ 8.21
ECGFiveDays	71.54 $\pm$ 13.30	70.27 $\pm$ 8.41	70.22 $\pm$ 9.52	69.38 $\pm$ 6.83	77.54 $\pm$ 3.65	72.94 $\pm$ 9.66	87.65 $\pm$ 8.42	75.22 $\pm$ 2.60	83.30 $\pm$ 11.51	75.61 $\pm$ 4.26	60.49 $\pm$ 6.59	70.96 $\pm$ 3.91
EOGHorizontalSignal	24.97 $\pm$ 1.88	32.71 $\pm$ 1.62	27.35 $\pm$ 3.43	29.56 $\pm$ 3.95	37.73 $\pm$ 6.23	36.92 $\pm$ 1.62	10.96 $\pm$ 1.15	38.40 $\pm$ 2.19	9.94 $\pm$ 1.90	35.17 $\pm$ 4.76	31.99 $\pm$ 1.89	46.35 $\pm$ 2.27
EOGVerticalSignal	18.18 $\pm$ 3.62	22.10 $\pm$ 3.22	18.78 $\pm$ 4.22	29.01 $\pm$ 2.45	38.23 $\pm$ 4.43	31.77 $\pm$ 5.00	8.47 $\pm$ 1.80	31.95 $\pm$ 3.43	10.39 $\pm$ 1.59	31.22 $\pm$ 4.29	22.82 $\pm$ 2.72	36.74 $\pm$ 2.24
Earthquakes	50.94 $\pm$ 12.41	50.22 $\pm$ 8.85	52.23 $\pm$ 11.61	56.83 $\pm$ 2.92	52.09 $\pm$ 5.05	56.35 $\pm$ 11.87	51.32 $\pm$ 9.11	56.12 $\pm$ 3.14	53.38 $\pm$ 4.56	54.68 $\pm$ 7.61	52.81 $\pm$ 9.99	54.68 $\pm$ 9.31
ElectricDevices	36.69 $\pm$ 5.52	36.63 $\pm$ 6.97	36.08 $\pm$ 5.74	23.65 $\pm$ 1.86	38.77 $\pm$ 8.33	38.73 $\pm$ 7.16	22.89 $\pm$ 3.05	25.09 $\pm$ 5.42	31.29 $\pm$ 6.08	38.61 $\pm$ 5.70	11.47 $\pm$ 4.02	47.06 $\pm$ 4.18
EthanolLevel	27.52 $\pm$ 1.92	26.36 $\pm$ 0.86	26.60 $\pm$ 1.66	27.88 $\pm$ 1.25	24.88 $\pm$ 0.18	28.40 $\pm$ 1.73	28.60 $\pm$ 2.78	26.53 $\pm$ 0.12	25.40 $\pm$ 0.75	29.53 $\pm$ 0.64	26.32 $\pm$ 1.07	25.92 $\pm$ 1.40
FaceAll	67.63 $\pm$ 4.66	67.75 $\pm$ 7.28	67.63 $\pm$ 4.66	53.96 $\pm$ 2.14	41.22 $\pm$ 3.29	55.76 $\pm$ 2.36	53.79 $\pm$ 5.19	55.44 $\pm$ 2.51	51.51 $\pm$ 5.08	54.30 $\pm$ 2.41	17.64 $\pm$ 1.73	54.54 $\pm$ 1.26
FaceFour	88.18 $\pm$ 2.85	61.82 $\pm$ 13.74	89.77 $\pm$ 4.62	78.86 $\pm$ 3.17	57.50 $\pm$ 8.82	86.36 $\pm$ 1.14	64.77 $\pm$ 2.27	87.88 $\pm$ 1.74	71.82 $\pm$ 1.24	85.61 $\pm$ 1.74	41.14 $\pm$ 6.88	75.68 $\pm$ 4.37
FacesUCR	78.67 $\pm$ 2.69	73.00 $\pm$ 4.70	76.25 $\pm$ 3.16	50.94 $\pm$ 2.11	43.00 $\pm$ 7.53	68.33 $\pm$ 3.31	58.46 $\pm$ 2.45	68.99 $\pm$ 1.59	60.62 $\pm$ 1.49	67.61 $\pm$ 2.78	27.20 $\pm$ 3.34	68.58 $\pm$ 2.57
FiftyWords	30.07 $\pm$ 3.14	27.91 $\pm$ 3.50	30.07 $\pm$ 3.14	47.03 $\pm$ 2.26	37.71 $\pm$ 3.50	52.23 $\pm$ 1.46	9.45 $\pm$ 3.91	56.19 $\pm$ 3.63	25.10 $\pm$ 2.19	49.38 $\pm$ 1.41	40.84 $\pm$ 2.35	61.58 $\pm$ 2.55
Fish	61.26 $\pm$ 15.63	48.69 $\pm$ 6.91	67.66 $\pm$ 6.55	56.80 $\pm$ 6.22	42.97 $\pm$ 5.32	68.76 $\pm$ 3.49	25.52 $\pm$ 2.93	56.57 $\pm$ 5.51	40.80 $\pm$ 3.78	67.05 $\pm$ 1.44	26.40 $\pm$ 8.24	64.57 $\pm$ 5.92
FordA	71.41 $\pm$ 10.22	71.94 $\pm$ 9.09	70.55 $\pm$ 10.29	50.33 $\pm$ 1.09	63.62 $\pm$ 6.82	50.28 $\pm$ 2.11	51.67 $\pm$ 1.61	49.67 $\pm$ 1.72	51.11 $\pm$ 2.04	49.82 $\pm$ 1.42	50.91 $\pm$ 1.25	49.82 $\pm$ 1.25
FordB	61.26 $\pm$ 2.75	58.96 $\pm$ 1.47	59.04 $\pm$ 3.16	50.00 $\pm$ 1.41	55.88 $\pm$ 4.39	50.86 $\pm$ 0.77	49.51 $\pm$ 0.81	48.81 $\pm$ 1.26	49.56 $\pm$ 0.80	50.21 <		

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	71.74 $\pm$ 3.41	64.53 $\pm$ 3.05	73.60 $\pm$ 2.12	65.35 $\pm$ 3.15	63.95 $\pm$ 8.94	65.89 $\pm$ 3.50	23.45 $\pm$ 1.21	73.64 $\pm$ 2.20	28.95 $\pm$ 4.06	76.16 $\pm$ 6.07	63.95 $\pm$ 8.94	63.84 $\pm$ 8.99
GesturePebbleZ2	67.34 $\pm$ 3.97	52.53 $\pm$ 17.40	73.04 $\pm$ 4.07	58.35 $\pm$ 6.18	48.10 $\pm$ 7.12	62.03 $\pm$ 7.30	19.41 $\pm$ 4.45	64.98 $\pm$ 9.05	24.81 $\pm$ 3.02	60.34 $\pm$ 8.06	48.10 $\pm$ 7.12	60.89 $\pm$ 8.06
GunPoint	82.53 $\pm$ 12.46	77.33 $\pm$ 12.19	79.87 $\pm$ 13.86	59.60 $\pm$ 4.87	73.07 $\pm$ 2.03	66.00 $\pm$ 6.36	55.11 $\pm$ 6.19	70.22 $\pm$ 2.52	65.87 $\pm$ 7.58	66.00 $\pm$ 2.91	62.40 $\pm$ 5.79	90.13 $\pm$ 10.49
GunPointAgeSpan	75.70 $\pm$ 7.16	69.81 $\pm$ 6.17	76.46 $\pm$ 8.05	75.51 $\pm$ 2.81	73.67 $\pm$ 12.74	76.79 $\pm$ 5.43	49.26 $\pm$ 6.34	75.74 $\pm$ 4.37	60.38 $\pm$ 3.16	69.20 $\pm$ 7.84	65.70 $\pm$ 10.14	81.52 $\pm$ 9.45
GunPointMaleVersusFemale	91.20 $\pm$ 8.64	96.65 $\pm$ 4.17	90.32 $\pm$ 9.07	61.14 $\pm$ 7.95	87.53 $\pm$ 12.91	74.37 $\pm$ 13.69	55.17 $\pm$ 0.48	75.00 $\pm$ 13.22	61.20 $\pm$ 4.73	77.95 $\pm$ 14.78	63.92 $\pm$ 13.57	94.62 $\pm$ 6.24
GunPointOldVersusYoung	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	75.87 $\pm$ 2.49	70.29 $\pm$ 8.33	70.79 $\pm$ 4.48	56.93 $\pm$ 3.97	73.23 $\pm$ 4.31	58.98 $\pm$ 5.43	70.48 $\pm$ 8.59	92.38 $\pm$ 8.67	80.44 $\pm$ 8.63
Ham	60.95 $\pm$ 6.21	60.00 $\pm$ 6.77	62.10 $\pm$ 4.59	62.86 $\pm$ 11.45	58.86 $\pm$ 7.48	60.00 $\pm$ 11.70	63.49 $\pm$ 0.55	56.51 $\pm$ 8.59	63.62 $\pm$ 3.26	66.35 $\pm$ 16.28	56.76 $\pm$ 9.56	60.57 $\pm$ 9.91
HandOutlines	46.86 $\pm$ 11.04	52.86 $\pm$ 13.07	67.24 $\pm$ 3.85	80.76 $\pm$ 4.40	74.49 $\pm$ 7.28	83.06 $\pm$ 3.92	47.84 $\pm$ 8.25	81.62 $\pm$ 4.95	35.95 $\pm$ 0.00	82.34 $\pm$ 4.91	75.57 $\pm$ 5.75	68.54 $\pm$ 6.21
Haptics	32.14 $\pm$ 2.79	28.25 $\pm$ 2.44	31.49 $\pm$ 1.92	35.39 $\pm$ 2.49	28.18 $\pm$ 3.44	36.58 $\pm$ 0.19	21.32 $\pm$ 3.65	37.12 $\pm$ 1.14	24.68 $\pm$ 3.46	37.12 $\pm$ 0.94	31.82 $\pm$ 7.39	26.95 $\pm$ 4.88
Herring	50.94 $\pm$ 5.70	55.00 $\pm$ 8.93	52.50 $\pm$ 7.46	51.88 $\pm$ 6.76	50.94 $\pm$ 6.68	48.96 $\pm$ 5.02	57.29 $\pm$ 4.51	51.56 $\pm$ 7.16	46.88 $\pm$ 13.58	49.48 $\pm$ 2.39	49.06 $\pm$ 7.13	45.00 $\pm$ 4.87
HouseTwenty	83.70 $\pm$ 6.31	81.18 $\pm$ 9.08	81.85 $\pm$ 9.95	58.99 $\pm$ 5.13	76.81 $\pm$ 7.55	73.67 $\pm$ 7.34	54.90 $\pm$ 2.57	66.95 $\pm$ 4.78	59.66 $\pm$ 6.85	72.83 $\pm$ 4.23	63.03 $\pm$ 4.28	84.37 $\pm$ 7.51
InlineSkate	20.15 $\pm$ 4.70	22.95 $\pm$ 2.93	19.75 $\pm$ 2.66	21.05 $\pm$ 2.12	18.98 $\pm$ 2.35	24.85 $\pm$ 1.07	16.00 $\pm$ 1.31	24.06 $\pm$ 0.93	16.51 $\pm$ 0.15	24.06 $\pm$ 0.90	19.20 $\pm$ 1.41	28.07 $\pm$ 1.19
InsectEPGRegularTrain	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	56.47 $\pm$ 1.83	78.96 $\pm$ 9.81	59.57 $\pm$ 2.67	38.96 $\pm$ 2.13	57.30 $\pm$ 1.41	53.17 $\pm$ 6.32	61.31 $\pm$ 3.42	89.88 $\pm$ 9.24	80.24 $\pm$ 3.80
InsectEPGSmallTrain	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	53.98 $\pm$ 1.90	77.03 $\pm$ 8.38	59.44 $\pm$ 2.44	52.61 $\pm$ 2.78	60.51 $\pm$ 1.98	55.66 $\pm$ 1.65	62.12 $\pm$ 1.62	83.13 $\pm$ 0.00	83.05 $\pm$ 2.80
InsectWingbeatSound	25.39 $\pm$ 1.91	24.63 $\pm$ 1.27	26.10 $\pm$ 5.93	53.74 $\pm$ 0.54	42.18 $\pm$ 1.58	56.25 $\pm$ 3.65	25.84 $\pm$ 1.75	58.96 $\pm$ 0.98	38.75 $\pm$ 2.42	57.51 $\pm$ 1.69	20.02 $\pm$ 5.68	41.44 $\pm$ 2.25
ItalyPowerDemand	87.66 $\pm$ 4.88	89.29 $\pm$ 4.25	88.36 $\pm$ 4.24	62.66 $\pm$ 11.58	93.39 $\pm$ 3.05	58.50 $\pm$ 7.42	89.57 $\pm$ 5.31	93.78 $\pm$ 2.95	95.72 $\pm$ 1.14	95.72 $\pm$ 1.14	56.64 $\pm$ 14.68	84.26 $\pm$ 5.94
LargeKitchenAppliances	60.00 $\pm$ 4.27	58.51 $\pm$ 9.39	62.67 $\pm$ 10.64	37.55 $\pm$ 2.23	60.48 $\pm$ 4.88	44.09 $\pm$ 3.07	36.98 $\pm$ 2.46	40.44 $\pm$ 3.60	37.71 $\pm$ 3.74	44.09 $\pm$ 6.61	37.17 $\pm$ 1.99	50.13 $\pm$ 3.07
Lightning2	64.92 $\pm$ 6.82	62.30 $\pm$ 7.42	67.87 $\pm$ 5.51	60.33 $\pm$ 3.91	64.92 $\pm$ 4.86	69.95 $\pm$ 2.50	50.82 $\pm$ 5.68	68.85 $\pm$ 5.68	61.64 $\pm$ 7.91	68.31 $\pm$ 3.41	59.02 $\pm$ 6.86	69.51 $\pm$ 10.47
Lightning7	58.90 $\pm$ 7.12	54.25 $\pm$ 7.09	60.55 $\pm$ 3.80	51.23 $\pm$ 2.84	50.96 $\pm$ 8.07	67.58 $\pm$ 3.45	18.26 $\pm$ 3.95	61.64 $\pm$ 3.62	30.41 $\pm$ 0.61	70.32 $\pm$ 2.85	22.19 $\pm$ 2.63	66.30 $\pm$ 3.83
Mallat	71.03 $\pm$ 13.07	70.33 $\pm$ 5.20	93.72 $\pm$ 2.08	88.67 $\pm$ 2.40	77.56 $\pm$ 9.50	94.66 $\pm$ 0.33	65.67 $\pm$ 5.01	87.51 $\pm$ 2.78	79.72 $\pm$ 5.50	90.06 $\pm$ 1.99	56.21 $\pm$ 5.78	90.88 $\pm$ 1.15
Meat	64.33 $\pm$ 14.22	50.00 $\pm$ 9.93	81.00 $\pm$ 8.47	85.33 $\pm$ 10.70	49.33 $\pm$ 15.48	76.67 $\pm$ 10.41	47.78 $\pm$ 15.03	37.22 $\pm$ 5.36	50.00 $\pm$ 11.61	88.89 $\pm$ 2.55	34.00 $\pm$ 1.49	64.00 $\pm$ 11.82
MedicalImages	45.61 $\pm$ 3.94	43.95 $\pm$ 4.74	42.63 $\pm$ 4.22	27.26 $\pm$ 7.75	32.26 $\pm$ 2.93	37.41 $\pm$ 0.59	36.84 $\pm$ 3.23	38.03 $\pm$ 3.53	40.16 $\pm$ 6.60	38.20 $\pm$ 1.38	5.24 $\pm$ 3.29	42.03 $\pm$ 2.75
MelbournePedestrian	76.79 $\pm$ 2.31	74.96 $\pm$ 2.28	77.84 $\pm$ 3.11	11.65 $\pm$ 1.29	58.25 $\pm$ 3.82	62.66 $\pm$ 1.83	70.73 $\pm$ 4.35	71.76 $\pm$ 3.72	70.23 $\pm$ 2.95	67.90 $\pm$ 5.27	58.25 $\pm$ 3.82	65.81 $\pm$ 3.77
MiddlePhalanxOutlineAgeGroup	47.01 $\pm$ 8.99	54.03 $\pm$ 8.97	50.78 $\pm$ 7.01	52.08 $\pm$ 8.63	49.09 $\pm$ 6.85	50.22 $\pm$ 4.61	51.73 $\pm$ 7.75	53.90 $\pm$ 7.87	55.06 $\pm$ 4.76	54.11 $\pm$ 9.46	31.82 $\pm$ 16.75	49.74 $\pm$ 4.72
MiddlePhalanxOutlineCorrect	53.20 $\pm$ 7.83	52.71 $\pm$ 15.62	58.14 $\pm$ 12.45	46.94 $\pm$ 12.76	48.93 $\pm$ 10.44	49.60 $\pm$ 3.67	58.65 $\pm$ 6.95	55.44 $\pm$ 15.07	51.68 $\pm$ 10.94	49.60 $\pm$ 9.23	50.93 $\pm$ 8.48	48.45 $\pm$ 7.93
MiddlePhalanxTW	45.84 $\pm$ 9.64	45.84 $\pm$ 3.45	45.71 $\pm$ 4.74	45.84 $\pm$ 4.10	43.12 $\pm$ 4.67	44.37 $\pm$ 0.37	40.04 $\pm$ 2.62	43.72 $\pm$ 4.56	44.16 $\pm$ 4.95	44.59 $\pm$ 2.28	22.86 $\pm$ 6.13	43.64 $\pm$ 5.71
MixedShapesRegularTrain	63.93 $\pm$ 9.77	61.94 $\pm$ 4.99	68.21 $\pm$ 8.35	68.04 $\pm$ 1.34	68.63 $\pm$ 3.58	67.15 $\pm$ 2.55	20.78 $\pm$ 0.21	70.05 $\pm$ 0.70	30.67 $\pm$ 6.49	67.15 $\pm$ 1.60	64.11 $\pm$ 1.37	76.68 $\pm$ 5.11
MixedShapesSmallTrain	66.24 $\pm$ 12.23	57.21 $\pm$ 4.26	72.37 $\pm$ 7.91	66.37 $\pm$ 2.62	66.13 $\pm$ 3.66	68.30 $\pm$ 1.42	22.61 $\pm$ 0.88	67.60 $\pm$ 3.67	25.23 $\pm$ 4.04	66.78 $\pm$ 1.26	60.00 $\pm$ 5.54	78.71 $\pm$ 1.70
MoteStrain	84.74 $\pm$ 4.04	83.85 $\pm$ 4.63	85.10 $\pm$ 2.03	80.19 $\pm$ 8.46	77.96 $\pm$ 3.58	85.73 $\pm$ 4.20	72.68 $\pm$ 2.25	80.09 $\pm$ 3.68	77.11 $\pm$ 6.04	85.36 $\pm$ 1.87	60.30 $\pm$ 15.64	80.37 $\pm$ 4.50
NonInvasiveFetalECGThorax1	40.80 $\pm$ 1.72	31.63 $\pm$ 3.55	40.80 $\pm$ 1.72	72.10 $\pm$ 1.96	51.79 $\pm$ 9.05	70.09 $\pm$ 3.02	22.04 $\pm$ 0.40	70.72 $\pm$ 1.13	32.89 $\pm$ 2.08	69.19 $\pm$ 0.16	28.56 $\pm$ 4.30	67.42 $\pm$ 0.56
NonInvasiveFetalECGThorax2	41.15 $\pm$ 5.46	43.13 $\pm$ 8.61	41.15 $\pm$ 5.46	78.55 $\pm$ 0.96	60.71 $\pm$ 3.55	75.84 $\pm$ 1.88	37.61 $\pm$ 1.37	75.40 $\pm$ 1.50	50.66 $\pm$ 2.46	74.93 $\pm$ 2.49	37.30 $\pm$ 2.25	75.31 $\pm$ 0.82
OSULeaf	57.85 $\pm$ 5.97	50.74 $\pm$ 10.99	72.23 $\pm$ 4.84	32.40 $\pm$ 1.86	35.37 $\pm$ 3.13	38.84 $\pm$ 1.09	16.80 $\pm$ 1.04	32.64 $\pm$ 2.71	24.05 $\pm$ 2.60	36.91 $\pm$ 1.04	23.06 $\pm$ 4.07	49.92 $\pm$ 6.46
OliveOil	59.33 $\pm$ 13.21	28.67 $\pm$ 13.66	74.00 $\pm$ 6.83	31.33 $\pm$ 9.60	26.67 $\pm$ 10.00	42.22 $\pm$ 25.02	40.00 $\pm$ 10.00	32.22 $\pm$ 13.47	40.00 $\pm$ 21.73	32.22 $\pm$ 22.19	26.67 $\pm$ 10.00	28.67 $\pm$ 11.69
PLAID	24.02 $\pm$ 4.96	24.51 $\pm$ 3.44	22.87 $\pm$ 2.95	30.06 $\pm$ 4.34	47.93 $\pm$ 3.93	36.56 $\pm$ 2.69	26.88 $\pm$ 2.01	40.16 $\pm$ 2.24	27.26 $\pm$ 3.48	37.93 $\pm$ 4.09	47.93 $\pm$ 3.93	52.96 $\pm$ 5.39
PhalangesOutlinesCorrect	56.22 $\pm$ 7.89	55.22 $\pm$ 9.82	56.85 $\pm$ 6.96	52.75 $\pm$ 11.57	53.15 $\pm$ 7.53	51.83 $\pm$ 11.55	50.35 $\pm$ 9.47	51.20 $\pm$ 13.21	53.05 $\pm$ 9.33	50.85 $\pm$ 11.66	53.61 $\pm$ 9.97	53.61 $\pm$ 8.69
Phoneme	19.05 $\pm$ 2.97	18.15 $\pm$ 2.47	19.05 $\pm$ 2.97	6.61 $\pm$ 0.25	9.00 $\pm$ 1.08	10.58 $\pm$ 0.95	6.87 $\pm$ 0.83	9.00 $\pm$ 0.29	9.61 $\pm$ 1.88	9.30 $\pm$ 1.34	8.21 $\pm$ 0.75	15.63 $\pm$ 1.56
PickupGestureWiimoteZ	56.00 $\pm$ 10.20	66.80 $\pm$ 5.93	73.60 $\pm$ 4.56	52.40 $\pm$ 1.67	56.80 $\pm$ 9.55	57.33 $\pm$ 3.06	27.33 $\pm$ 1.15	58.00 $\pm$ 3.46	42.00 $\pm$ 2.00	58.67 $\pm$ 1.15	56.80 $\pm$ 9.55	68.00 $\pm$ 2.45
PigAirwayPressure	25.00 $\pm$ 2.61	13.37 $\pm$ 2.05	48.17 $\pm$ 0.40	5.10 $\pm$ 0.26	4.62 $\pm$ 0.55	4.81 $\pm$ 0.00	4.17 $\pm$ 2.22	6.73 $\pm$ 0.48	4.04 $\pm$ 0.87	5.93 $\pm$ 0.56	4.42 $\pm$ 0.86	9.81 $\pm$ 0.73
PigArtPressure	50.96 $\pm$ 5.81	20.77 $\pm$ 3.29	68.17 $\pm$ 4.63	8.46 $\pm$ 0.65	5.10 $\pm$ 1.82	8.97 $\pm$ 1.21	6.57 $\pm$ 2.00	8.97 $\pm$ 0.73	6.92 $\pm$ 1.65	9.29 $\pm$ 1.11	18.17 $\pm$ 2.24	33.08 $\pm$ 1.33
PigCVP	29.33 $\pm$ 2.90	18.75 $\pm$ 2.40	62.98 $\pm$ 1.73	6.83 $\pm$ 0.22	3.37 $\pm$ 1.32	7.53 $\pm$ 0.56	4.65 $\pm$ 1.11	7.53 $\pm$ 0.28	3.75 $\pm$ 1.24	4.81 $\pm$ 0.48	10.38 $\pm$ 1.21	19.52 $\pm$ 0.55
Plane	100.00 $\pm$ 0.00	98.10 $\pm$ 2.43	99.62 $\pm$ 0.52	88.95 $\pm$ 7.93	95.24 $\pm$ 3.16	97.78 $\pm$ 0.55	96.51 $\pm$ 1.98	95.87 $\pm$ 1.45	97.14 $\pm$ 1.51	95.87 $\pm$ 0.55	23.05 $\pm$ 11.32	99.24 $\pm$ 1.24
PowerCons	80.56 $\pm$ 3.36	84.44 $\pm$ 5.02	84.00 $\pm$ 6.30	70.11 $\pm$ 5.26	81.11 $\pm$ 10.74	73.37 $\pm$ 5.45	54.26 $\pm$ 3.90	80.93 $\pm$ 10.72	52.22 $\pm$ 4.39	78.52 $\pm$ 10.32	68.89 $\pm$ 15.97	76.11 $\pm$ 6.55
ProximalPhalanxOutlineAgeGroup	73.37 $\pm$ 8.19	81.76 $\pm$ 3.12	79.90 $\pm$ 3.95	65.56 $\pm$ 9.78	75.41 $\pm$ 5.61	79.02 $\pm$ 7.19	79.02 $\pm$					

Table 9: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	37.11 $\pm$ 4.58	44.09 $\pm$ 1.20	35.60 $\pm$ 3.13	50.76 $\pm$ 1.90	49.02 $\pm$ 0.83	46.67 $\pm$ 2.12	27.78 $\pm$ 0.59	51.19 $\pm$ 2.74	30.62 $\pm$ 1.93	41.48 $\pm$ 6.03	36.62 $\pm$ 4.30	54.31 $\pm$ 3.37
ShakeGestureWiimoteZ	80.40 $\pm$ 4.34	91.60 $\pm$ 2.61	89.20 $\pm$ 1.10	56.40 $\pm$ 1.67	72.80 $\pm$ 5.22	62.00 $\pm$ 0.00	35.33 $\pm$ 1.15	60.67 $\pm$ 3.06	44.00 $\pm$ 2.45	55.33 $\pm$ 2.31	72.80 $\pm$ 5.22	92.00 $\pm$ 0.00
ShapeletSim	82.78 $\pm$ 10.96	99.67 $\pm$ 0.75	92.56 $\pm$ 6.55	49.44 $\pm$ 2.69	88.22 $\pm$ 10.06	55.93 $\pm$ 2.31	49.81 $\pm$ 2.80	49.63 $\pm$ 4.21	50.67 $\pm$ 1.82	49.44 $\pm$ 3.33	49.67 $\pm$ 3.98	92.00 $\pm$ 3.88
ShapesAll	57.57 $\pm$ 3.68	48.33 $\pm$ 5.58	57.57 $\pm$ 3.68	58.13 $\pm$ 1.68	51.50 $\pm$ 3.69	64.67 $\pm$ 1.64	9.22 $\pm$ 0.79	65.83 $\pm$ 3.88	13.83 $\pm$ 3.47	59.22 $\pm$ 2.28	56.47 $\pm$ 1.46	76.20 $\pm$ 1.83
SmallKitchenAppliances	59.47 $\pm$ 8.51	57.39 $\pm$ 6.38	62.35 $\pm$ 6.85	39.04 $\pm$ 4.11	57.55 $\pm$ 2.52	57.51 $\pm$ 6.62	38.67 $\pm$ 2.54	39.56 $\pm$ 5.26	46.67 $\pm$ 3.53	56.62 $\pm$ 3.96	44.00 $\pm$ 7.15	51.63 $\pm$ 12.25
SmoothSubspace	86.80 $\pm$ 2.13	82.00 $\pm$ 2.83	88.67 $\pm$ 2.36	33.87 $\pm$ 1.19	57.07 $\pm$ 4.58	65.78 $\pm$ 2.04	69.33 $\pm$ 3.71	61.11 $\pm$ 0.77	77.07 $\pm$ 5.51	84.00 $\pm$ 4.81	37.07 $\pm$ 6.74	88.00 $\pm$ 2.87
SonyAIBORobotSurface1	93.94 $\pm$ 2.22	95.71 $\pm$ 1.41	92.85 $\pm$ 2.46	61.83 $\pm$ 10.72	84.73 $\pm$ 2.02	71.21 $\pm$ 3.16	67.39 $\pm$ 1.20	71.60 $\pm$ 6.11	64.96 $\pm$ 2.40	64.89 $\pm$ 3.03	54.54 $\pm$ 12.34	84.86 $\pm$ 2.67
SonyAIBORobotSurface2	92.51 $\pm$ 3.32	91.14 $\pm$ 0.89	91.54 $\pm$ 4.19	78.15 $\pm$ 2.59	85.83 $\pm$ 4.10	79.96 $\pm$ 1.73	83.11 $\pm$ 2.73	81.78 $\pm$ 2.73	80.76 $\pm$ 4.57	83.21 $\pm$ 2.88	61.15 $\pm$ 14.84	77.21 $\pm$ 3.67
StarLightCurves	81.52 $\pm$ 8.85	60.67 $\pm$ 16.70	76.44 $\pm$ 1.22	74.70 $\pm$ 4.25	76.72 $\pm$ 3.78	75.63 $\pm$ 4.10	28.48 $\pm$ 9.45	75.33 $\pm$ 3.39	27.99 $\pm$ 0.00	75.69 $\pm$ 5.23	75.95 $\pm$ 2.54	81.86 $\pm$ 7.93
Strawberry	63.51 $\pm$ 2.99	64.22 $\pm$ 4.90	60.22 $\pm$ 9.83	58.49 $\pm$ 2.90	63.03 $\pm$ 7.09	61.89 $\pm$ 4.92	67.21 $\pm$ 4.90	59.82 $\pm$ 1.80	61.03 $\pm$ 3.64	65.77 $\pm$ 7.40	52.92 $\pm$ 6.34	58.05 $\pm$ 7.04
SwedishLeaf	79.81 $\pm$ 1.70	76.00 $\pm$ 4.97	79.81 $\pm$ 1.70	47.97 $\pm$ 1.68	65.66 $\pm$ 4.54	62.61 $\pm$ 4.53	49.23 $\pm$ 4.26	67.15 $\pm$ 4.00	56.93 $\pm$ 3.43	62.29 $\pm$ 1.68	15.90 $\pm$ 6.80	79.78 $\pm$ 2.56
Symbols	88.78 $\pm$ 7.99	88.74 $\pm$ 7.74	91.24 $\pm$ 0.96	83.44 $\pm$ 1.29	84.92 $\pm$ 3.31	84.29 $\pm$ 1.01	36.11 $\pm$ 0.23	86.40 $\pm$ 1.63	58.19 $\pm$ 7.40	83.75 $\pm$ 0.90	27.56 $\pm$ 4.43	88.04 $\pm$ 1.30
SyntheticControl	87.80 $\pm$ 2.38	96.33 $\pm$ 0.33	92.60 $\pm$ 1.01	48.53 $\pm$ 3.68	92.73 $\pm$ 2.94	73.22 $\pm$ 3.10	52.11 $\pm$ 3.20	66.22 $\pm$ 3.66	56.27 $\pm$ 3.45	72.78 $\pm$ 3.66	20.93 $\pm$ 4.87	95.07 $\pm$ 1.62
ToeSegmentation1	85.96 $\pm$ 7.05	77.37 $\pm$ 5.14	81.58 $\pm$ 5.14	54.74 $\pm$ 3.58	68.16 $\pm$ 3.79	55.56 $\pm$ 2.16	53.95 $\pm$ 2.74	51.61 $\pm$ 0.91	53.25 $\pm$ 3.38	54.82 $\pm$ 1.32	48.77 $\pm$ 4.31	70.53 $\pm$ 10.39
ToeSegmentation2	74.77 $\pm$ 12.53	<u>77.85 $\pm$ 7.33</u>	<u>74.77 $\pm$ 8.79</u>	50.77 $\pm$ 2.88	68.46 $\pm$ 6.25	58.97 $\pm$ 1.60	50.26 $\pm$ 0.44	59.49 $\pm$ 6.72	37.54 $\pm$ 3.14	64.10 $\pm$ 2.70	52.31 $\pm$ 9.12	78.00 $\pm$ 6.36
Trace	88.60 $\pm$ 7.37	<u>99.00 $\pm$ 1.00</u>	93.00 $\pm$ 8.00	53.00 $\pm$ 4.42	76.40 $\pm$ 5.77	51.67 $\pm$ 4.93	53.33 $\pm$ 2.08	49.67 $\pm$ 3.79	59.80 $\pm$ 2.59	55.67 $\pm$ 8.50	48.80 $\pm$ 10.71	100.00 $\pm$ 0.00
TwoLeadECG	97.98 $\pm$ 4.17	95.47 $\pm$ 5.89	75.84 $\pm$ 5.58	58.75 $\pm$ 2.38	82.55 $\pm$ 14.28	62.16 $\pm$ 1.52	60.81 $\pm$ 0.10	56.34 $\pm$ 2.59	61.88 $\pm$ 1.65	68.51 $\pm$ 9.61	52.10 $\pm$ 2.73	96.68 $\pm$ 5.58
TwoPatterns	64.62 $\pm$ 3.64	74.98 $\pm$ 5.04	74.94 $\pm$ 1.43	38.09 $\pm$ 3.07	51.91 $\pm$ 7.48	40.83 $\pm$ 3.89	26.00 $\pm$ 0.92	39.83 $\pm$ 3.80	26.50 $\pm$ 1.83	37.54 $\pm$ 1.63	27.50 $\pm$ 4.13	74.22 $\pm$ 2.81
UMD	87.08 $\pm$ 10.10	91.53 $\pm$ 5.28	96.67 $\pm$ 2.05	56.53 $\pm$ 3.69	74.44 $\pm$ 17.00	61.57 $\pm$ 1.45	62.96 $\pm$ 5.82	68.52 $\pm$ 10.79	71.81 $\pm$ 2.17	70.14 $\pm$ 10.69	48.47 $\pm$ 7.32	95.78 $\pm$ 2.33
UWaveGestureLibraryAll	42.53 $\pm$ 1.15	39.33 $\pm$ 5.11	44.53 $\pm$ 1.84	77.02 $\pm$ 5.21	57.17 $\pm$ 2.24	78.18 $\pm$ 3.39	26.42 $\pm$ 1.62	<u>77.47 $\pm$ 3.61</u>	35.67 $\pm$ 1.53	76.96 $\pm$ 4.32	65.94 $\pm$ 6.48	48.36 $\pm$ 0.30
UWaveGestureLibraryX	39.64 $\pm$ 3.12	46.31 $\pm$ 4.46	46.23 $\pm$ 4.40	56.88 $\pm$ 2.64	50.95 $\pm$ 2.22	58.42 $\pm$ 2.37	19.27 $\pm$ 3.18	58.58 $\pm$ 2.34	31.22 $\pm$ 3.55	57.86 $\pm$ 1.47	26.48 $\pm$ 5.59	47.36 $\pm$ 0.30
UWaveGestureLibraryY	37.00 $\pm$ 1.20	38.44 $\pm$ 2.57	40.23 $\pm$ 3.63	52.69 $\pm$ 2.24	48.41 $\pm$ 3.12	51.42 $\pm$ 3.10	21.22 $\pm$ 1.11	51.72 $\pm$ 2.98	30.92 $\pm$ 5.43	50.81 $\pm$ 4.53	19.79 $\pm$ 3.07	51.36 $\pm$ 0.31
UWaveGestureLibraryZ	38.74 $\pm$ 2.41	44.95 $\pm$ 4.51	40.85 $\pm$ 4.13	48.35 $\pm$ 3.41	47.76 $\pm$ 4.10	50.34 $\pm$ 2.37	20.23 $\pm$ 1.68	49.48 $\pm$ 1.66	31.23 $\pm$ 4.16	48.05 $\pm$ 3.36	25.20 $\pm$ 6.78	55.54 $\pm$ 2.61
Wafer	76.83 $\pm$ 9.15	69.99 $\pm$ 14.88	71.79 $\pm$ 15.46	55.60 $\pm$ 16.77	73.34 $\pm$ 19.66	80.42 $\pm$ 16.85	76.25 $\pm$ 9.34	83.86 $\pm$ 16.37	75.80 $\pm$ 12.54	83.26 $\pm$ 17.85	64.01 $\pm$ 22.29	71.65 $\pm$ 9.58
Wine	57.41 $\pm$ 13.03	48.15 $\pm$ 3.93	61.11 $\pm$ 5.40	55.93 $\pm$ 3.56	53.33 $\pm$ 5.62	47.53 $\pm$ 8.35	54.94 $\pm$ 8.55	50.00 $\pm$ 0.00	46.67 $\pm$ 6.47	37.65 $\pm$ 19.80	50.00 $\pm$ 0.00	51.48 $\pm$ 2.03
WordSynonyms	25.08 $\pm$ 1.54	21.22 $\pm$ 1.56	25.08 $\pm$ 1.54	29.56 $\pm$ 2.06	30.38 $\pm$ 2.74	45.87 $\pm$ 4.63	7.99 $\pm$ 1.24	44.67 $\pm$ 4.83	18.81 $\pm$ 2.07	41.33 $\pm$ 4.67	22.26 $\pm$ 2.64	47.05 $\pm$ 3.87
Worms	57.14 $\pm$ 9.76	55.58 $\pm$ 5.91	62.86 $\pm$ 4.55	24.68 $\pm$ 4.68	40.26 $\pm$ 5.11	22.94 $\pm$ 0.75	19.91 $\pm$ 3.27	25.97 $\pm$ 0.00	16.62 $\pm$ 2.32	22.08 $\pm$ 1.30	22.08 $\pm$ 5.11	44.42 $\pm$ 6.26
WormsTwoClass	68.05 $\pm$ 7.21	59.74 $\pm$ 6.69	62.08 $\pm$ 7.97	48.83 $\pm$ 3.13	51.17 $\pm$ 8.09	52.38 $\pm$ 2.70	42.86 $\pm$ 3.44	51.08 $\pm$ 5.25	48.83 $\pm$ 3.26	52.81 $\pm$ 3.27	48.83 $\pm$ 5.92	54.81 $\pm$ 7.59
Yoga	55.71 $\pm$ 3.37	53.57 $\pm$ 4.72	<u>55.19 $\pm$ 2.57</u>	50.13 $\pm$ 2.22	53.89 $\pm$ 3.78	49.79 $\pm$ 3.21	52.94 $\pm$ 1.57	47.50 $\pm$ 3.55	51.67 $\pm$ 1.99	49.68 $\pm$ 4.28	48.87 $\pm$ 2.05	51.31 $\pm$ 4.34
Avg. Acc.	61.27	59.47	<u>63.55</u>	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	64.14
Avg. Rank	4.77	5.34	<u>4.23</u>	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	4.19
#Best	21	13	<u>30</u>	6	4	8	4	7	2	8	1	36

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
ACSF1	85.40 $\pm$ 2.19	76.80 $\pm$ 3.35	80.80 $\pm$ 0.84	38.60 $\pm$ 0.89	58.80 $\pm$ 2.59	64.67 $\pm$ 0.58	42.67 $\pm$ 1.53	67.00 $\pm$ 2.00	53.60 $\pm$ 2.51	67.00 $\pm$ 2.00	36.80 $\pm$ 3.27	81.33 $\pm$ 1.53
Adiac	62.46 $\pm$ 4.53	56.98 $\pm$ 3.84	62.46 $\pm$ 4.53	45.32 $\pm$ 1.06	47.72 $\pm$ 5.48	56.01 $\pm$ 2.77	48.68 $\pm$ 0.78	47.91 $\pm$ 6.68	71.71 $\pm$ 1.94	48.00 $\pm$ 3.58	12.84 $\pm$ 1.42	76.42 $\pm$ 1.68
AllGestureWiimoteX	49.37 $\pm$ 3.15	43.11 $\pm$ 2.57	39.26 $\pm$ 2.50	23.91 $\pm$ 1.60	43.54 $\pm$ 2.85	37.14 $\pm$ 0.52	15.00 $\pm$ 0.87	35.19 $\pm$ 1.33	25.80 $\pm$ 2.24	31.00 $\pm$ 1.57	43.54 $\pm$ 2.85	52.31 $\pm$ 2.26
AllGestureWiimoteY	54.89 $\pm$ 2.09	50.74 $\pm$ 4.62	40.17 $\pm$ 2.24	27.26 $\pm$ 1.11	48.03 $\pm$ 1.58	42.57 $\pm$ 3.19	16.95 $\pm$ 2.22	42.57 $\pm$ 3.81	30.37 $\pm$ 4.65	41.24 $\pm$ 2.84	48.03 $\pm$ 1.58	56.34 $\pm$ 3.54
AllGestureWiimoteZ	49.34 $\pm$ 1.38	46.03 $\pm$ 2.76	39.80 $\pm$ 2.97	22.43 $\pm$ 1.80	34.54 $\pm$ 4.20	31.29 $\pm$ 1.13	16.67 $\pm$ 1.15	29.67 $\pm$ 3.32	21.71 $\pm$ 1.69	32.19 $\pm$ 0.59	34.54 $\pm$ 4.20	44.43 $\pm$ 2.91
ArrowHead	72.23 $\pm$ 5.22	63.89 $\pm$ 7.06	65.49 $\pm$ 9.09	64.34 $\pm$ 0.65	62.86 $\pm$ 7.57	60.38 $\pm$ 1.19	48.19 $\pm$ 2.58	59.62 $\pm$ 2.88	57.83 $\pm$ 5.80	63.43 $\pm$ 3.48	45.83 $\pm$ 13.23	81.49 $\pm$ 3.22
BME	64.27 $\pm$ 9.29	69.33 $\pm$ 1.56	78.67 $\pm$ 4.64	64.53 $\pm$ 0.30	97.60 $\pm$ 3.29	71.11 $\pm$ 2.34	78.89 $\pm$ 1.02	88.44 $\pm$ 1.02	91.33 $\pm$ 2.31	86.67 $\pm$ 0.00	46.40 $\pm$ 7.46	99.20 $\pm$ 0.30
Beef	51.33 $\pm$ 6.91	44.00 $\pm$ 6.83	52.00 $\pm$ 2.98	61.33 $\pm$ 4.47	46.00 $\pm$ 6.41	70.00 $\pm$ 3.33	51.11 $\pm$ 5.09	54.44 $\pm$ 3.85	64.00 $\pm$ 7.23	80.00 $\pm$ 0.00	37.33 $\pm$ 4.35	68.00 $\pm$ 1.83
BeetleFly	71.00 $\pm$ 8.94	74.00 $\pm$ 6.52	75.00 $\pm$ 6.12	86.00 $\pm$ 4.18	86.00 $\pm$ 2.24	81.67 $\pm$ 2.89	48.33 $\pm$ 2.89	83.33 $\pm$ 2.89	49.00 $\pm$ 11.94	75.00 $\pm$ 5.00	70.00 $\pm$ 12.25	96.67 $\pm$ 2.89
BirdChicken	84.00 $\pm$ 12.45	92.00 $\pm$ 5.70	94.00 $\pm$ 4.18	68.00 $\pm$ 2.74	81.00 $\pm$ 9.62	71.67 $\pm$ 7.64	41.67 $\pm$ 7.64	68.33 $\pm$ 16.07	56.00 $\pm$ 5.48	56.67 $\pm$ 7.64	56.00 $\pm$ 4.18	89.00 $\pm$ 5.48
CBF	98.96 $\pm$ 0.23	98.27 $\pm$ 0.96	99.24 $\pm$ 0.32	67.49 $\pm$ 2.09	95.98 $\pm$ 4.64	90.07 $\pm$ 1.09	39.37 $\pm$ 1.51	91.44 $\pm$ 1.93	51.24 $\pm$ 7.49	90.11 $\pm$ 0.88	49.31 $\pm$ 9.05	97.71 $\pm$ 0.69
Car	57.67 $\pm$ 10.84	58.67 $\pm$ 3.21	65.33 $\pm$ 2.17	73.00 $\pm$ 2.98	62.67 $\pm$ 5.22	70.00 $\pm$ 2.89	27.22 $\pm$ 5.85	65.56 $\pm$ 0.96	37.33 $\pm$ 3.84	72.22 $\pm$ 1.92	50.33 $\pm$ 8.28	76.33 $\pm$ 5.45
Chinatown	98.19 $\pm$ 0.13	98.13 $\pm$ 0.16	98.54 $\pm$ 0.21	54.58 $\pm$ 24.80	97.73 $\pm$ 0.75	94.75 $\pm$ 2.10	95.63 $\pm$ 1.01	97.28 $\pm$ 0.61	96.62 $\pm$ 0.60	96.60 $\pm$ 1.10	63.56 $\pm$ 20.21	94.56 $\pm$ 3.09
ChlorineConcentration	41.18 $\pm$ 3.66	34.41 $\pm$ 8.63	37.86 $\pm$ 5.20	34.42 $\pm$ 4.40	35.51 $\pm$ 6.32	35.06 $\pm$ 1.39	38.85 $\pm$ 1.62	32.99 $\pm$ 4.01	39.05 $\pm$ 4.08	39.50 $\pm$ 4.07	33.01 $\pm$ 11.68	34.77 $\pm$ 6.93
CinCECGTorso	54.03 $\pm$ 3.03	65.94 $\pm$ 5.62	64.64 $\pm$ 5.11	45.36 $\pm$ 1.09	80.65 $\pm$ 7.24	81.59 $\pm$ 1.76	35.82 $\pm$ 2.41	84.42 $\pm$ 2.94	63.93 $\pm$ 1.27	85.19 $\pm$ 1.24	52.77 $\pm$ 4.47	94.71 $\pm$ 4.65
Coffee	97.14 $\pm$ 6.39	97.14 $\pm$ 1.60	100.00 $\pm$ 0.00	96.43 $\pm$ 4.37	97.86 $\pm$ 1.96	91.67 $\pm$ 2.06	86.90 $\pm$ 16.50	92.86 $\pm$ 0.00	98.57 $\pm$ 3.19	90.48 $\pm$ 2.06	56.43 $\pm$ 7.74	95.71 $\pm$ 2.99
Computers	66.48 $\pm$ 3.03	63.20 $\pm$ 6.81	68.08 $\pm$ 4.62	48.88 $\pm$ 2.52	56.40 $\pm$ 10.98	52.13 $\pm$ 9.60	52.13 $\pm$ 6.02	50.93 $\pm$ 12.73	49.68 $\pm$ 5.04	53.87 $\pm$ 12.94	51.52 $\pm$ 3.69	65.60 $\pm$ 6.66
CricketX	60.72 $\pm$ 2.35	55.18 $\pm$ 4.36	51.28 $\pm$ 3.40	30.15 $\pm$ 2.84	42.92 $\pm$ 1.99	46.50 $\pm$ 1.21	14.27 $\pm$ 1.71	45.38 $\pm$ 2.45	20.87 $\pm$ 2.94	45.38 $\pm$ 1.94	19.33 $\pm$ 2.95	66.92 $\pm$ 2.71
CricketY	56.92 $\pm$ 2.81	49.44 $\pm$ 1.94	47.69 $\pm$ 2.43	32.82 $\pm$ 4.21	42.15 $\pm$ 5.17	44.62 $\pm$ 0.44	9.32 $\pm$ 1.04	44.02 $\pm$ 2.09	17.90 $\pm$ 2.44	41.11 $\pm$ 0.53	23.95 $\pm$ 3.97	57.95 $\pm$ 1.33
CricketZ	63.18 $\pm$ 1.66	57.38 $\pm$ 2.58	48.82 $\pm$ 4.98	30.56 $\pm$ 3.52	43.18 $\pm$ 5.27	46.32 $\pm$ 1.71	12.91 $\pm$ 2.15	47.78 $\pm$ 1.41	18.26 $\pm$ 2.08	45.30 $\pm$ 1.50	21.95 $\pm$ 3.02	65.30 $\pm$ 1.16
Crop	57.92 $\pm$ 1.22	54.52 $\pm$ 1.67	57.92 $\pm$ 1.22	42.46 $\pm$ 2.10	44.96 $\pm$ 1.35	54.14 $\pm$ 2.07	52.84 $\pm$ 0.15	54.82 $\pm$ 1.79	54.74 $\pm$ 0.79	54.60 $\pm$ 1.45	25.02 $\pm$ 2.10	57.97 $\pm$ 1.25
DiatomSizeReduction	87.32 $\pm$ 3.50	71.05 $\pm$ 13.58	96.80 $\pm$ 0.91	44.77 $\pm$ 1.16	62.68 $\pm$ 4.64	87.04 $\pm$ 1.80	66.88 $\pm$ 2.86	51.96 $\pm$ 5.03	48.89 $\pm$ 18.63	87.47 $\pm$ 0.19	30.13 $\pm$ 0.15	92.61 $\pm$ 2.00
DistalPhalanxOutlineAgeGroup	68.20 $\pm$ 2.98	68.78 $\pm$ 4.24	70.50 $\pm$ 1.61	69.50 $\pm$ 1.66	69.50 $\pm$ 2.81	66.67 $\pm$ 1.10	65.71 $\pm$ 3.62	70.50 $\pm$ 1.25	68.78 $\pm$ 2.72	66.91 $\pm$ 1.25	48.06 $\pm$ 11.76	70.65 $\pm$ 2.18
DistalPhalanxOutlineCorrect	64.78 $\pm$ 12.49	65.07 $\pm$ 11.29	63.62 $\pm$ 12.87	50.00 $\pm$ 13.55	60.14 $\pm$ 9.92	58.94 $\pm$ 4.04	67.15 $\pm$ 2.41	55.43 $\pm$ 14.84	62.39 $\pm$ 9.46	61.71 $\pm$ 6.72	47.61 $\pm$ 8.80	60.51 $\pm$ 11.85
DistalPhalanxTW	55.83 $\pm$ 3.65	59.14 $\pm$ 2.12	60.58 $\pm$ 3.58	47.34 $\pm$ 6.42	57.55 $\pm$ 6.08	53.96 $\pm$ 3.14	53.00 $\pm$ 2.91	54.92 $\pm$ 0.83	52.52 $\pm$ 4.35	54.92 $\pm$ 2.91	32.37 $\pm$ 14.50	54.10 $\pm$ 3.11
DodgerLoopDay	49.25 $\pm$ 3.71	47.25 $\pm$ 4.28	39.75 $\pm$ 3.11	41.50 $\pm$ 2.24	47.00 $\pm$ 4.11	49.58 $\pm$ 5.64	28.75 $\pm$ 2.50	45.42 $\pm$ 1.91	29.00 $\pm$ 2.40	48.75 $\pm$ 1.25	47.00 $\pm$ 4.11	48.75 $\pm$ 1.77
DodgerLoopGame	68.70 $\pm$ 6.09	83.33 $\pm$ 5.28	75.07 $\pm$ 2.64	79.28 $\pm$ 0.97	88.84 $\pm$ 1.67	76.81 $\pm$ 1.26	48.79 $\pm$ 3.42	81.40 $\pm$ 1.51	53.48 $\pm$ 2.83	78.99 $\pm$ 1.45	88.84 $\pm$ 1.67	83.62 $\pm$ 0.65
DodgerLoopWeekend	94.06 $\pm$ 1.39	94.20 $\pm$ 2.11	96.81 $\pm$ 0.83	97.83 $\pm$ 0.00	96.81 $\pm$ 0.65	96.86 $\pm$ 0.42	72.71 $\pm$ 1.82	97.58 $\pm$ 0.42	76.38 $\pm$ 1.82	97.10 $\pm$ 0.00	96.81 $\pm$ 0.65	97.83 $\pm$ 0.00
ECG200	80.00 $\pm$ 4.00	83.00 $\pm$ 2.00	80.40 $\pm$ 4.28	77.60 $\pm$ 4.34	79.80 $\pm$ 6.53	79.33 $\pm$ 3.21	78.00 $\pm$ 3.61	81.00 $\pm$ 5.00	82.00 $\pm$ 2.92	82.00 $\pm$ 3.61	63.00 $\pm$ 15.26	77.40 $\pm$ 4.88
ECG5000	83.56 $\pm$ 5.09	78.23 $\pm$ 2.20	80.72 $\pm$ 4.37	88.37 $\pm$ 1.22	76.31 $\pm$ 5.83	79.59 $\pm$ 2.75	73.64 $\pm$ 2.21	84.75 $\pm$ 1.16	79.28 $\pm$ 5.11	82.19 $\pm$ 1.79	24.56 $\pm$ 15.15	73.78 $\pm$ 6.18
ECGFiveDays	95.40 $\pm$ 1.88	88.99 $\pm$ 4.44	97.56 $\pm$ 1.20	74.87 $\pm$ 4.61	94.61 $\pm$ 2.00	83.82 $\pm$ 7.05	96.44 $\pm$ 1.42	91.64 $\pm$ 1.90	97.51 $\pm$ 1.12	86.18 $\pm$ 8.29	57.47 $\pm$ 4.38	75.28 $\pm$ 1.71
EOGHorizontalSignal	30.61 $\pm$ 3.99	43.15 $\pm$ 5.84	30.94 $\pm$ 3.48	29.89 $\pm$ 1.60	38.78 $\pm$ 6.14	39.59 $\pm$ 1.67	8.01 $\pm$ 1.27	41.62 $\pm$ 1.42	10.61 $\pm$ 3.44	40.06 $\pm$ 3.72	39.56 $\pm$ 4.17	49.17 $\pm$ 1.62
EOGVerticalSignal	21.71 $\pm$ 2.53	28.23 $\pm$ 2.47	19.83 $\pm$ 3.08	31.55 $\pm$ 2.19	40.06 $\pm$ 1.98	36.00 $\pm$ 1.12	8.47 $\pm$ 0.84	37.75 $\pm$ 1.67	11.99 $\pm$ 1.33	37.02 $\pm$ 1.99	30.55 $\pm$ 3.17	42.21 $\pm$ 2.26
Earthquakes	53.81 $\pm$ 8.62	53.96 $\pm$ 9.50	51.37 $\pm$ 3.47	52.52 $\pm$ 6.49	54.82 $\pm$ 2.85	60.91 $\pm$ 4.63	49.16 $\pm$ 8.34	59.71 $\pm$ 6.86	54.68 $\pm$ 5.26	61.87 $\pm$ 3.60	47.91 $\pm$ 6.66	54.68 $\pm$ 7.63
ElectricDevices	40.12 $\pm$ 4.42	38.65 $\pm$ 3.96	36.00 $\pm$ 4.64	28.04 $\pm$ 1.63	39.08 $\pm$ 8.26	48.59 $\pm$ 6.21	30.07 $\pm$ 3.26	33.74 $\pm$ 3.43	37.44 $\pm$ 1.49	48.14 $\pm$ 2.51	16.08 $\pm$ 5.57	49.42 $\pm$ 5.02
EthanolLevel	26.56 $\pm$ 1.96	25.76 $\pm$ 1.05	26.16 $\pm$ 0.71	29.28 $\pm$ 1.78	25.92 $\pm$ 1.70	28.07 $\pm$ 2.55	25.20 $\pm$ 3.34	26.80 $\pm$ 0.69	25.60 $\pm$ 1.13	29.87 $\pm$ 2.47	25.56 $\pm$ 1.05	28.24 $\pm$ 0.50
FaceAll	77.73 $\pm$ 3.48	80.30 $\pm$ 1.67	77.73 $\pm$ 3.48	67.09 $\pm$ 2.40	54.49 $\pm$ 5.41	67.46 $\pm$ 1.82	59.68 $\pm$ 1.83	67.32 $\pm$ 2.82	60.69 $\pm$ 2.63	64.06 $\pm$ 2.32	32.89 $\pm$ 2.71	63.01 $\pm$ 1.34
FaceFour	87.95 $\pm$ 3.07	59.32 $\pm$ 7.51	92.95 $\pm$ 2.03	74.77 $\pm$ 0.95	66.36 $\pm$ 10.77	85.98 $\pm$ 1.31	65.91 $\pm$ 2.27	88.26 $\pm$ 0.66	69.55 $\pm$ 1.69	83.71 $\pm$ 1.31	34.55 $\pm$ 3.82	79.09 $\pm$ 1.90
FacesUCR	87.80 $\pm$ 1.60	84.26 $\pm$ 2.06	82.41 $\pm$ 2.48	63.80 $\pm$ 1.79	54.40 $\pm$ 3.05	80.54 $\pm$ 1.15	70.11 $\pm$ 1.05	83.69 $\pm$ 1.13	72.08 $\pm$ 1.16	79.51 $\pm$ 1.53	34.67 $\pm$ 3.28	79.08 $\pm$ 1.41
FiftyWords	45.36 $\pm$ 2.39	36.70 $\pm$ 1.89	45.36 $\pm$ 2.39	53.67 $\pm$ 1.47	50.46 $\pm$ 2.83	64.32 $\pm$ 1.87	16.85 $\pm$ 0.67	67.99 $\pm$ 1.84	37.05 $\pm$ 3.62	57.36 $\pm$ 1.54	56.40 $\pm$ 1.98	71.34 $\pm$ 2.10
Fish	83.31 $\pm$ 3.78	56.00 $\pm$ 4.28	71.54 $\pm$ 6.28	65.60 $\pm$ 3.49	62.74 $\pm$ 2.15	68.95 $\pm$ 4.29	24.38 $\pm$ 3.80	71.05 $\pm$ 1.44	57.14 $\pm$ 3.45	75.05 $\pm$ 2.31	38.17 $\pm$ 5.05	78.29 $\pm$ 3.92
FordA	78.26 $\pm$ 5.14	75.61 $\pm$ 0.71	78.70 $\pm$ 3.26	49.09 $\pm$ 0.93	67.88 $\pm$ 3.90	51.99 $\pm$ 2.52	52.22 $\pm$ 0.95	48.81 $\pm$ 1.31	53.05 $\pm$ 1.09	48.21 $\pm$ 1.67	50.06 $\pm$ 2.65	57.15 $\pm$ 1.85
FordB	63.83 $\pm$ 5.74	62.17 $\pm$ 3.33	61.70 $\pm$ 3.19	48.54 $\pm$ 0.72	59.06 $\pm$ 1.85	46.75 $\pm$ 0.61	49.79 $\pm$ 1.45	47.24 $\pm$ 1.24	5			

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	81.40 ± 2.53	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	75.57 ± 4.64
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	97.47 ± 2.23
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	92.72 ± 5.80
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	98.54 ± 0.94
GunPointOldVersusYoung	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	71.24 ± 5.65	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	82.65 ± 3.66	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	39.22 ± 4.29
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	59.69 ± 3.39	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	90.92 ± 1.10	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	33.49 ± 2.77
InsectEPGRegularTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	61.67 ± 1.18	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	95.82 ± 0.32	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	67.95 ± 4.05	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	77.05 ± 4.18
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	75.34 ± 2.37	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	95.78 ± 0.57	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	87.00 ± 7.94	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	56.61 ± 3.74	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MedbournePedestrian	84.60 ± 1.98	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	58.66 ± 1.35	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	65.02 ± 2.98	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	49.74 ± 5.84	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	79.60 ± 1.74
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	81.61 ± 2.47
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	90.65 ± 0.48	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	81.90 ± 0.59	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	86.79 ± 1.25	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	72.64 ± 5.62	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	66.67 ± 9.72	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	59.89 ± 3.28
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	58.78 ± 4.01	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	28.10 ± 1.26	25.96 ± 1.60	28.10 ± 1.26	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	73.60 ± 2.19	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	50.58 ± 0.92	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	69.81 ± 3.29	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	61.35 ± 2.42	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	100.00 ± 0.00	99.43 ± 1.28	100.00 ± 0.00	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	88.11 ± 1.91	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	78.98 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	81.63 ± 4.89	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	66.55 ± 3.51	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	72.39 ± 1.88	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	66.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	44.62 ± 2.22	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	88.00 ± 0.00	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	40.53 ± 1.87	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	74.93 ± 6.24	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	45.56 ± 2.93

Continued on next page

Table 10: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	ChronoMorph
SemgHandSubjectCh2	43.56 $\pm$ 2.78	45.20 $\pm$ 5.06	40.22 $\pm$ 5.31	62.80 $\pm$ 4.54	54.13 $\pm$ 4.47	60.37 $\pm$ 0.71	34.22 $\pm$ 3.27	68.07 $\pm$ 2.65	39.96 $\pm$ 2.48	60.96 $\pm$ 1.10	38.18 $\pm$ 2.42	66.71 $\pm$ 4.45
ShakeGestureWiimoteZ	82.00 $\pm$ 6.16	91.20 $\pm$ 2.28	88.80 $\pm$ 1.10	55.60 $\pm$ 2.97	68.40 $\pm$ 7.67	60.00 $\pm$ 4.00	33.33 $\pm$ 3.06	62.67 $\pm$ 3.06	45.20 $\pm$ 7.29	56.67 $\pm$ 1.15	68.40 $\pm$ 7.67	91.60 $\pm$ 2.97
ShapeletSim	86.33 $\pm$ 6.80	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	48.78 $\pm$ 0.82	95.22 $\pm$ 4.26	59.07 $\pm$ 5.56	48.70 $\pm$ 1.40	50.19 $\pm$ 2.80	51.00 $\pm$ 1.98	49.81 $\pm$ 1.95	52.33 $\pm$ 4.64	96.67 $\pm$ 1.96
ShapesAll	68.60 $\pm$ 2.31	66.63 $\pm$ 1.82	68.60 $\pm$ 2.31	62.47 $\pm$ 0.34	65.47 $\pm$ 2.49	72.11 $\pm$ 0.69	12.89 $\pm$ 4.16	73.28 $\pm$ 0.38	31.00 $\pm$ 5.80	65.56 $\pm$ 1.08	70.00 $\pm$ 1.20	83.13 $\pm$ 0.32
SmallKitchenAppliances	67.20 $\pm$ 8.07	61.87 $\pm$ 4.21	64.69 $\pm$ 7.08	36.59 $\pm$ 1.57	60.00 $\pm$ 3.85	55.11 $\pm$ 4.59	41.51 $\pm$ 2.40	38.76 $\pm$ 2.56	47.25 $\pm$ 3.08	51.02 $\pm$ 1.54	41.33 $\pm$ 5.03	57.60 $\pm$ 5.56
SmoothSubspace	92.00 $\pm$ 1.56	87.07 $\pm$ 3.67	92.00 $\pm$ 3.02	38.13 $\pm$ 6.59	67.73 $\pm$ 10.31	79.56 $\pm$ 2.78	80.67 $\pm$ 0.67	60.00 $\pm$ 11.02	86.13 $\pm$ 3.38	90.67 $\pm$ 4.00	43.60 $\pm$ 9.96	89.47 $\pm$ 2.60
SonyAIBORobotSurface1	95.11 $\pm$ 2.42	96.01 $\pm$ 0.87	92.45 $\pm$ 1.40	42.96 $\pm$ 0.07	87.95 $\pm$ 1.92	64.28 $\pm$ 1.83	67.05 $\pm$ 1.01	76.43 $\pm$ 0.69	64.86 $\pm$ 0.43	67.83 $\pm$ 1.06	45.09 $\pm$ 4.84	86.09 $\pm$ 3.97
SonyAIBORobotSurface2	96.73 $\pm$ 3.11	92.82 $\pm$ 3.91	94.67 $\pm$ 2.40	77.50 $\pm$ 0.71	90.95 $\pm$ 2.30	81.25 $\pm$ 2.70	83.07 $\pm$ 2.21	86.08 $\pm$ 2.22	82.71 $\pm$ 1.50	85.73 $\pm$ 2.18	61.15 $\pm$ 12.88	84.11 $\pm$ 2.29
StarLightCurves	71.47 $\pm$ 14.56	82.56 $\pm$ 3.87	75.10 $\pm$ 5.06	74.16 $\pm$ 3.22	77.51 $\pm$ 4.09	75.29 $\pm$ 4.53	29.60 $\pm$ 9.11	76.31 $\pm$ 3.24	30.20 $\pm$ 4.95	74.55 $\pm$ 5.89	70.71 $\pm$ 14.11	85.26 $\pm$ 3.54
Strawberry	79.73 $\pm$ 8.80	76.70 $\pm$ 3.26	76.70 $\pm$ 4.55	60.81 $\pm$ 1.94	79.89 $\pm$ 6.30	61.98 $\pm$ 1.58	80.63 $\pm$ 1.13	59.10 $\pm$ 1.09	77.51 $\pm$ 4.36	68.11 $\pm$ 2.15	50.54 $\pm$ 10.66	75.84 $\pm$ 0.99
SwedishLeaf	88.03 $\pm$ 1.39	83.68 $\pm$ 1.26	88.03 $\pm$ 1.39	66.11 $\pm$ 1.36	73.47 $\pm$ 2.54	75.95 $\pm$ 1.36	60.69 $\pm$ 4.66	81.39 $\pm$ 2.49	68.80 $\pm$ 1.87	74.72 $\pm$ 1.82	28.96 $\pm$ 3.20	85.89 $\pm$ 1.27
Symbols	93.83 $\pm$ 2.55	89.89 $\pm$ 7.08	92.52 $\pm$ 1.13	75.94 $\pm$ 0.93	85.87 $\pm$ 1.87	83.32 $\pm$ 0.52	34.77 $\pm$ 0.30	85.96 $\pm$ 1.03	61.21 $\pm$ 7.54	82.51 $\pm$ 0.70	21.11 $\pm$ 2.93	89.33 $\pm$ 0.82
SyntheticControl	95.60 $\pm$ 1.14	97.87 $\pm$ 0.61	97.60 $\pm$ 0.43	59.07 $\pm$ 3.98	94.20 $\pm$ 1.64	90.78 $\pm$ 1.64	59.44 $\pm$ 1.95	79.44 $\pm$ 4.86	68.40 $\pm$ 3.43	90.11 $\pm$ 1.35	22.60 $\pm$ 7.62	97.33 $\pm$ 0.85
ToeSegmentation1	93.25 $\pm$ 1.60	89.56 $\pm$ 3.33	91.58 $\pm$ 1.68	55.53 $\pm$ 3.14	68.95 $\pm$ 9.85	56.73 $\pm$ 5.97	50.29 $\pm$ 6.71	55.85 $\pm$ 3.55	49.39 $\pm$ 2.18	55.70 $\pm$ 4.98	50.53 $\pm$ 1.40	80.53 $\pm$ 2.75
ToeSegmentation2	81.38 $\pm$ 10.25	87.08 $\pm$ 4.53	85.85 $\pm$ 2.64	50.15 $\pm$ 3.62	73.38 $\pm$ 4.70	68.21 $\pm$ 10.44	56.67 $\pm$ 7.86	58.46 $\pm$ 6.30	46.77 $\pm$ 3.41	61.54 $\pm$ 3.35	43.54 $\pm$ 14.75	80.31 $\pm$ 4.70
Trace	92.20 $\pm$ 4.60	100.00 $\pm$ 0.00	95.20 $\pm$ 5.07	55.80 $\pm$ 5.72	86.20 $\pm$ 10.35	72.67 $\pm$ 1.53	55.67 $\pm$ 4.51	61.33 $\pm$ 4.62	74.80 $\pm$ 3.27	78.00 $\pm$ 4.36	46.80 $\pm$ 8.17	100.00 $\pm$ 0.00
TwoLeadECG	99.86 $\pm$ 0.08	98.79 $\pm$ 1.25	97.56 $\pm$ 2.37	58.05 $\pm$ 1.68	91.89 $\pm$ 5.00	70.85 $\pm$ 3.05	75.62 $\pm$ 1.63	65.26 $\pm$ 1.91	78.86 $\pm$ 1.56	78.99 $\pm$ 2.23	54.01 $\pm$ 3.73	99.68 $\pm$ 0.24
TwoPatterns	70.89 $\pm$ 4.77	77.56 $\pm$ 2.96	78.86 $\pm$ 3.59	42.28 $\pm$ 1.08	74.73 $\pm$ 6.48	42.55 $\pm$ 1.37	26.22 $\pm$ 0.85	42.35 $\pm$ 1.82	29.82 $\pm$ 3.35	37.95 $\pm$ 1.72	25.60 $\pm$ 4.32	70.06 $\pm$ 0.70
UMD	92.78 $\pm$ 3.35	98.33 $\pm$ 1.05	98.75 $\pm$ 0.58	64.17 $\pm$ 3.05	91.53 $\pm$ 4.77	76.62 $\pm$ 2.44	75.93 $\pm$ 7.23	89.58 $\pm$ 4.22	93.06 $\pm$ 1.47	87.27 $\pm$ 3.21	47.50 $\pm$ 8.96	98.67 $\pm$ 7.42
UWaveGestureLibraryAll	48.45 $\pm$ 3.31	48.15 $\pm$ 2.07	49.06 $\pm$ 3.81	80.57 $\pm$ 3.71	68.35 $\pm$ 4.39	84.19 $\pm$ 2.90	27.54 $\pm$ 1.68	84.63 $\pm$ 3.19	42.77 $\pm$ 5.45	83.32 $\pm$ 3.39	81.20 $\pm$ 2.71	52.41 $\pm$ 1.70
UWaveGestureLibraryX	51.98 $\pm$ 1.26	57.08 $\pm$ 2.26	52.99 $\pm$ 2.93	60.99 $\pm$ 2.04	57.69 $\pm$ 1.76	62.37 $\pm$ 1.85	17.30 $\pm$ 0.19	62.30 $\pm$ 0.55	36.29 $\pm$ 8.88	61.26 $\pm$ 1.95	56.81 $\pm$ 2.90	53.50 $\pm$ 1.91
UWaveGestureLibraryY	45.58 $\pm$ 1.39	47.42 $\pm$ 0.67	46.02 $\pm$ 2.85	55.44 $\pm$ 1.29	53.52 $\pm$ 3.23	56.83 $\pm$ 0.99	23.04 $\pm$ 3.42	56.65 $\pm$ 1.12	32.16 $\pm$ 4.98	55.22 $\pm$ 1.56	46.61 $\pm$ 1.40	52.03 $\pm$ 0.97
UWaveGestureLibraryZ	48.34 $\pm$ 2.78	51.60 $\pm$ 2.75	44.60 $\pm$ 1.67	53.18 $\pm$ 1.58	53.45 $\pm$ 2.14	54.99 $\pm$ 0.25	20.01 $\pm$ 2.19	54.36 $\pm$ 2.47	34.40 $\pm$ 4.42	51.61 $\pm$ 2.72	46.80 $\pm$ 1.81	60.65 $\pm$ 2.84
Wafer	84.60 $\pm$ 6.19	78.03 $\pm$ 11.28	81.53 $\pm$ 4.78	69.44 $\pm$ 4.07	84.77 $\pm$ 11.10	82.09 $\pm$ 7.52	76.30 $\pm$ 12.19	83.57 $\pm$ 7.70	81.81 $\pm$ 10.33	81.67 $\pm$ 9.12	78.96 $\pm$ 10.53	82.53 $\pm$ 8.58
Wine	68.15 $\pm$ 6.33	52.59 $\pm$ 5.80	64.07 $\pm$ 8.03	50.74 $\pm$ 1.66	50.00 $\pm$ 0.00	58.64 $\pm$ 5.35	50.62 $\pm$ 1.07	50.00 $\pm$ 3.70	50.00 $\pm$ 0.00	62.96 $\pm$ 7.41	47.78 $\pm$ 4.97	49.26 $\pm$ 2.11
WordSynonyms	35.14 $\pm$ 2.03	30.72 $\pm$ 3.50	35.14 $\pm$ 2.03	35.86 $\pm$ 2.07	39.44 $\pm$ 4.12	51.78 $\pm$ 3.11	15.15 $\pm$ 0.48	51.10 $\pm$ 4.21	25.64 $\pm$ 4.61	49.01 $\pm$ 4.47	33.92 $\pm$ 1.82	58.90 $\pm$ 2.04
Worms	50.39 $\pm$ 8.29	51.95 $\pm$ 5.35	45.71 $\pm$ 7.71	23.38 $\pm$ 5.11	39.74 $\pm$ 4.27	23.38 $\pm$ 6.49	26.84 $\pm$ 3.97	27.27 $\pm$ 9.09	25.45 $\pm$ 3.74	25.97 $\pm$ 9.09	20.78 $\pm$ 5.35	48.57 $\pm$ 4.91
WormsTwoClass	64.94 $\pm$ 5.51	63.12 $\pm$ 4.37	61.30 $\pm$ 7.20	49.09 $\pm$ 4.63	47.27 $\pm$ 1.97	50.65 $\pm$ 6.49	48.05 $\pm$ 4.50	48.05 $\pm$ 4.68	48.05 $\pm$ 3.56	51.95 $\pm$ 4.50	46.49 $\pm$ 5.15	56.36 $\pm$ 3.85
Yoga	59.91 $\pm$ 2.90	59.97 $\pm$ 4.98	59.88 $\pm$ 3.11	55.72 $\pm$ 1.71	57.07 $\pm$ 1.49	58.79 $\pm$ 3.68	53.03 $\pm$ 3.90	59.02 $\pm$ 1.75	51.43 $\pm$ 2.65	59.87 $\pm$ 3.99	51.89 $\pm$ 2.29	57.73 $\pm$ 2.05
Avg. Acc.	66.24	64.93	<u>67.18</u>	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	69.16
Avg. Rank	4.83	5.03	<u>4.55</u>	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	3.91
#Best	25	15	<u>28</u>	6	5	7	3	5	1	8	1	40

Table 11: Key few-shot adaptation settings for ChronoMorph.

Component	Setting
Base LLM	meta-llama/Llama-3.2-1B
Vision backbone	facebook/dinov2-base
Morphology-map construction	Overlapping-window grid transform with per-sequence robust normalization
Patch length / stride	16 / 8
Hidden size	128
Default downstream checkpoint	Stage II (TSQA transfer)
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	$2 \times 10^{-4} / 1 \times 10^{-4} / 1 \times 10^{-4}$
Language-side adaptation	LoRA enabled in Phase II ($r = 16, \alpha = 32$)
Tokenizer training mode	class_rows
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Freeze vision backbone	true
Data augmentation	none

B Additional Experimental Details

We summarize the key implementation choices of ChronoMorph for the few-shot benchmark in Table 11. These settings are fixed across the main benchmark reported in Section 4.

B.1 Pretraining Stages and Data Sources

Table 12 summarizes the three training stages discussed in the main paper and clarifies which stage provides the default downstream checkpoint.

After stage-wise pretraining, downstream few-shot adaptation uses only the sampled support labels of the target dataset. For UCR targets contained in the Stage I pool, unlabeled TRAIN sequences may already have contributed to the encoder pretraining.

B.2 Code Release and Assets

We will release an anonymized code package with the submission. The release includes the ChronoMorph implementation, the few-shot adaptation and evaluation scripts, baseline launchers, reporting scripts for the tables in this paper, environment files, and README instructions for preparing the public datasets used in our experiments. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The released scripts document which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.3 Asset and License Table

Table 13 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release. For assets whose official source does not state a separate dataset license, we cite the official repository or archive page and do not redistribute the raw data.

Table 13: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR 2018	Archive Main few-shot benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists active Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dau et al.	Not redistributed; expect users to download from the original archive or cited mirror.

Table 13: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
TSQA / MQA	Time-series and Stage II time-series question answering	raw Hugging Face dataset card	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; scripts load from the original dataset source.
M4 Competition dataset	Stage I raw and optional captioning / semantic enrichment data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; pre-processing scripts require a local PhysioNet download.
Sleep-EDF Database Expanded	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; pre-processing scripts require a local PhysioNet download.
PhysioNet/CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; pre-processing scripts require a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; loaded from the original model repository.
OpenTSLM base code	Starting code-base and time-series language-model components	Project repository and release package	MIT license in the repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.
Python libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	ML Training, adaptation, evaluation, and reporting	Version constraints are listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

437 B.4 Computational Resource Reporting

438 All reported downstream experiments were run on a local server with six NVIDIA GeForce RTX
439 3090 GPUs (24GB each), two Intel Xeon Silver 4314 CPUs, and 251GiB system memory. Unless
440 otherwise stated, each training job used one GPU, and the batch launchers executed up to six
441 single-GPU workers in parallel. Table 14 summarizes the compute used by the main experiment
442 groups. GPU-hours are summed over single-GPU jobs; wall-clock time can be lower when multiple
443 workers run concurrently. The efficiency measurements in Appendix B.7 additionally report peak
444 memory, train-step latency, inference latency, throughput, total parameters, and downstream-updated
445 parameters on the representative runtime subset.

446 Retained launcher summaries also include failed or exploratory jobs. These runs account for less
447 than 2 additional GPU-hours in the preserved logs, and dry-run checks add a negligible amount of
448 compute.

Table 12: Stage-wise data sources and roles for ChronoMorph.

Stage	Training data	Validation / monitoring data	Role in the main UCR few-shot benchmark
Stage I	Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool.	Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used.	Learns dual-view representations before the LLM interface; corresponds to <code>stage0_encoder_ssl</code> in the implementation.
Stage II	TSQA train split only, with standard language-model training on time-series question answering.	TSQA validation/test splits.	Provides the default downstream checkpoint used by all main few-shot results in Section 4; corresponds to <code>stage1_tsqa_transfer</code> .
Stage III	Optional mixture of TSQA, M4 captioning, and synthetic attribute supervision derived from TSQA/M4 raw series.	Corresponding validation/test splits of the same mixture.	Used only for additional semantic enrichment or caption specialization; not used for the default main benchmark initialization unless explicitly stated. This corresponds to <code>stage2_semantic_bridge</code> and later caption specialization in code.

Table 14: Compute-resource budget for the reported experiment groups. Downstream jobs use single RTX 3090 workers unless noted otherwise.

Experiment group	Scope	Compute workers	Mean per run	Total hours	GPU-	Notes
Stage I–II pretraining	Curriculum pretraining checkpoint used to initialize downstream adaptation	RTX 3090 workers	–	–		Uses the same local GPU server; stage-wise timing is recorded with the pretraining checkpoint logs.
Optional Stage III enrichment	Semantic-enrichment training used outside the default UCR results	RTX 3090 workers	–	–		Not used for the default UCR benchmark.
ChronoMorph main UCR benchmark	128 UCR datasets, 1/2/5/10-shot, five runs per setting; 2,507 completed downstream runs	6 single-GPU workers	0.67 h/run	GPU-	1,677	Batch size 8, gradient accumulation 4, 60 downstream epochs.
UCR baseline runs	ResNet, TapNet, COSCO-ResNet, PatchTST, GPT4TS, and TSLib-family baselines under the same few-shot protocol	6 single-GPU workers	0.40–0.49 h/run	GPU-	10,937	Includes the retained main UCR baseline logs; GPT4TS coverage is reported for the completed 60-dataset run.
External-dataset ChronoMorph runs	MIT-BIH, SleepEDF, and CinC2017 under 10/20/30-shot protocols	single-GPU workers	1.69 GPU-h/run		153	Same downstream adaptation recipe as the external few-shot experiments.
External-dataset baseline runs	Reported baseline families on MIT-BIH, SleepEDF, and CinC2017	single-GPU workers	0.68 GPU-h/job		22	Launcher jobs cover all three external datasets and baseline families.
Top-level ablations	No-pretraining, temporal-only, visual-only, class-token decoding, and no-LLM backbone ablations	6 single-GPU workers	1.67 GPU-h/run		3,895	Includes the two retained no-LLM linear runs added together.
Semantic-prior ablation	External semantic-label interface runs	single-GPU workers	1.69 GPU-h/run		31	Uses the same 10/20/30-shot external evaluation protocol.
Efficiency probe	Representative UCR runtime subset with parameter, memory, latency, and throughput measurements	one RTX 3090	–	–		Detailed runtime and memory values are reported in Tables 4, 17, and 19.

449 B.5 External Dataset Results

450 Table 15 reports the full 10-, 20-, and 30-shot results on MIT-BIH, SleepEDF, and CinC2017. All
451 methods use the same external few-shot settings and reported run count; we use these datasets as
452 transfer checks beyond the primary UCR benchmark rather than as a claim of uniform dominance
453 across external domains.

454 B.6 Semantic-Prior Decision Ablation

455 Table 16 expands the semantic-prior comparison summarized in Table 6. Both variants use one-pass
456 class-token scoring; the semantic-prior variant additionally initializes and regularizes class-token

Table 15: Few-shot classification accuracy (%) on three external datasets. Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
ChronoMorph	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	37.98 $\pm$ 3.69	48.80 $\pm$ 2.31	54.69 $\pm$ 3.67	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
ResNet	37.97 $\pm$ 14.13	55.66 $\pm$ 8.58	57.07 $\pm$ 9.05	60.27 $\pm$ 7.78	70.59 $\pm$ 4.14	<u>69.64 $\pm$ 3.64</u>	30.95 $\pm$ 12.41	<u>33.73 $\pm$ 7.43</u>	36.20 $\pm$ 4.84
TapNet	31.94 $\pm$ 7.18	40.15 $\pm$ 10.28	44.11 $\pm$ 11.14	<u>58.75 $\pm$ 9.67</u>	<u>70.50 $\pm$ 5.64</u>	70.86 $\pm$ 5.23	30.05 $\pm$ 6.73	32.80 $\pm$ 8.81	31.01 $\pm$ 11.54
COSCO-ResNet	32.33 $\pm$ 14.25	35.08 $\pm$ 13.75	41.17 $\pm$ 8.94	57.86 $\pm$ 11.29	63.04 $\pm$ 8.19	67.32 $\pm$ 3.13	31.41 $\pm$ 4.48	31.04 $\pm$ 6.97	31.98 $\pm$ 8.37
PatchTST	38.94 $\pm$ 14.35	55.36 $\pm$ 8.84	57.08 $\pm$ 9.40	25.21 $\pm$ 11.89	29.47 $\pm$ 9.76	32.08 $\pm$ 8.05	26.07 $\pm$ 6.24	30.35 $\pm$ 7.51	31.21 $\pm$ 3.36
OneFitsAll	32.92 $\pm$ 16.63	42.48 $\pm$ 10.82	54.87 $\pm$ 10.04	26.98 $\pm$ 3.35	27.57 $\pm$ 3.81	26.60 $\pm$ 3.59	30.80 $\pm$ 6.07	27.56 $\pm$ 4.33	29.23 $\pm$ 3.03
TimesNet	51.01 $\pm$ 5.84	54.21 $\pm$ 7.81	56.15 $\pm$ 8.77	21.11 $\pm$ 2.91	23.42 $\pm$ 2.34	25.17 $\pm$ 1.82	27.25 $\pm$ 2.81	28.41 $\pm$ 4.24	29.43 $\pm$ 2.44
DLinear	<u>55.57 $\pm$ 9.85</u>	57.45 $\pm$ 7.95	<u>58.84 $\pm$ 8.69</u>	19.57 $\pm$ 1.32	19.94 $\pm$ 1.73	19.92 $\pm$ 1.69	24.92 $\pm$ 1.16	25.17 $\pm$ 1.22	24.53 $\pm$ 1.70
Autoformer	38.05 $\pm$ 5.65	42.49 $\pm$ 8.58	44.39 $\pm$ 7.25	26.25 $\pm$ 7.52	24.55 $\pm$ 4.24	28.64 $\pm$ 8.05	28.37 $\pm$ 5.44	27.67 $\pm$ 2.76	26.30 $\pm$ 4.36
Crossformer	53.48 $\pm$ 9.36	<u>58.55 $\pm$ 10.43</u>	58.24 $\pm$ 9.71	23.75 $\pm$ 1.74	24.53 $\pm$ 3.34	26.49 $\pm$ 3.02	24.02 $\pm$ 6.10	25.50 $\pm$ 2.86	27.35 $\pm$ 2.35
FEDformer	58.88 $\pm$ 5.77	56.80 $\pm$ 9.98	55.12 $\pm$ 10.09	15.15 $\pm$ 8.43	24.73 $\pm$ 6.55	25.47 $\pm$ 5.56	20.35 $\pm$ 8.32	26.90 $\pm$ 3.54	27.91 $\pm$ 4.87
Informer	53.19 $\pm$ 6.03	54.20 $\pm$ 7.61	52.35 $\pm$ 10.80	20.35 $\pm$ 12.77	22.18 $\pm$ 5.40	24.81 $\pm$ 8.94	26.93 $\pm$ 9.30	28.14 $\pm$ 4.01	30.12 $\pm$ 5.95

Table 16: Semantic-prior class-token ablation on three external datasets (accuracy, %). Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH			SleepEDF			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
Anonymous labels	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	37.98 $\pm$ 3.69	48.80 $\pm$ 2.31	54.69 $\pm$ 3.67	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
Semantic prior	57.05 $\pm$ 10.96	66.62 $\pm$ 6.54	68.87 $\pm$ 11.23	42.56 $\pm$ 3.24	52.69 $\pm$ 5.05	57.24 $\pm$ 4.34	34.72 $\pm$ 3.15	35.24 $\pm$ 1.24	38.86 $\pm$ 7.00
Semantic prior – Anonymous	+2.02	+2.67	+4.72	+4.58	+3.89	+2.55	+1.92	-1.74	+4.73

rows from label-verbalizer prototypes. The semantic prior improves the averaged performance on all three datasets, while individual shot-level deltas remain dataset-dependent.

B.7 Efficiency Summary

Table 17 reports computational efficiency and adaptation cost under the 10-shot UCR protocol. Runtime and memory are measured on the predefined representative UCR subset with batch size 8 on a single RTX 3090, while parameter counts are reported separately for total deployed parameters and downstream-updated parameters.

B.8 Baseline Training Recipes and Hyperparameters

Table 18 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

B.9 Computational Efficiency Probe Details

The efficiency probe is implemented by the repository’s `efficiency_probe.py` script. It uses the same model families as the main benchmark and omits ChronoMorph-only ablations. The default runtime subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1, PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long UCR series as well as different class counts. The script writes the LaTeX tables, CSV summary, and accuracy-versus-updated-parameters plot to the efficiency-report directory.

C Additional Plot

Figure 2 visualizes the shot-scaling trend over the 128 UCR datasets. ChronoMorph remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

Table 17: Computational efficiency and adaptation cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters optimized during downstream adaptation. Adapter size estimates the per-dataset trainable-state footprint. Runtime is measured on representative UCR datasets with batch size 8 on a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ChronoMorph	1.28B	12.3M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

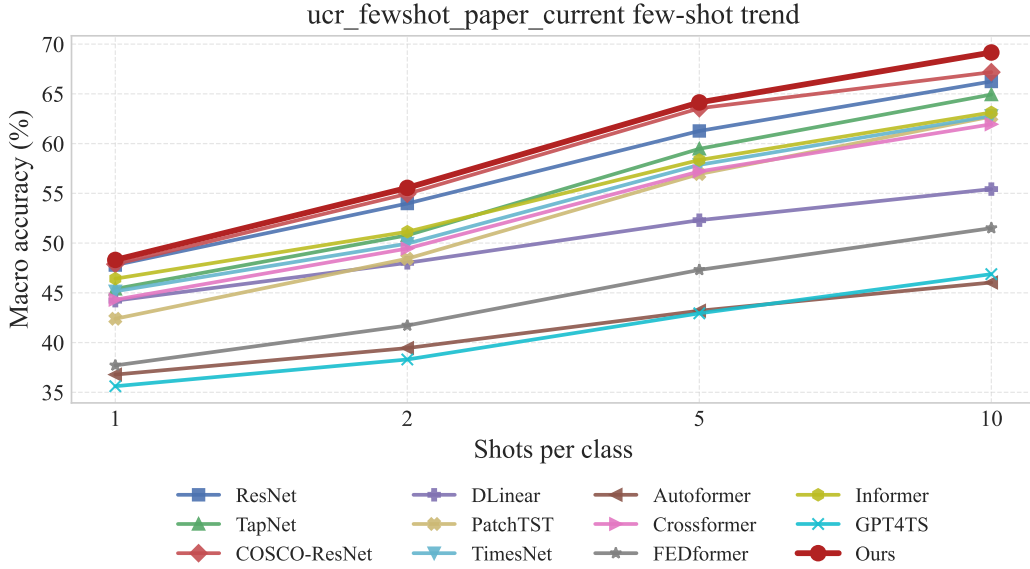


Figure 2: Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. ChronoMorph remains the strongest method across all four shot settings.

Table 18: Few-shot training recipes for ChronoMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
ChronoMorph	AdamW	5 warm-up + 55 joint	$8 / 2 \times 10^{-4}$ (encoder), 1×10^{-4} (projector, LoRA)	Report the last Phase II check-point	No	No	Class-token rows remain trainable; language-side adaptation uses LoRA in Phase II.
ResNet, Tap-Net	AdamW	Fixed schedule in the few-shot runner (60 in the benchmark configs)	model-specific / benchmark configs use 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	Fixed schedule (100 in the benchmark configs)	128 / 10^{-2} by default	Final model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	Two-phase fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs)	16 / separate backbone-head LR	Report the last Phase II check-point	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLINER, AUTO-FORMER, CROSS-FORMER, FED-FORMER, INFORMER, TIMESNET) GPT4TS	RAdam + cosine schedule	Fixed model-specific schedule from the TSLib profile or benchmark override	usually 16 / 10^{-3}	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter profile than the other TSLib baselines.
	RAdam (default runner setting)	Fixed schedule (60 in the benchmark configs)	64 / 10^{-4} in the benchmark configs	Report the last checkpoint	No	No	Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.

Table 19: Available per-dataset runtime details for the computational-efficiency probe. The full machine-readable runtime file is produced by the efficiency script; rows with missing measurements are omitted from the paper appendix.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	ChronoMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

Table 20: Ablation of aggregate curriculum pretraining on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94	3.35
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

D Ablation Tables

Table 5 in the main paper reports the compact ablation summary. Tables 20–22 expand that summary into three focused views under the same few-shot protocol. Table 23 summarizes the LLM-backbone ablation, and Table 24 reports its full per-dataset comparison on the 120 UCR datasets covered by both runs. Because UCR class identifiers are anonymous, the label-space ablation should be interpreted as testing closed-set class-token scoring; the semantic-prior class-token ablation is reported on external datasets with meaningful class names in Appendix B.6.

Table 21: Ablation of dual-view time-series-to-language modeling on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
Temporal only	43.51	50.55	58.40	64.39	54.21	5.08
Visual only	45.88	50.02	60.86	63.58	55.09	4.20
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

Table 22: Ablation of closed-set class-token label-space scoring on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full ChronoMorph model. Since UCR labels are anonymous, this ablation does not evaluate semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14	4.15
ChronoMorph	48.31	55.55	64.14	69.16	59.29	–

Table 23: Ablation of the LLM backbone on the 120 shared UCR datasets covered by both runs. Each value is benchmark-level mean accuracy (%) over the available runs recorded in the result files. Drop denotes the average accuracy decrease relative to full ChronoMorph.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
ChronoMorph	47.93	54.98	63.27	68.48	58.67	–
w/o LLM Backbone	42.95	48.20	57.06	61.01	52.31	6.36

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets. Each value is mean $\pm$ standard deviation accuracy (%) over the available runs recorded in the result files.

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
ACSF1	41.40 $\pm$ 10.01	54.60 $\pm$ 3.58	62.80 $\pm$ 4.32	71.80 $\pm$ 2.17	36.00 $\pm$ 10.37	42.80 $\pm$ 3.96	51.60 $\pm$ 5.46	62.60 $\pm$ 3.44
Adiac	45.17 $\pm$ 5.28	56.01 $\pm$ 1.25	69.51 $\pm$ 2.42	76.42 $\pm$ 1.68	26.85 $\pm$ 1.68	21.53 $\pm$ 2.11	36.37 $\pm$ 4.24	33.66 $\pm$ 2.69
AllGestureWiimoteX	19.91 $\pm$ 1.48	26.51 $\pm$ 4.36	39.40 $\pm$ 2.86	52.31 $\pm$ 2.26	20.23 $\pm$ 4.10	23.34 $\pm$ 2.88	31.00 $\pm$ 1.71	35.34 $\pm$ 1.51
AllGestureWiimoteY	25.54 $\pm$ 4.42	29.43 $\pm$ 1.69	48.11 $\pm$ 2.89	56.34 $\pm$ 3.54	23.11 $\pm$ 3.11	26.46 $\pm$ 2.53	38.97 $\pm$ 1.87	44.00 $\pm$ 2.17
AllGestureWiimoteZ	22.26 $\pm$ 3.96	27.91 $\pm$ 5.21	36.14 $\pm$ 0.97	44.43 $\pm$ 2.91	22.26 $\pm$ 4.37	25.74 $\pm$ 5.60	31.91 $\pm$ 0.57	36.06 $\pm$ 1.75
ArrowHead	48.80 $\pm$ 4.79	57.03 $\pm$ 2.81	73.60 $\pm$ 3.71	81.49 $\pm$ 3.22	44.57 $\pm$ 6.72	47.20 $\pm$ 4.58	61.03 $\pm$ 6.78	58.74 $\pm$ 3.82
BME	66.40 $\pm$ 3.88	79.47 $\pm$ 11.40	97.73 $\pm$ 1.53	99.20 $\pm$ 0.30	66.80 $\pm$ 2.92	74.53 $\pm$ 11.00	93.47 $\pm$ 5.34	93.60 $\pm$ 0.37
Beef	30.00 $\pm$ 8.16	43.33 $\pm$ 5.27	51.33 $\pm$ 6.91	68.00 $\pm$ 1.83	31.33 $\pm$ 8.69	38.67 $\pm$ 6.06	36.67 $\pm$ 7.07	48.00 $\pm$ 3.80
BeetleFly	55.00 $\pm$ 14.58	61.00 $\pm$ 12.45	57.00 $\pm$ 11.51	59.00 $\pm$ 4.18	56.00 $\pm$ 5.48	58.00 $\pm$ 4.47	51.00 $\pm$ 9.62	59.00 $\pm$ 4.18
BirdChicken	68.00 $\pm$ 14.83	69.00 $\pm$ 22.19	85.00 $\pm$ 11.73	89.00 $\pm$ 5.48	74.00 $\pm$ 12.94	76.00 $\pm$ 9.62	86.00 $\pm$ 8.94	88.00 $\pm$ 5.70
CBF	67.51 $\pm$ 4.85	85.91 $\pm$ 4.23	93.76 $\pm$ 3.90	97.71 $\pm$ 0.69	69.18 $\pm$ 4.62	78.13 $\pm$ 11.07	87.40 $\pm$ 4.28	90.04 $\pm$ 1.83
Car	48.00 $\pm$ 5.06	58.67 $\pm$ 3.80	69.67 $\pm$ 3.98	76.33 $\pm$ 5.45	43.33 $\pm$ 12.86	53.00 $\pm$ 7.21	63.33 $\pm$ 8.50	66.33 $\pm$ 3.80
Chinatown	67.17 $\pm$ 13.59	85.31 $\pm$ 4.44	82.45 $\pm$ 9.21	88.05 $\pm$ 3.11	57.26 $\pm$ 10.98	72.36 $\pm$ 12.11	78.02 $\pm$ 7.25	75.39 $\pm$ 3.06
ChlorineConcentration	34.73 $\pm$ 3.81	34.25 $\pm$ 5.73	36.45 $\pm$ 2.05	33.98 $\pm$ 7.65	34.41 $\pm$ 3.06	34.99 $\pm$ 5.96	36.05 $\pm$ 1.89	36.44 $\pm$ 3.56
CinCECGTorso	32.32 $\pm$ 4.12	44.68 $\pm$ 5.88	49.61 $\pm$ 6.52	71.09 $\pm$ 3.22	31.30 $\pm$ 5.37	40.86 $\pm$ 7.99	37.54 $\pm$ 6.84	49.80 $\pm$ 0.39
Coffee	75.71 $\pm$ 13.69	88.57 $\pm$ 4.66	94.29 $\pm$ 1.96	95.71 $\pm$ 2.99	68.57 $\pm$ 11.68	82.14 $\pm$ 15.15	72.86 $\pm$ 20.45	87.86 $\pm$ 9.65
Computers	46.00 $\pm$ 10.27	52.72 $\pm$ 10.03	55.92 $\pm$ 7.37	59.36 $\pm$ 8.46	53.20 $\pm$ 7.38	58.96 $\pm$ 7.32	58.00 $\pm$ 7.81	64.00 $\pm$ 4.03
CricketX	26.26 $\pm$ 5.29	37.33 $\pm$ 3.78	45.74 $\pm$ 2.60	56.77 $\pm$ 3.51	25.59 $\pm$ 5.37	31.90 $\pm$ 2.09	38.97 $\pm$ 3.38	47.74 $\pm$ 2.29
CricketY	25.54 $\pm$ 2.45	34.51 $\pm$ 6.10	45.64 $\pm$ 1.99	54.31 $\pm$ 1.68	25.13 $\pm$ 3.56	33.28 $\pm$ 2.21	36.62 $\pm$ 1.65	41.95 $\pm$ 1.59
CricketZ	29.28 $\pm$ 4.82	33.59 $\pm$ 5.51	49.59 $\pm$ 2.62	57.38 $\pm$ 4.03	28.36 $\pm$ 1.29	29.44 $\pm$ 4.90	42.87 $\pm$ 1.47	46.62 $\pm$ 1.96
DiatomSizeReduction	85.82 $\pm$ 8.55	90.39 $\pm$ 5.62	93.66 $\pm$ 1.97	92.61 $\pm$ 2.00	86.73 $\pm$ 8.56	85.62 $\pm$ 7.19	77.78 $\pm$ 3.00	75.23 $\pm$ 1.32
DistalPhalanxOutlineAgeGroup	40.72 $\pm$ 17.98	67.77 $\pm$ 3.39	69.35 $\pm$ 1.20	70.65 $\pm$ 2.18	43.31 $\pm$ 18.16	70.22 $\pm$ 4.60	70.65 $\pm$ 1.56	71.80 $\pm$ 1.79
DistalPhalanxOutlineCorrect	53.91 $\pm$ 13.35	55.80 $\pm$ 14.95	63.77 $\pm$ 11.63	65.51 $\pm$ 11.85	60.58 $\pm$ 6.43	51.23 $\pm$ 12.26	64.49 $\pm$ 7.76	59.06 $\pm$ 13.65
DistalPhalanxTW	54.53 $\pm$ 1.72	51.37 $\pm$ 8.97	56.40 $\pm$ 4.87	54.10 $\pm$ 3.11	51.51 $\pm$ 3.83	47.48 $\pm$ 13.90	57.27 $\pm$ 3.58	55.11 $\pm$ 1.40
DodgerLoopGame	56.96 $\pm$ 11.69	61.01 $\pm$ 13.03	72.61 $\pm$ 10.25	83.62 $\pm$ 0.65	50.29 $\pm$ 8.64	54.93 $\pm$ 10.24	60.58 $\pm$ 4.46	73.19 $\pm$ 1.36
DodgerLoopWeekend	85.36 $\pm$ 4.30	84.64 $\pm$ 2.73	93.62 $\pm$ 2.37	97.83 $\pm$ 0.00	87.68 $\pm$ 3.80	84.35 $\pm$ 2.83	88.26 $\pm$ 2.14	92.32 $\pm$ 0.40
ECG200	60.60 $\pm$ 6.91	73.00 $\pm$ 3.46	67.40 $\pm$ 8.91	77.40 $\pm$ 4.88	62.80 $\pm$ 12.30	67.80 $\pm$ 5.22	68.60 $\pm$ 2.97	77.60 $\pm$ 2.70
ECG5000	67.18 $\pm$ 10.57	67.68 $\pm$ 9.76	73.89 $\pm$ 8.21	73.78 $\pm$ 6.18	64.82 $\pm$ 0.00	39.33 $\pm$ 0.00	85.18 $\pm$ 0.00	80.11 $\pm$ 0.00
ECGFiveDays	61.77 $\pm$ 6.61	59.35 $\pm$ 9.08	70.96 $\pm$ 3.91	75.28 $\pm$ 1.71	58.07 $\pm$ 0.00	71.31 $\pm$ 0.00	70.85 $\pm$ 0.00	76.42 $\pm$ 0.00
EOGHorizontalSignal	29.06 $\pm$ 4.16	37.35 $\pm$ 4.07	46.35 $\pm$ 2.27	49.17 $\pm$ 1.62	31.22 $\pm$ 0.00	33.15 $\pm$ 0.00	45.30 $\pm$ 0.00	44.20 $\pm$ 0.00
EOGVerticalSignal	27.35 $\pm$ 1.90	31.82 $\pm$ 3.47	36.74 $\pm$ 2.24	42.21 $\pm$ 2.26	22.10 $\pm$ 0.00	33.70 $\pm$ 0.00	34.53 $\pm$ 0.00	38.95 $\pm$ 0.00
Earthquakes	55.54 $\pm$ 11.57	55.25 $\pm$ 14.33	54.68 $\pm$ 9.31	54.68 $\pm$ 7.63	50.36 $\pm$ 0.00	61.87 $\pm$ 0.00	39.57 $\pm$ 0.00	42.45 $\pm$ 0.00
ElectricDevices	33.14 $\pm$ 10.96	47.14 $\pm$ 3.99	47.06 $\pm$ 4.18	49.42 $\pm$ 5.02	17.31 $\pm$ 0.00	46.54 $\pm$ 0.00	47.30 $\pm$ 0.00	47.89 $\pm$ 0.00
EthanolLevel	26.68 $\pm$ 1.86	27.00 $\pm$ 3.11	25.92 $\pm$ 1.40	28.24 $\pm$ 0.50	26.20 $\pm$ 0.00	27.20 $\pm$ 0.00	27.00 $\pm$ 0.00	24.00 $\pm$ 0.00
FaceAll	32.84 $\pm$ 6.86	39.09 $\pm$ 1.28	54.54 $\pm$ 1.26	63.01 $\pm$ 1.34	32.49 $\pm$ 0.00	28.58 $\pm$ 0.00	40.30 $\pm$ 0.00	44.02 $\pm$ 0.00
FaceFour	54.77 $\pm$ 8.25	64.55 $\pm$ 4.91	75.68 $\pm$ 4.37	79.09 $\pm$ 1.90	46.59 $\pm$ 0.00	53.41 $\pm$ 0.00	65.91 $\pm$ 0.00	73.86 $\pm$ 0.00
FacesUCR	50.37 $\pm$ 2.98	55.08 $\pm$ 2.32	68.58 $\pm$ 2.57	79.08 $\pm$ 1.41	46.83 $\pm$ 0.00	46.68 $\pm$ 0.00	54.83 $\pm$ 0.00	58.29 $\pm$ 0.00
FiftyWords	36.48 $\pm$ 2.66	47.30 $\pm$ 2.89	61.58 $\pm$ 2.55	71.34 $\pm$ 2.10	31.87 $\pm$ 0.00	33.41 $\pm$ 0.00	45.05 $\pm$ 0.00	54.51 $\pm$ 0.00
Fish	34.74 $\pm$ 4.91	48.91 $\pm$ 10.25	64.57 $\pm$ 5.92	78.29 $\pm$ 3.92	36.00 $\pm$ 0.00	34.29 $\pm$ 0.00	54.86 $\pm$ 0.00	61.71 $\pm$ 0.00
FordA	50.71 $\pm$ 0.98	53.35 $\pm$ 2.84	54.00 $\pm$ 2.22	57.15 $\pm$ 1.85	51.44 $\pm$ 0.00	51.36 $\pm$ 0.00	50.61 $\pm$ 0.00	48.26 $\pm$ 0.00
FordB	49.75 $\pm$ 2.56	50.42 $\pm$ 2.72	52.32 $\pm$ 2.78	55.60 $\pm$ 1.91	52.84 $\pm$ 0.00	49.75 $\pm$ 0.00	51.60 $\pm$ 0.00	52.84 $\pm$ 0.00
FreezerRegularTrain	61.90 $\pm$ 22.25	79.14 $\pm$ 7.94	85.11 $\pm$ 3.83	89.20 $\pm$ 5.65	76.35 $\pm$ 0.00	69.68 $\pm$ 0.00	81.09 $\pm$ 0.00	78.95 $\pm$ 0.00
FreezerSmallTrain	66.53 $\pm$ 10.64	64.30 $\pm$ 13.28	75.05 $\pm$ 3.10	82.86 $\pm$ 3.49	67.30 $\pm$ 0.00	75.09 $\pm$ 0.00	77.82 $\pm$ 0.00	82.63 $\pm$ 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Fungi	79.35 ± 5.30	80.97 ± 3.24	81.18 ± 3.16	79.57 ± 1.20	40.32 ± 0.00	37.10 ± 0.00	40.86 ± 0.00	51.61 ± 0.00
GestureMidAirD1	32.00 ± 3.94	44.92 ± 6.31	60.31 ± 2.35	67.08 ± 1.00	22.31 ± 0.00	36.92 ± 0.00	44.62 ± 0.00	50.00 ± 0.00
GestureMidAirD2	33.85 ± 3.03	43.85 ± 4.03	54.62 ± 1.96	55.69 ± 1.85	25.38 ± 0.00	34.62 ± 0.00	44.62 ± 0.00	42.31 ± 0.00
GestureMidAirD3	18.46 ± 2.61	21.08 ± 1.85	32.15 ± 3.33	33.69 ± 2.63	15.38 ± 0.00	20.00 ± 0.00	20.00 ± 0.00	23.08 ± 0.00
GesturePebbleZ1	35.81 ± 4.48	37.91 ± 6.35	63.84 ± 8.99	75.12 ± 5.19	40.70 ± 0.00	36.05 ± 0.00	49.42 ± 0.00	64.53 ± 0.00
GesturePebbleZ2	33.29 ± 5.85	41.39 ± 4.24	60.89 ± 8.06	75.57 ± 4.64	24.68 ± 0.00	47.47 ± 0.00	54.43 ± 0.00	67.72 ± 0.00
GunPoint	58.00 ± 7.79	79.33 ± 16.25	90.13 ± 10.49	97.47 ± 2.23	60.67 ± 0.00	60.67 ± 0.00	85.33 ± 0.00	90.00 ± 0.00
GunPointAgeSpan	60.51 ± 10.32	64.49 ± 7.90	81.52 ± 9.45	92.72 ± 5.80	61.71 ± 0.00	52.22 ± 0.00	71.52 ± 0.00	71.84 ± 0.00
GunPointMaleVersusFemale	81.01 ± 10.12	87.66 ± 5.66	94.62 ± 6.24	98.54 ± 0.94	64.56 ± 0.00	77.53 ± 0.00	98.10 ± 0.00	97.15 ± 0.00
GunPointOldVersusYoung	57.97 ± 10.78	70.92 ± 5.05	80.44 ± 8.63	86.79 ± 4.71	54.29 ± 0.00	71.11 ± 0.00	76.83 ± 0.00	80.32 ± 0.00
Ham	52.76 ± 9.30	50.86 ± 8.07	60.57 ± 9.91	62.86 ± 3.37	61.90 ± 0.00	64.76 ± 0.00	65.71 ± 0.00	58.10 ± 0.00
HandOutlines	61.73 ± 6.31	64.59 ± 10.14	68.54 ± 6.21	71.51 ± 5.35	62.43 ± 0.00	66.49 ± 0.00	72.16 ± 0.00	55.68 ± 0.00
Haptics	25.65 ± 4.99	26.62 ± 4.64	26.95 ± 4.88	39.22 ± 4.29	19.81 ± 0.00	27.92 ± 0.00	32.79 ± 0.00	33.77 ± 0.00
Herring	50.00 ± 10.00	55.00 ± 4.34	45.00 ± 4.87	55.31 ± 5.01	53.12 ± 0.00	59.38 ± 0.00	45.31 ± 0.00	62.50 ± 0.00
HouseTwenty	62.18 ± 8.04	71.93 ± 10.30	84.37 ± 7.51	89.24 ± 2.81	65.55 ± 0.00	56.30 ± 0.00	62.18 ± 0.00	85.71 ± 0.00
InlineSkate	17.13 ± 2.24	18.98 ± 2.54	28.07 ± 1.19	33.49 ± 2.77	13.45 ± 0.00	19.64 ± 0.00	23.27 ± 0.00	25.45 ± 0.00
InsectEPGRegularTrain	55.34 ± 14.47	68.76 ± 9.60	80.24 ± 3.80	86.59 ± 4.06	54.22 ± 0.00	57.03 ± 0.00	75.90 ± 0.00	77.51 ± 0.00
InsectEPGSmallTrain	50.76 ± 7.86	67.79 ± 2.45	83.05 ± 2.80	85.06 ± 2.08	34.14 ± 0.00	67.47 ± 0.00	71.49 ± 0.00	77.51 ± 0.00
InsectWingbeatSound	25.27 ± 2.20	31.49 ± 3.72	41.44 ± 2.25	51.36 ± 2.05	25.86 ± 0.00	30.00 ± 0.00	35.30 ± 0.00	38.43 ± 0.00
ItalyPowerDemand	58.52 ± 15.27	65.03 ± 13.70	84.26 ± 5.94	92.19 ± 2.62	86.98 ± 0.00	53.35 ± 0.00	85.33 ± 0.00	92.03 ± 0.00
LargeKitchenAppliances	45.44 ± 5.60	42.56 ± 5.00	50.13 ± 3.07	55.09 ± 5.30	44.53 ± 0.00	43.20 ± 0.00	50.93 ± 0.00	50.67 ± 0.00
Lightning2	59.67 ± 8.87	63.28 ± 8.25	69.51 ± 10.47	77.05 ± 4.18	55.74 ± 0.00	63.93 ± 0.00	62.30 ± 0.00	68.85 ± 0.00
Lightning7	41.64 ± 7.60	46.03 ± 6.17	66.30 ± 3.83	66.85 ± 7.34	47.95 ± 0.00	58.90 ± 0.00	63.01 ± 0.00	69.86 ± 0.00
Mallat	68.24 ± 7.54	80.67 ± 4.06	90.88 ± 1.15	89.72 ± 1.50	58.68 ± 0.00	70.02 ± 0.00	66.10 ± 0.00	60.00 ± 0.00
Meat	55.67 ± 9.62	57.67 ± 17.30	64.00 ± 11.82	64.00 ± 6.08	51.67 ± 0.00	56.67 ± 0.00	75.00 ± 0.00	73.33 ± 0.00
MedicalImages	27.66 ± 4.41	31.63 ± 5.59	42.03 ± 2.75	54.68 ± 2.10	25.53 ± 0.00	35.39 ± 0.00	38.68 ± 0.00	53.29 ± 0.00
MelbournePedestrian	39.16 ± 5.22	49.78 ± 4.68	65.81 ± 3.77	70.88 ± 1.49	37.15 ± 0.00	41.94 ± 0.00	52.85 ± 0.00	65.23 ± 0.00
MiddlePhalanxOutlineAgeGroup	39.48 ± 7.40	42.99 ± 12.95	49.74 ± 4.72	51.04 ± 7.11	43.51 ± 0.00	31.17 ± 0.00	60.39 ± 0.00	52.60 ± 0.00
MiddlePhalanxOutlineCorrect	51.68 ± 5.00	48.87 ± 3.56	48.45 ± 7.93	59.18 ± 5.65	41.92 ± 0.00	54.98 ± 0.00	62.20 ± 0.00	64.26 ± 0.00
MiddlePhalanxTW	37.14 ± 16.84	44.81 ± 7.78	43.64 ± 5.71	44.42 ± 6.27	27.27 ± 0.00	46.75 ± 0.00	22.08 ± 0.00	48.05 ± 0.00
MixedShapesRegularTrain	56.06 ± 5.34	67.98 ± 4.06	76.68 ± 5.11	79.60 ± 1.74	48.62 ± 0.00	69.44 ± 0.00	68.49 ± 0.00	74.68 ± 0.00
MixedShapesSmallTrain	63.30 ± 3.34	68.35 ± 3.26	78.71 ± 1.70	81.61 ± 2.47	64.54 ± 0.00	65.81 ± 0.00	61.90 ± 0.00	71.88 ± 0.00
MoteStrain	54.36 ± 12.77	58.56 ± 15.12	80.37 ± 4.50	89.23 ± 1.53	68.61 ± 0.00	50.00 ± 0.00	82.27 ± 0.00	83.39 ± 0.00
NonInvasiveFetalECGThorax1	39.16 ± 2.28	50.58 ± 2.13	67.42 ± 0.56	76.84 ± 1.40	24.53 ± 0.00	36.49 ± 0.00	46.41 ± 0.00	51.30 ± 0.00
NonInvasiveFetalECGThorax2	47.48 ± 2.06	59.72 ± 2.76	75.31 ± 0.82	82.53 ± 1.33	39.64 ± 0.00	46.87 ± 0.00	55.73 ± 0.00	66.77 ± 0.00
OSULeaf	29.83 ± 5.82	36.28 ± 6.97	49.92 ± 6.46	59.42 ± 3.58	33.47 ± 0.00	29.75 ± 0.00	43.80 ± 0.00	49.17 ± 0.00
OliveOil	28.00 ± 14.26	23.33 ± 9.72	28.67 ± 11.69	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00	16.67 ± 0.00	40.00 ± 0.00
PLAID	32.10 ± 5.72	40.19 ± 5.06	52.96 ± 5.39	59.89 ± 3.28	21.60 ± 0.00	36.13 ± 0.00	53.63 ± 0.00	58.66 ± 0.00
PhalangesOutlinesCorrect	51.93 ± 9.54	50.84 ± 2.93	53.61 ± 8.69	55.73 ± 8.07	61.89 ± 0.00	54.55 ± 0.00	36.60 ± 0.00	65.27 ± 0.00
Phoneme	6.87 ± 1.15	10.02 ± 0.59	15.63 ± 1.56	21.95 ± 1.27	7.12 ± 0.00	10.50 ± 0.00	15.82 ± 0.00	17.25 ± 0.00
PickupGestureWiimoteZ	47.20 ± 7.29	58.80 ± 3.03	68.00 ± 2.45	70.40 ± 2.61	26.00 ± 0.00	54.00 ± 0.00	60.00 ± 0.00	52.00 ± 0.00
PigAirwayPressure	5.48 ± 2.19	9.81 ± 0.43	9.81 ± 0.73	9.62 ± 0.48	5.77 ± 0.00	7.21 ± 0.00	7.69 ± 0.00	4.81 ± 0.00
PigArtPressure	15.38 ± 4.07	30.67 ± 1.38	33.08 ± 1.33	31.73 ± 1.92	17.79 ± 0.00	25.00 ± 0.00	23.56 ± 0.00	24.52 ± 0.00
PigCVP	11.44 ± 2.08	19.33 ± 0.99	19.52 ± 0.55	19.52 ± 1.34	10.58 ± 0.00	16.83 ± 0.00	15.87 ± 0.00	18.27 ± 0.00

Table 24: Per-dataset LLM-backbone ablation on 120 shared UCR datasets (continued).

Dataset	ChronoMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
Plane	93.90 $\pm$ 6.92	99.24 $\pm$ 0.80	99.24 $\pm$ 1.24	99.24 $\pm$ 1.24	91.43 $\pm$ 0.00	90.48 $\pm$ 0.00	100.00 $\pm$ 0.00	99.05 $\pm$ 0.00
PowerCons	60.89 $\pm$ 11.09	66.78 $\pm$ 15.52	76.11 $\pm$ 6.55	82.33 $\pm$ 5.46	77.78 $\pm$ 0.00	48.89 $\pm$ 0.00	52.22 $\pm$ 0.00	78.33 $\pm$ 0.00
ProximalPhalanxOutlineAgeGroup	48.98 $\pm$ 23.27	66.24 $\pm$ 17.89	76.88 $\pm$ 5.79	79.80 $\pm$ 3.30	78.54 $\pm$ 0.00	81.46 $\pm$ 0.00	83.90 $\pm$ 0.00	76.10 $\pm$ 0.00
ProximalPhalanxOutlineCorrect	52.23 $\pm$ 12.03	59.93 $\pm$ 12.82	65.84 $\pm$ 4.70	60.07 $\pm$ 4.20	59.79 $\pm$ 0.00	64.26 $\pm$ 0.00	62.89 $\pm$ 0.00	62.89 $\pm$ 0.00
ProximalPhalanxTW	55.22 $\pm$ 6.63	57.76 $\pm$ 7.59	66.24 $\pm$ 3.19	69.66 $\pm$ 3.47	60.98 $\pm$ 0.00	39.02 $\pm$ 0.00	73.17 $\pm$ 0.00	45.37 $\pm$ 0.00
RefrigerationDevices	29.76 $\pm$ 11.02	33.39 $\pm$ 7.13	40.96 $\pm$ 5.59	41.23 $\pm$ 4.90	31.20 $\pm$ 0.00	26.40 $\pm$ 0.00	41.60 $\pm$ 0.00	40.00 $\pm$ 0.00
Rock	39.20 $\pm$ 13.75	64.00 $\pm$ 7.07	76.00 $\pm$ 3.16	76.40 $\pm$ 2.97	20.00 $\pm$ 0.00	66.00 $\pm$ 0.00	56.00 $\pm$ 0.00	64.00 $\pm$ 0.00
ScreenType	32.53 $\pm$ 4.84	35.63 $\pm$ 4.40	34.51 $\pm$ 4.14	37.12 $\pm$ 3.93	33.60 $\pm$ 0.00	33.60 $\pm$ 0.00	35.73 $\pm$ 0.00	33.33 $\pm$ 0.00
SemgHandGenderCh2	51.90 $\pm$ 14.92	59.73 $\pm$ 10.87	66.80 $\pm$ 7.25	63.80 $\pm$ 10.73	42.00 $\pm$ 0.00	68.17 $\pm$ 0.00	60.33 $\pm$ 0.00	70.67 $\pm$ 0.00
SemgHandMovementCh2	28.13 $\pm$ 4.95	31.16 $\pm$ 4.70	36.31 $\pm$ 5.92	45.56 $\pm$ 2.93	28.89 $\pm$ 0.00	26.00 $\pm$ 0.00	32.00 $\pm$ 0.00	37.78 $\pm$ 0.00
SemgHandSubjectCh2	37.42 $\pm$ 8.82	43.20 $\pm$ 6.72	54.31 $\pm$ 3.37	66.71 $\pm$ 4.45	41.56 $\pm$ 0.00	44.22 $\pm$ 0.00	37.33 $\pm$ 0.00	56.89 $\pm$ 0.00
ShakeGestureWiimoteZ	60.40 $\pm$ 7.92	81.60 $\pm$ 2.61	92.00 $\pm$ 0.00	91.60 $\pm$ 2.97	70.00 $\pm$ 0.00	72.00 $\pm$ 0.00	82.00 $\pm$ 0.00	84.00 $\pm$ 0.00
ShapeletSim	62.44 $\pm$ 7.44	67.44 $\pm$ 10.92	92.00 $\pm$ 3.88	96.67 $\pm$ 1.96	52.78 $\pm$ 0.00	61.11 $\pm$ 0.00	86.67 $\pm$ 0.00	96.67 $\pm$ 0.00
ShapesAll	50.20 $\pm$ 3.03	63.03 $\pm$ 2.22	76.20 $\pm$ 1.83	83.13 $\pm$ 0.32	46.50 $\pm$ 0.00	52.67 $\pm$ 0.00	64.00 $\pm$ 0.00	71.83 $\pm$ 0.00
SmallKitchenAppliances	42.88 $\pm$ 15.27	55.68 $\pm$ 8.74	51.63 $\pm$ 12.25	57.60 $\pm$ 5.56	54.93 $\pm$ 0.00	56.00 $\pm$ 0.00	45.07 $\pm$ 0.00	54.40 $\pm$ 0.00
SmoothSubspace	68.53 $\pm$ 8.95	85.07 $\pm$ 5.20	88.00 $\pm$ 2.87	89.47 $\pm$ 2.60	58.67 $\pm$ 0.00	77.33 $\pm$ 0.00	76.00 $\pm$ 0.00	87.33 $\pm$ 0.00
SonyAIBORobotSurface1	57.20 $\pm$ 8.62	71.85 $\pm$ 8.28	84.86 $\pm$ 2.67	86.09 $\pm$ 3.97	67.22 $\pm$ 0.00	69.72 $\pm$ 0.00	70.88 $\pm$ 0.00	78.04 $\pm$ 0.00
SonyAIBORobotSurface2	66.09 $\pm$ 10.78	68.21 $\pm$ 5.28	77.21 $\pm$ 3.67	84.11 $\pm$ 2.29	62.12 $\pm$ 0.00	67.68 $\pm$ 0.00	73.87 $\pm$ 0.00	75.87 $\pm$ 0.00
StarLightCurves	63.03 $\pm$ 14.23	74.47 $\pm$ 15.08	81.86 $\pm$ 7.93	85.26 $\pm$ 3.54	64.24 $\pm$ 0.00	77.25 $\pm$ 0.00	85.65 $\pm$ 0.00	82.39 $\pm$ 0.00
Strawberry	51.46 $\pm$ 5.21	59.41 $\pm$ 6.38	58.05 $\pm$ 7.04	75.84 $\pm$ 0.99	54.05 $\pm$ 0.00	54.59 $\pm$ 0.00	52.97 $\pm$ 0.00	59.73 $\pm$ 0.00
SwedishLeaf	57.98 $\pm$ 2.72	64.99 $\pm$ 3.83	79.78 $\pm$ 2.56	85.89 $\pm$ 1.27	52.96 $\pm$ 0.00	56.48 $\pm$ 0.00	72.64 $\pm$ 0.00	74.24 $\pm$ 0.00
Symbols	80.94 $\pm$ 7.54	82.69 $\pm$ 3.43	88.04 $\pm$ 1.30	89.33 $\pm$ 0.82	86.03 $\pm$ 0.00	80.10 $\pm$ 0.00	91.16 $\pm$ 0.00	76.98 $\pm$ 0.00
SyntheticControl	84.20 $\pm$ 6.54	87.93 $\pm$ 3.95	95.07 $\pm$ 1.62	97.33 $\pm$ 0.85	75.00 $\pm$ 0.00	80.67 $\pm$ 0.00	90.33 $\pm$ 0.00	90.67 $\pm$ 0.00
ToeSegmentation1	58.16 $\pm$ 3.70	60.26 $\pm$ 8.54	70.53 $\pm$ 10.39	80.53 $\pm$ 2.75	67.11 $\pm$ 0.00	67.98 $\pm$ 0.00	67.54 $\pm$ 0.00	76.75 $\pm$ 0.00
ToeSegmentation2	75.23 $\pm$ 12.13	76.15 $\pm$ 12.44	78.00 $\pm$ 6.36	80.31 $\pm$ 4.70	69.23 $\pm$ 0.00	70.00 $\pm$ 0.00	66.92 $\pm$ 0.00	70.77 $\pm$ 0.00
Trace	99.20 $\pm$ 1.79	98.40 $\pm$ 3.05	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	97.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
UWaveGestureLibraryZ	46.44 $\pm$ 2.61	52.88 $\pm$ 2.03	55.54 $\pm$ 2.61	60.65 $\pm$ 2.84	38.11 $\pm$ 0.00	41.57 $\pm$ 0.00	51.20 $\pm$ 0.00	56.31 $\pm$ 0.00
Wafer	46.27 $\pm$ 7.11	48.01 $\pm$ 15.62	71.65 $\pm$ 9.58	82.53 $\pm$ 8.58	68.14 $\pm$ 0.00	67.07 $\pm$ 0.00	86.02 $\pm$ 0.00	80.50 $\pm$ 0.00
Wine	51.11 $\pm$ 7.48	48.89 $\pm$ 5.34	51.48 $\pm$ 2.03	49.26 $\pm$ 2.11	61.11 $\pm$ 0.00	57.41 $\pm$ 0.00	50.00 $\pm$ 0.00	48.15 $\pm$ 0.00
WordSynonyms	20.91 $\pm$ 2.73	32.92 $\pm$ 2.64	47.05 $\pm$ 3.87	58.90 $\pm$ 2.04	13.64 $\pm$ 0.00	21.79 $\pm$ 0.00	25.24 $\pm$ 0.00	30.09 $\pm$ 0.00
Worms	24.68 $\pm$ 8.95	38.70 $\pm$ 4.63	44.42 $\pm$ 6.26	48.57 $\pm$ 4.91	42.86 $\pm$ 0.00	35.06 $\pm$ 0.00	42.86 $\pm$ 0.00	44.16 $\pm$ 0.00
WormsTwoClass	47.53 $\pm$ 10.69	54.03 $\pm$ 7.72	54.81 $\pm$ 7.59	56.36 $\pm$ 3.85	54.55 $\pm$ 0.00	45.45 $\pm$ 0.00	51.95 $\pm$ 0.00	55.84 $\pm$ 0.00
Yoga	50.42 $\pm$ 4.51	50.64 $\pm$ 4.90	51.31 $\pm$ 4.34	57.73 $\pm$ 2.05	51.50 $\pm$ 0.00	55.20 $\pm$ 0.00	53.20 $\pm$ 0.00	53.63 $\pm$ 0.00
Average	47.93	54.98	63.27	68.48	46.95	51.20	57.06	61.01

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes].

Justification: The abstract, introduction, experiments, and limitations state that the main claims concern few-shot UCR classification, dual-view modeling, curriculum transfer, and class-token scoring; external and semantic-prior analyses are reported as additional transfer checks.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The paper includes a Limitations section discussing modest gains over strong CNN/prototype baselines, larger inference cost, dataset-dependent semantic-prior effects, and deployment caveats.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A].

Justification: The paper is empirical and does not present theoretical theorems or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: The experimental protocol, data splits, support-set sampling, pretraining data usage, baseline families, metrics, implementation details, and full per-dataset results are described in the main paper and appendix.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes].

Justification: We will release an anonymized code package with the submission, including the training and evaluation scripts used for ChronoMorph and the reported baselines, environment files, and data-access instructions. Appendix B.2 summarizes the release contents and notes that raw third-party datasets and gated pretrained model weights should be obtained from their original providers.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: Training/test splits, shot construction, optimizer and checkpoint-selection rules, model hyperparameters, and baseline recipes are specified in Section 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: The main protocol averages five independent support-set samples; full appendix tables report mean and standard deviation over those runs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes].

Justification: The paper reports the local compute environment, GPU type, worker setup, and project-level GPU-hour budget for the main UCR benchmark, baseline runs, external-dataset runs, ablations, and efficiency probe in Appendix B.4. Section 4.3 and Appendix B.7 additionally provide peak memory, latency, throughput, updated-parameter counts, and per-dataset runtime details for the efficiency measurements.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes].

Justification: The study uses public time-series datasets and pretrained backbones for empirical evaluation, reports limitations, and does not involve deceptive deployment, human-subject experiments, or release of high-risk generated content.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes].

Justification: The Limitations section discusses possible positive uses in scarce-label time-series domains and cautions that clinical or safety-critical deployment would require additional validation, privacy review, and failure analysis.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release a high-risk generative model, scraped dataset, or other asset requiring special misuse safeguards.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes].

Justification: Appendix B.3 lists the datasets, pretrained models, and external code assets used in the paper, together with their source URLs, versions where available, licenses or stated terms, and redistribution status. We cite the original creators and do not redistribute third-party raw datasets or gated pretrained model weights.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#).

Justification: The new code asset will be released as an anonymized research artifact under the MIT license, with README instructions, environment files, experiment scripts, and documentation of the supported datasets and limitations; see Appendix B.2.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#).

Justification: The work does not involve crowdsourcing experiments or direct human-subject studies.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

801 **15. Institutional review board (IRB) approvals or equivalent for research with human**
802 **subjects**

803 Question: Does the paper describe potential risks incurred by study participants, whether
804 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
805 approvals (or an equivalent approval/review based on the requirements of your country or
806 institution) were obtained?

807 Answer: [N/A].

808 Justification: The work does not involve direct human-subject research requiring IRB
809 approval in this paper.

810 Guidelines:

- 811 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
812 with human subjects.
- 813 • Depending on the country in which research is conducted, IRB approval (or equivalent)
814 may be required for any human subjects research. If you obtained IRB approval, you
815 should clearly state this in the paper.
- 816 • We recognize that the procedures for this may vary significantly between institutions
817 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
818 guidelines for their institution.
- 819 • For initial submissions, do not include any information that would break anonymity (if
820 applicable), such as the institution conducting the review.

821 **16. Declaration of LLM usage**

822 Question: Does the paper describe the usage of LLMs if it is an important, original, or
823 non-standard component of the core methods in this research? Note that if the LLM is used
824 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
825 scientific rigor, or originality of the research, declaration is not required.

826 Answer: [Yes].

827 Justification: LLM usage is central to the proposed method and is described throughout the
828 method and experimental sections, including the Llama-3.2-1B backbone and class-token
829 scoring interface.

830 Guidelines:

- 831 • The answer [N/A] means that the core method development in this research does not
832 involve LLMs as any important, original, or non-standard components.
- 833 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
834 be described.